

# ANALYTIC AND ALGEBRAIC OKA-1 APPROXIMATION FOR SMOOTH PROJECTIVE MORPHISMS WITH RATIONALLY CONNECTED FIBERS

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**ABSTRACT.** Let  $\pi : Z \rightarrow Y$  be a smooth projective morphism of complex manifolds with connected rationally connected fibers. We prove holomorphic approximation on arbitrary compact sets and finite-jet interpolation on arbitrary closed discrete sets for continuous liftings defined on open Riemann surfaces and holomorphic near those sets. For smooth projective morphisms of smooth complex algebraic varieties and algebraic base maps from smooth affine curves, the approximating liftings can be chosen algebraic, with interpolation on any finite set. In both cases the resulting lifting is homotopic to the initial one through continuous liftings of the fixed base map. For connected smooth projective complex manifolds, this gives the equivalence between the algebraic Oka-1 property and rational connectedness. Every rationally connected smooth projective complex manifold is also Oka-1.

## 1. INTRODUCTION

We study approximation and finite-jet interpolation for liftings of maps from complex curves through smooth projective morphisms. Rational connectedness of the fibers suffices for a holomorphic endpoint on an open Riemann surface and, when the source and morphisms are algebraic, for an algebraic endpoint on a smooth affine curve. The resulting lifting is homotopic to the given one through continuous liftings of the fixed base map. The homotopy is obtained by a topological argument on the open source. The algebraic construction also uses a cutoff homotopy on the projective completion of that source.

Alarcón–Forstnerič conjectured that every rationally connected smooth projective manifold is Oka-1 [1, Conjecture 9.1]. Forstnerič–Lárusson introduced the algebraic Oka-1 property and proved that rational connectedness is necessary in the smooth projective case [8, Definition 1.5 and Proposition 1.7]. We prove the relative analytic and algebraic approximation statements below. Specializing Theorems 1.5 and 1.7 to  $Y = \{\text{pt}\}$  settles the analytic conjecture and proves that rational connectedness also suffices for the algebraic Oka-1 property.

Forstnerič–Lárusson proved that compact rational manifolds and algebraically elliptic projective manifolds are algebraically Oka-1 [8, Corollary 3.5 and Theorem 4.1]. The algebraic Oka-1 property implies the Oka-1 property [8, Proposition 1.9]. Our analytic argument isolates the first-jet sphere families supplied by rational connectedness; the algebraic argument uses the same section equation after a separate construction of a bundle with vanishing first cohomology.

**1.1. Conventions and lifting properties.** Throughout this paper, open Riemann surfaces are assumed to be connected. By a *compact bordered Riemann surface* we mean the closure of a relatively compact smoothly bounded domain in an open Riemann surface. A map is holomorphic near a compact set if it is holomorphic on an open neighborhood of that set. We call a compact subset  $K$  of an open Riemann surface  $R$  *Runge* if  $R \setminus K$  has no relatively compact connected component. The term *closed discrete* means that  $A \subset R$  is closed and every point of  $R$  has a neighborhood meeting  $A$  in finitely many points.

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If two holomorphic maps  $f$  and  $F$  are defined near  $a \in R$  and have the same value at  $a$ , choose a source coordinate  $z$  with  $z(a) = 0$  and a holomorphic target chart  $\phi$  at  $f(a) = F(a)$ . We write  $j_a^k F = j_a^k f$  if

$$\left. \frac{d^j}{dz^j} (\phi \circ F - \phi \circ f) \right|_{z=0} = 0 \quad (0 \leq j \leq k).$$

Equivalently, each component of  $\phi \circ F - \phi \circ f$  is divisible by  $z^{k+1}$ . The jet-equality condition is independent of the coordinates. A jet of order  $k$  is equivalently a restriction to the infinitesimal neighborhood of length  $k+1$ ; in particular, a first jet corresponds to the divisor  $2[a]$ . For a closed discrete set  $A$ , an *order function* is a map  $q : A \rightarrow \mathbb{N}$ , where  $\mathbb{N}$  denotes the set of nonnegative integers. Fix a smooth Hermitian metric on every complex manifold used as a target and denote the associated distance by  $d$  with a subscript when needed. All algebraic varieties are over  $\mathbb{C}$ , unless another ground field is specified. Distances and holomorphic jets of algebraic maps are understood on their analytifications. Unless otherwise stated, a smooth algebraic curve is connected, and hence integral.

Rational connectedness of a connected smooth projective manifold  $X$  means that there is a nonempty Zariski-open subset  $U \subset X$  such that any  $x, y \in U$  lie on the image of some morphism  $h : \mathbb{P}^1 \rightarrow X$ .

**Definition 1.1.** A holomorphic map  $\pi : Z \rightarrow Y$  between complex manifolds is called *smooth projective* if  $\pi$  is a holomorphic submersion and every point of  $Y$  has an open neighborhood  $U$  for which there exist an integer  $N$  and a closed holomorphic embedding

$$Z|_U \hookrightarrow U \times \mathbb{P}^N$$

over  $U$ , where  $Z|_U = \pi^{-1}(U)$ . Thus projectivity in this analytic definition is local on the base; no global relatively ample line bundle is chosen. In particular,  $\pi$  is proper and every nonempty fiber  $Z_y$  is a smooth projective manifold. Throughout, a rationally connected fiber is understood to be nonempty, so the morphisms in the main theorems are surjective.

For a continuous map  $q : E \rightarrow B$ , the *Serre fibration* condition can be expressed as follows for every CW complex  $T$ . Given continuous maps  $a : T \rightarrow E$  and  $H : T \times [0, 1] \rightarrow B$  with  $H(t, 0) = q(a(t))$ , there is a continuous map  $\tilde{H} : T \times [0, 1] \rightarrow E$  satisfying

$$q \circ \tilde{H} = H, \quad \tilde{H}(t, 0) = a(t).$$

The following lifting property is the Oka-1 map condition of [1, Definition 7.7].

**Definition 1.2.** We call a holomorphic map  $\pi : Z \rightarrow Y$  an *Oka-1 map* if it is a Serre fibration with the following lifting property. Let  $R$  be an open Riemann surface, let  $K \subset R$  be compact and Runge, let  $A \subset K$  be finite, and let  $g : R \rightarrow Y$  be holomorphic. Suppose that  $f : R \rightarrow Z$  is a continuous lifting of  $g$ , meaning  $\pi \circ f = g$ , and is holomorphic on a neighborhood of  $K$ . Given  $\epsilon > 0$  and an integer  $k_a \geq 0$  for every  $a \in A$ , there are a holomorphic lifting  $F : R \rightarrow Z$  of  $g$  and a homotopy from  $f$  to  $F$  through continuous liftings  $f_t$  of  $g$ , with  $t \in [0, 1]$ ,  $\pi \circ f_t = g$ ,  $f_0 = f$ , and  $f_1 = F$ , such that

$$\sup_{z \in K} d_Z(F(z), f(z)) < \epsilon, \quad j_a^{k_a} F = j_a^{k_a} f \quad (a \in A).$$

For a point base, this is the Oka-1 property of the target manifold [1, Proposition 2.7 and Definition 7.7].

Definition 7.7 of [1] requires interpolation only at finitely many points of  $K$ . That definition does not require the intermediate maps in the homotopy to preserve the prescribed jets. Theorem 1.4 gives a stronger endpoint interpolation statement on arbitrary closed discrete subsets, together with approximation on arbitrary compact sets.

**1.2. The analytic criterion and rationally connected fibers.** Fix an affine coordinate  $w$  centered at  $0 \in \mathbb{P}^1$ . The following property is the geometric input to the analytic construction.

**Definition 1.3.** For a holomorphic submersion  $p : X \rightarrow S$  of complex manifolds, the *relative first-jet sphere-family property* is the following requirement: for every  $(x, b) \in T_{X/S}$ , there are an open neighborhood  $\Omega \subset T_{X/S}$  of  $(x, b)$  and a jointly holomorphic map

$$G : \mathbb{P}^1 \times \Omega \longrightarrow X$$

such that, for every  $(x', b') \in \Omega$ ,

$$p(G(w; x', b')) = p(x'), \quad G(0; x', b') = x', \quad \partial_w G(0; x', b') = b'. \quad (1.1)$$

Here  $T_{X/S} = \ker dp$  is the vertical tangent bundle. The requirement includes  $b = 0$ ; a realizing sphere need not be constant when its derivative at 0 vanishes.

**Theorem 1.4.** *Let  $\pi : Z \rightarrow Y$  be a surjective proper holomorphic submersion of complex manifolds with connected and simply connected fibers. Suppose that  $\pi$  has the relative first-jet sphere-family property. Then  $\pi$  is an Oka-1 map. If  $R$  is an open Riemann surface,  $K \subset R$  is compact,  $A \subset R$  is closed and discrete, and  $g : R \rightarrow Y$  is holomorphic, and if a continuous lifting  $f : R \rightarrow Z$  of  $g$  is holomorphic near  $K \cup A$ , then for every  $\epsilon > 0$  and every order function  $q : A \rightarrow \mathbb{N}$ , the lifting  $f$  is homotopic through continuous liftings of  $g$  to a holomorphic lifting  $F$  satisfying*

$$\sup_K d_Z(F, f) < \epsilon, \quad j_a^{q(a)} F = j_a^{q(a)} f \quad (a \in A).$$

For smooth projective morphisms, rational connectedness supplies the sphere families and simple connectivity of the fibers.

**Theorem 1.5.** *Let  $\pi : Z \rightarrow Y$  be a smooth projective morphism whose fibers are connected and rationally connected. Then all the conclusions of Theorem 1.4 hold. In particular,  $\pi$  is an Oka-1 map, with approximation on arbitrary compact sets and finite-jet interpolation on arbitrary closed discrete subsets of the source.*

Approximation and interpolation are required at the holomorphic endpoint; the intermediate liftings need only be continuous. The compact set  $K$  need not be Runge, and neither  $A \subset K$  nor  $K \subset A$  is required.

**1.3. Algebraic approximation.** For algebraic source data we obtain an algebraic lifting. We formulate a relative version of the algebraic Oka-1 property introduced by Forstnerič and Lárusson in [8, Definition 1.5], using the lifting formulation of Oka-1 maps in [1, Definition 7.7].

**Definition 1.6.** An *algebraic Oka-1 map*, or *aOka-1 map*, is a morphism  $\pi : Z \rightarrow Y$  of smooth complex algebraic varieties for which  $\pi^{\text{an}}$  is a Serre fibration and the following lifting property holds. Let  $R$  be a smooth connected affine algebraic curve, let  $K \subset R^{\text{an}}$  be compact and Runge, let  $A \subset K$  be finite, let  $g : R \rightarrow Y$  be algebraic, and let  $f : R^{\text{an}} \rightarrow Z^{\text{an}}$  be a continuous lifting of  $g^{\text{an}}$  which is holomorphic near  $K$ . Given  $\epsilon > 0$  and integers  $k_a \geq 0$ , there are an algebraic lifting  $F : R \rightarrow Z$  and a homotopy from  $f$  to  $F^{\text{an}}$  through continuous liftings of  $g^{\text{an}}$  such that

$$\sup_K d_Z(F, f) < \epsilon, \quad j_a^{k_a} F = j_a^{k_a} f \quad (a \in A).$$

When  $Y$  is a point, the preceding property is the algebraic Oka-1 property of [8, Definition 1.5]. The common order in that definition is a positive integer; for nonempty finite  $A$  one may take  $k = \max\{1, \max_{a \in A} k_a\}$ , and conversely, point-dependent interpolation includes the choice  $k_a = k$  for every  $a$ . For  $A = \emptyset$  there is no interpolation condition.

The same rational-connectedness hypothesis gives an algebraic lifting when the source and the base map are algebraic.

**Theorem 1.7.** *Let*

$$\pi : Z \longrightarrow Y$$

*be a smooth projective morphism of smooth complex algebraic varieties whose fibers are connected and rationally connected. Then  $\pi$  is an aOka-1 map. More precisely, let  $R$  be a smooth connected affine complex algebraic curve, let  $K \subset R^{\text{an}}$  be compact, let  $A \subset R$  be finite, and let  $g : R \rightarrow Y$  be algebraic. Suppose that  $f : R^{\text{an}} \rightarrow Z^{\text{an}}$  is a continuous lifting of  $g^{\text{an}}$  which is holomorphic on a neighborhood of  $K \cup A$ . Given  $\epsilon > 0$  and an integer  $k_a \geq 0$  for each  $a \in A$ , there is an algebraic lifting  $F : R \rightarrow Z$  such that*

$$\sup_K d_Z(F, f) < \epsilon, \quad j_a^{k_a} F = j_a^{k_a} f \quad (a \in A).$$

*Moreover,  $F^{\text{an}}$  and  $f$  are homotopic through continuous liftings of  $g^{\text{an}}$ .*

The following characterization follows by taking a point as base in Theorem 1.7 and using the necessary condition in [8, Proposition 1.7].

**Corollary 1.8.** *A connected smooth projective complex manifold  $X$  is algebraically Oka-1 if and only if  $X$  is rationally connected. Every rationally connected smooth projective complex manifold is also an Oka-1 manifold.*

*Proof.* If  $X$  is algebraically Oka-1, [8, Proposition 1.7] implies that  $X$  is rationally connected. If  $X$  is rationally connected, apply Theorem 1.7 to  $X \rightarrow \{\text{pt}\}$ . Definition 1.6 with a point as base gives the algebraic Oka-1 property of  $X$ . Applying Theorem 1.5 to  $X \rightarrow \{\text{pt}\}$  gives the Oka-1 assertion.  $\square$

**1.4. Relation to previous work and proof strategy.** Benoist–Wittenberg prove algebraic approximation for rationally simply connected projective varieties and tight approximation for such varieties over function fields of complex curves [2, Theorems 1.2 and 1.4]. The relative formulation of Benoist–Wittenberg starts with a regular projective model over a projective curve and a holomorphic section on an arbitrary open subset; that formulation includes approximation on compact subsets and finite jets, allowing bad fibers of the model [2, Definitions 3.1 and 3.3 and Remarks 3.4]. Theorem 1.7 assumes rational connectedness, with a smooth family over the affine source and a continuous global lifting holomorphic near the approximation and interpolation sets. In the projective model used in the proof of Theorem 1.7, bad fibers can occur only outside the affine source; the theorem does not assert the full bad-reduction formulation of tight approximation.

Alarcón–Forstnerič [1, §9] observe that Gournay’s Runge theorem, taken at face value, implies the approximation part of their conjecture, but report that they were unable to understand its proof. Gournay describes two relevant difficulties: dependence of the linearized equation on derivatives of the grafted map and control of the target chart used for matching [10, §§1.3 and 3.1]. Lemma 4.2 fixes the principal Dolbeault operator, and Lemma 5.2 supplies uniform fiber-derivative bounds and a positive target-chart margin. The replacements change only the zero-order term. Both the residual and the difference between the linearization and the fixed Dolbeault operator vanish outside measurable sets whose areas tend to zero. A fixed right inverse for the principal part then yields uniformly bounded right inverses for the full linearizations; the operator calculation is given in Lemma 4.5.

Lemma 4.5 solves the semilinear equation

$$\mathcal{F}_r(v) = \bar{\partial}_E v + A_r(z, v)$$

on a fixed holomorphic bundle with  $H^1(\Sigma, E) = 0$ , once both  $\mathcal{F}_r(0)$  and  $d\mathcal{F}_r(0) - \bar{\partial}_E$  are small in fixed Sobolev norms. The geometric construction supplies these two smallness conditions by replacing sections on disjoint rational disks and matching on thin annuli.

In the analytic construction a positive auxiliary bundle removes the Dolbeault obstruction outside the bordered reconstruction region. Bordered approximation is followed by an exhaustion

retaining the prescribed jets; connectedness and simple connectivity of the fibers give extension and homotopy of sections over the open source.

The algebraic proof works on a projective completion of the source. An algebraic section with vanishing twisted first cohomology provides the fixed bundle. A smooth bundle isomorphism, holomorphic near the prescribed data and near the omitted fibers, transfers the section equation to this bundle. The constant and linear gauge defects are then controlled before Lemma 4.5 is applied; global reconstruction and Chow's theorem give the algebraic endpoint.

Section 2 contains the analytic and topological tools. Sphere families are constructed in Section 3. The section equation and the solution lemma are treated in Section 4; the analytic criterion follows in Section 5. Algebraic preparation is in Section 6, and compact approximation and algebraic liftings are in Section 7.

## 2. ANALYTIC AND TOPOLOGICAL TOOLS

**2.1. Dolbeault estimates and holomorphic frames.** Fix smooth metrics and an area form whenever Sobolev norms are used. For a complex vector bundle  $E$  on a Riemann surface  $\Sigma$ , write  $L_{0,1}^p(\Sigma, E) = L^p(\Sigma, \Lambda^{0,1}T^*\Sigma \otimes E)$ . When  $\Sigma$  is compact, we use the Sobolev embeddings

$$W^{1,4} \hookrightarrow C^{0,1/2}, \quad W^{2,4} \hookrightarrow C^{1,1/2}$$

for sections of  $E$  [18, Theorem B.1.11]. In particular, for  $v \in W^{1,4}(\Sigma, E)$  and a bounded coefficient  $a$  vanishing outside a measurable set  $S$ , the estimates used below are

$$\|v\|_{C^0} \leq C_S \|v\|_{W^{1,4}}, \quad \|a\|_{L^4} \leq \|a\|_{L^\infty} \text{area}(S)^{1/4}. \quad (2.1)$$

Here  $C_S$  is the norm of the fixed Sobolev embedding, determined by the surface, bundle, metrics, and area form; the subscript stands for Sobolev.

For  $R > 0$ , the Cauchy–Green operator on  $D_R = \{z \in \mathbb{C} : |z| < R\}$  is

$$T_R f(z) = \frac{1}{\pi} \int_{D_R} \frac{f(\zeta)}{z - \zeta} dA(\zeta), \quad \partial_{\bar{z}} T_R f = f. \quad (2.2)$$

Here  $dA$  is Euclidean area measure, and the differential identity holds in the sense of distributions. For  $1 < p < \infty$ , the operator maps  $L^p(D_R)$  to  $W^{1,p}(D_R)$ ; for bounded  $f$ , the same integral is continuous on  $\overline{D_R}$ . The corresponding estimates are

$$\begin{aligned} \|T_R f\|_{W^{1,p}(D_R)} &\leq C_{p,R} \|f\|_{L^p(D_R)}, \\ \|T_R f\|_{C^0(\overline{D_R})} &\leq CR \|f\|_{L^\infty(D_R)}. \end{aligned} \quad (2.3)$$

The formula is given in [6, Lemma 13.1]; the  $L^p$  estimate follows from the Calderón–Zygmund estimate [18, Theorem B.2.7]. We also use the corresponding interior higher-order estimates and smooth regularity.

For a holomorphic bundle  $E$  on a compact Riemann surface  $\Sigma$ , Dolbeault Fredholm theory identifies the kernel and cokernel of

$$\bar{\partial}_E : W^{k+1,p}(\Sigma, E) \longrightarrow W_{0,1}^{k,p}(\Sigma, E)$$

with  $H^0(\Sigma, E)$  and  $H^1(\Sigma, E)$ , for  $k \geq 0$  and  $1 < p < \infty$ . Here  $W_{0,1}^{k,p}$  denotes Sobolev sections of  $\Lambda^{0,1}T^*\Sigma \otimes E$ . The range is closed, and

$$\|u\|_{W^{k+1,p}} \leq C(\|\bar{\partial}_E u\|_{W^{k,p}} + \|u\|_{L^p}). \quad (2.4)$$

We also use the local version of this estimate and smoothness of weak solutions with smooth right-hand side [18, Theorem C.1.10, Lemma C.2.1, and Theorem C.2.3]. Thus  $H^1(\Sigma, E) = 0$  gives a bounded right inverse of  $\bar{\partial}_E$ . Serre duality identifies  $H^1(\Sigma, E)^*$  with  $H^0(\Sigma, K_\Sigma \otimes E^*)$ , where  $K_\Sigma$  is the canonical bundle [18, Section C.1 and Remark C.1.12].

A complex-linear Cauchy–Riemann operator on a smooth bundle over a Riemann surface is equivalently a holomorphic structure. We retain the local argument to fix the frame convention used for the section equation.

**Lemma 2.1.** *Let  $E$  be a smooth complex vector bundle on a Riemann surface  $\Sigma$ , and let*

$$D : C^\infty(\Sigma, E) \longrightarrow C^\infty(\Sigma, \Lambda^{0,1} T^* \Sigma \otimes E)$$

*be a complex-linear first-order operator satisfying*

$$D(fu) = \bar{\partial}f \otimes u + fDu.$$

*Then  $D$  defines a holomorphic structure on  $E$ : there are local smooth frames in which  $D = \bar{\partial}$ , and the transition functions of these frames are holomorphic.*

*Proof.* Put  $n = \text{rank } E$ . In a smooth frame on a disk, write  $D = \bar{\partial} + A(z) d\bar{z}$ , where  $A$  is a smooth matrix. Use the operator norm on matrices. A new frame represented by an invertible matrix  $T$  is  $D$ -holomorphic exactly when  $\partial_{\bar{z}} T = -AT$ . Let  $T_R$  act entrywise on matrices. On a sufficiently small disk, consider

$$T = \text{Id} - T_R(AT).$$

The  $C^0$  estimate in (2.3) makes the right-hand side a contraction on the Banach space  $C^0(\bar{D}_R, \text{Mat}_{n \times n}(\mathbb{C}))$  when  $CR\|A\|_\infty < 1/4$ . Put  $\kappa = CR\|A\|_\infty < 1/4$ . The fixed point  $T$  satisfies

$$\|T\|_\infty \leq 1 + \kappa\|T\|_\infty, \quad \|T - \text{Id}\|_\infty \leq \frac{\kappa}{1 - \kappa} < \frac{1}{2}.$$

Thus every  $T(z)$  is invertible. Applying the  $L^p$  estimate in (2.3) to  $AT$  first gives  $T \in W^{1,p}$  and then, from the equation and interior regularity,  $T \in C^\infty$ . If  $T_1, T_2$  are two such frames, applying  $D$  to  $T_1 = T_2(T_2^{-1}T_1)$  shows that  $\bar{\partial}(T_2^{-1}T_1) = 0$ . The transition matrix  $T_2^{-1}T_1$  is therefore holomorphic.  $\square$

**2.2. Surfaces, bundles, and sections.** An open Riemann surface has an exhaustion by connected compact bordered surfaces  $C_j \Subset \text{int } C_{j+1}$ , whose boundaries can be chosen to avoid a prescribed closed discrete set and with  $C_1$  containing a prescribed compact set in its interior. One obtains such an exhaustion by taking regular levels of compactly supported smooth functions, joining the required compact pieces by finitely many arcs, and avoiding the finitely many forbidden levels at each stage. Smooth triangulations are available by [19, Theorem 10.6].

We use the isothermal-coordinate theorem [16, Theorem 3.11.1]: an oriented smooth surface with a smooth positive-definite metric has a unique complex structure compatible with its conformal class. We also use the fact that an open Riemann surface deformation retracts onto a countable locally finite graph [24, Lemma 2.1]. A proof in terms of a triangulation is given in [21, Theorem 0.1 and its proof]. The isothermal-coordinate theorem gives the compact-neighborhood construction below; the graph retraction is used for bundle trivializations and section homotopies.

**Lemma 2.2.** *Let  $R$  be an open Riemann surface and let  $C \subset R$  be compact. There exist an open neighborhood  $W \subset R$  of  $C$ , a compact Riemann surface  $\Sigma$ , and a holomorphic embedding  $W \hookrightarrow \Sigma$ . The embedding identifies  $W$  with an open subset of  $\Sigma$  and preserves the complex structure that  $W$  inherits from  $R$ .*

*Proof.* Choose connected bordered smooth surfaces  $D_0, D_1 \subset R$  with  $C \subset \text{int } D_0$  and  $D_0 \subset \text{int } D_1$ . Form the smooth double  $M$  of  $D_1$ : take a second copy with the opposite orientation, identify corresponding boundary points, and use a boundary collar to give the identification a smooth structure. In a collar, the signed distance normal to the boundary is the transverse coordinate on the double. Thus  $M$  is a compact oriented smooth surface without boundary, containing  $\text{int } D_1$  as an open subset.

Choose a smooth Hermitian metric for the original complex structure on  $R$ . Let  $g_0$  be its restriction near  $D_0$ , and choose a smooth positive-definite metric  $g_1$  on  $M$ . Take  $\chi \in C^\infty(M, [0, 1])$  equal to one near  $D_0$  and supported where  $g_0$  is defined inside  $\text{int } D_1$ . Set

$$g = \chi g_0 + (1 - \chi)g_1$$

on that open set and  $g = g_1$  outside  $\text{supp } \chi$ . The formulas agree where  $\chi = 0$ ; positivity of  $g_0$  and  $g_1$  makes  $g$  positive definite. The isothermal-coordinate theorem [16, Theorem 3.11.1] gives a complex structure on  $M$ . Near  $D_0$ , the conformal class of  $g$  and the orientation of  $M$  are those of the original Riemann surface, so the two complex structures agree. Write  $\Sigma$  for  $M$  with the complex structure determined by  $g$ . Any sufficiently small neighborhood  $W$  of  $C$  contained in  $\text{int } D_0$  has the required embedding.  $\square$

Ehresmann's theorem [4] makes a proper surjective smooth submersion a locally trivial smooth fiber bundle. Over a paracompact base it is a Hurewicz fibration, hence a Serre fibration [12, p. 379]. Pullback by a smooth map is a smooth fiber bundle; pullback by a continuous map is a topological fiber bundle.

We use the following relative smoothing statement for sections. Let  $p : E \rightarrow M$  be a smooth fiber bundle, let  $C \subset M$  be closed, and let  $s : M \rightarrow E$  be a continuous section that is smooth near  $C$ . Fix a metric  $d_E$  inducing the topology of  $E$ . Given a neighborhood  $U$  of  $C$  and a positive continuous function  $\delta$  on  $M$ , there is a smooth section  $\tilde{s}$  equal to  $s$  near  $C$ , with  $d_E(\tilde{s}(x), s(x)) < \delta(x)$ , homotopic to  $s$  through sections by a homotopy fixed near  $C$ . The statement also holds for manifolds with boundary or corners. Apply the finite-dimensional section case of Wockel's relative Steenrod approximation theorem [25, Theorem 11] on each connected component of  $M$ . To obtain the stated form, choose  $C \subset V \subset \bar{V} \subset U$  with  $\bar{V}$  in the region where  $s$  is smooth. In Wockel's theorem, take  $L = M$ , allow modifications on  $M \setminus \bar{V}$ , and use the neighborhood

$$\{e \in E : d_E(e, s(p(e))) < \delta(p(e))\}.$$

An open Riemann surface is Stein by the Behnke–Stein theorem [6, Corollary 26.8]. Grauert's Oka principle for vector bundles [11, Satz I] implies that a topologically trivial complex vector bundle with a holomorphic structure on a Stein manifold is holomorphically trivial. In particular this applies to every holomorphic vector bundle on an open Riemann surface: a graph retraction reduces its topological trivialization to extending frames along edges, using path connectedness of  $\text{GL}_n(\mathbb{C})$ , where  $n$  is the bundle rank.

The next two lemmas are the topological steps in exhaustion and in the passage from a holomorphic endpoint to a homotopy through liftings.

**Lemma 2.3.** *Let  $p : E \rightarrow M$  be a smooth locally trivial fiber bundle over a smooth real surface without boundary, with connected and simply connected fibers. Suppose that  $M_0 \subset M_1 \subset M$  are compact smooth subsurfaces with boundary and  $M_0 \subset \text{int } M_1$ . Fix pairwise disjoint closed coordinate disks  $D_1, \dots, D_s \subset \text{int}(M_1 \setminus M_0)$ . Given a smooth section  $s_*$  on a neighborhood of*

$$M_0 \cup D_1 \cup \dots \cup D_s,$$

*there is a smooth section on a neighborhood of  $M_1$  that agrees with  $s_*$  on a possibly smaller neighborhood of this union.*

*Proof.* Let  $s_* : O \rightarrow E$  be the prescribed section, where  $O$  is a neighborhood of  $M_0 \cup D_1 \cup \dots \cup D_s$ . Choose a compact smoothly bounded, possibly disconnected surface  $N$  such that

$$M_0 \cup D_1 \cup \dots \cup D_s \subset \text{int } N \subset N \subset O \cap \text{int } M_1.$$

Also choose a compact smoothly bounded neighborhood  $M_2$  of  $M_1$  in  $M$ , with  $M_1 \subset \text{int } M_2$ . It is enough to construct the section on  $M_2$ .

Triangulate  $M_2$  so that  $N$  is a subcomplex. First triangulate the compact one-manifolds  $\partial N$  and  $\partial M_2$ . The relative triangulation theorem in [19, Theorem 10.6] extends the boundary triangulations first over  $N$  and then over the closure of  $M_2 \setminus N$ ; the latter is a compact smooth surface with boundary  $\partial N \sqcup \partial M_2$ . The two triangulations agree on  $\partial N$  and hence glue to a finite smooth triangulation of  $M_2$  in which  $N$  is a subcomplex. Choose a finite cover of  $M_2$  by bundle trivialization domains. After sufficiently many barycentric subdivisions, the mesh is smaller than a Lebesgue number for this cover. Each closed simplex then lies in one trivialization domain, which is an open neighborhood of that simplex. Put  $\hat{s} = s_*|_N$  and extend  $\hat{s}$  over the remaining simplices in increasing dimension.

For each vertex  $a$  outside  $N$ , choose  $\hat{s}(a) \in p^{-1}(a)$ . For an edge  $e = [a, b]$  not contained in  $N$ , the endpoint values are now prescribed. Choose a trivialization  $\tau_e : p^{-1}(e) \rightarrow e \times F_e$ , and let  $\alpha_e, \beta_e \in F_e$  be the fiber coordinates of  $\hat{s}(a), \hat{s}(b)$ . Connectedness of the fiber manifold  $F_e$  gives a path from  $\alpha_e$  to  $\beta_e$ . Parametrize  $e$  by  $[0, 1]$  and apply  $\tau_e^{-1}$  to the graph of that path to extend  $\hat{s}$  over  $e$  without changing the prescribed values at  $a$  and  $b$ .

For a two-simplex  $\Delta$  not contained in  $N$ , the section is now defined on  $\partial\Delta$ . A trivialization  $\tau_\Delta : p^{-1}(\Delta) \rightarrow \Delta \times F_\Delta$  identifies the boundary section with a map  $b_\Delta : \partial\Delta \rightarrow F_\Delta$ . The loop  $b_\Delta$  is null-homotopic because  $\pi_1(F_\Delta) = 0$ . A null-homotopy descends to the cone on  $\partial\Delta$ , which is a closed disk, and hence gives an extension  $\tilde{b}_\Delta : \Delta \rightarrow F_\Delta$  with  $\tilde{b}_\Delta|_{\partial\Delta} = b_\Delta$ . Set  $\hat{s}(x) = \tau_\Delta^{-1}(x, \tilde{b}_\Delta(x))$  for  $x \in \Delta$ . The extensions agree on common faces, so finite gluing gives a continuous section on  $M_2$ . There are no simplices of higher dimension.

The resulting  $\hat{s}$  is continuous on  $M_2$  and agrees with  $s_*$  on  $N$ . It is therefore smooth on  $\text{int } N$ , a neighborhood of  $M_0 \cup D_1 \cup \dots \cup D_s$ . Apply the relative smoothing statement above on  $M_2$ , relative to a smaller compact smoothly bounded neighborhood

$$M_0 \cup D_1 \cup \dots \cup D_s \subset N' \Subset \text{int } N.$$

Use  $C = N'$  and  $U = \text{int } N$  in that statement (with any positive continuous tolerance  $\delta$ ). It gives a smooth section on  $M_2$  which is unchanged near  $N'$ . Since  $M_1 \subset \text{int } M_2$ , its restriction is the required smooth section on a neighborhood of  $M_1$ .  $\square$

**Lemma 2.4.** *Let  $p : E \rightarrow R$  be a locally trivial fiber bundle over an open Riemann surface. If the fibers of  $p$  are connected and simply connected, any two continuous sections of  $p$  are homotopic through sections.*

*Proof.* Let  $s_0, s_1 : R \rightarrow E$  be the sections and put  $I = [0, 1]$ . By the graph-retraction statement above, there is a deformation retraction  $d_t : R \rightarrow R$ ,  $t \in [0, 1]$ , onto a countable locally finite graph  $\Gamma \subset R$ . Write  $i : \Gamma \hookrightarrow R$  and  $r : R \rightarrow \Gamma$  for the inclusion and retraction, so  $d_0 = \text{id}_R$  and  $d_1 = i \circ r$ .

First construct a homotopy through sections over  $\Gamma$ . Subdivide the edges of  $\Gamma$  so that each closed edge lies in a bundle chart. At each vertex  $a$ , choose a path in  $E_a = p^{-1}(a)$  from  $s_0(a)$  to  $s_1(a)$ . On the boundary of an edge square  $e \times I$ , prescribe  $s_0$  and  $s_1$  on the two horizontal sides and the chosen vertex paths on the two vertical sides. In a trivialization over  $e$ , the four prescribed sides form a loop in the model fiber. Simple connectivity fills the loop over the square. The fillings agree on common sides, and local finiteness gives a continuous map

$$H_\Gamma : \Gamma \times I \rightarrow E, \quad p(H_\Gamma(a, u)) = i(a), \quad H_\Gamma(a, j) = s_j(i(a)) \quad (j = 0, 1).$$

We now return to  $R$  while retaining the original endpoints. A fiber bundle has the relative homotopy-lifting property for every CW pair [12, Proposition 4.48]. Triangulate  $R$ , give  $R \times I$  the product CW structure, and apply the relative homotopy-lifting property to the CW pair  $(R \times I, R \times \partial I)$ , using  $v \in I$  as the homotopy-lifting variable;  $u$  remains the section-homotopy parameter. The base homotopy is

$$B(x, u, v) = d_{1-v}(x).$$

At  $v = 0$  prescribe the lift  $H_\Gamma(r(x), u)$ ; on the two sides  $u = j$  prescribe  $s_j(d_{1-v}(x))$  for  $j = 0, 1$ . On the intersections  $v = 0$ ,  $u = j$ , the two prescriptions agree because

$$H_\Gamma(r(x), j) = s_j(i(r(x))) = s_j(d_1(x)).$$

Both prescribed lifts project to  $B$ . Relative homotopy lifting therefore extends the prescribed boundary data to a continuous lift  $\tilde{B} : R \times I \times I \rightarrow E$ . Set

$$H(x, u) = \tilde{B}(x, u, 1).$$

Then  $p(H(x, u)) = d_0(x) = x$ ,  $H(x, 0) = s_0(x)$ , and  $H(x, 1) = s_1(x)$ . Thus  $H$  is the required homotopy through sections. The relative lifting step retains both prescribed endpoints while returning the base map to  $\text{id}_R$ .  $\square$

### 3. FIRST-JET SPHERE FAMILIES

We verify the geometric hypothesis of Theorem 1.4 for rationally connected fibers. Kollár's finite-jet theorem [17, Definition–Theorem 2.1(5)] and simple-connectivity result [17, Proposition 2.3] give the following assertions, respectively.

**Theorem 3.1.** *Let  $X$  be a connected smooth projective rationally connected complex manifold.*

- (i) *Let  $p_1, \dots, p_\ell$  be distinct points of  $\mathbb{P}^1$ , let  $m_1, \dots, m_\ell$  be positive integers, and prescribe at each  $p_i$  a holomorphic map jet of length  $m_i$  (that is, derivatives through order  $m_i - 1$ ), with values in  $X$ . There exists a morphism  $h : \mathbb{P}^1 \rightarrow X$  realizing these jets and satisfying*

$$H^1\left(\mathbb{P}^1, h^*TX\left(-\sum_i m_i p_i\right)\right) = 0.$$

- (ii) *The topological fundamental group of  $X$  is trivial. Equivalently, every continuous loop in  $X$  is homotopic to a constant loop.*

To vary the base point and the prescribed first jet holomorphically, we need relative spaces of holomorphic maps. Fix a smooth projective morphism  $\pi : X \rightarrow S$  of complex manifolds, a smooth projective curve  $C$ , and a finite effective Cartier divisor  $D \subset C$ , possibly empty. For analytic spaces  $T \rightarrow S$ , consider the functors

$$T \mapsto \text{Mor}_S(C \times T, X), \quad T \mapsto \text{Mor}_S(D \times T, X),$$

where  $\text{Mor}_S$  denotes holomorphic maps over  $S$ . The following theorem establishes their representability. We write

$$\mathcal{H} = \text{Hom}_S(C \times S, X), \quad \mathcal{J}_D = \text{Hom}_S(D \times S, X)$$

for the representing spaces, with  $\mathcal{J}_D = S$  when  $D$  is empty, and  $\rho_D : \mathcal{H} \rightarrow \mathcal{J}_D$  for restriction to  $D$ . For  $s_0 \in S$  and  $h : C \rightarrow X_{s_0}$ , put  $E = h^*T_{X/S}$  and write  $[h]$  for the point of  $\mathcal{H}$  represented by  $h$ .

**Theorem 3.2.** *The two functors above, including their values on nonreduced analytic spaces, are represented by complex analytic spaces  $\mathcal{H}$  and  $\mathcal{J}_D$  with universal evaluations  $C \times \mathcal{H} \rightarrow X$  and  $D \times \mathcal{J}_D \rightarrow X$ . These representing properties are compatible with analytic base change. The restriction map  $\rho_D$  is holomorphic, and  $\mathcal{J}_D$  is smooth over  $S$ .*

*At  $[h]$ , the relative tangent space of  $\mathcal{H} \rightarrow S$  is  $H^0(C, E)$ , and  $H^1(C, E)$  is a complete obstruction space for lifting maps over a prescribed infinitesimal extension of the base point. For  $\rho_D$ , the corresponding relative tangent and complete obstruction spaces are*

$$H^0(C, E(-D)), \quad H^1(C, E(-D)).$$

In particular, if  $H^1(C, E(-D)) = 0$ , then  $\mathcal{H}$  is nonsingular at  $[h]$  and  $\rho_D$  is a holomorphic submersion at  $[h]$ . Thus  $\rho_D$  has a local holomorphic section through  $[h]$ . More explicitly, if  $\gamma : S' \rightarrow \mathcal{J}_D$  is a holomorphic family of jets and  $\gamma(s'_0) = \rho_D([h])$ , then the fiber product

$$\mathcal{H}_\gamma = \mathcal{H} \times_{\mathcal{J}_D} S'$$

is smooth over  $S'$  near  $([h], s'_0)$ .

*Proof.* The proof has three steps: analytic representability, the relative Čech obstruction calculation, and the analytic submersion criterion. Pourcin's relative Douady theorem [20, Théorème 2], applied to  $C \times X \rightarrow S$  and the structure sheaf, gives a relative Douady space  $\mathcal{Q} \rightarrow S$  and a universal proper flat family

$$\mathcal{U} \subset \mathcal{Q} \times_S (C \times X).$$

Let

$$q : \mathcal{U} \longrightarrow \mathcal{Q} \times C$$

be the projection. Both  $\mathcal{U}$  and  $\mathcal{Q} \times C$  are proper and flat over  $\mathcal{Q}$ . By Douady [3, §10.1, Proposition 1], the set of  $u \in \mathcal{Q}$  for which  $q_u$  is an isomorphism is open, and  $q$  itself is an isomorphism after restriction to that open set. Denote the open set by  $\mathcal{H}$ . Writing  $\mathcal{U}_{\mathcal{H}}$  for the restriction of  $\mathcal{U}$ , the universal evaluation is

$$\text{ev}(c, h') = \text{pr}_X((q|_{\mathcal{U}_{\mathcal{H}}})^{-1}(h', c)), \quad (c, h') \in C \times \mathcal{H}.$$

We verify the representing property, including nonreduced test spaces. For every analytic space  $T \rightarrow S$ , a  $T$ -point of  $\mathcal{H}$  pulls the universal family back to a graph and hence gives an  $S$ -map  $C \times T \rightarrow X$ . Conversely, the graph of any such map is a closed analytic subspace of  $T \times_S (C \times X)$ , isomorphic to  $C \times T$  and therefore proper and flat over  $T$ . Pourcin's universal property gives a unique classifying map  $c_T : T \rightarrow \mathcal{Q}$ . Since the pullback of  $q$  to  $T$  is an isomorphism,  $c_T$  factors through the open set  $\mathcal{H}$ . The two operations are inverse and functorial in  $T$ . The correspondence between maps and graphs is precisely the argument of Douady [3, §10.2, Théorème 1], now applied to the relative Douady space. Replacing  $C$  by the compact, possibly nonreduced space  $D$  in the relative Douady construction and taking the open locus of graphs gives  $\mathcal{J}_D$ . Pourcin works with analytic spaces that need not be reduced, and [20, §5, Remark 3] gives compatibility with arbitrary analytic base change. Thus the asserted functors, universal evaluations, and restriction morphism have the stated strength. No algebraicity of  $S$  is used here.

Write  $D = \sum_{i=1}^{\ell} m_i a_i$  and let  $n$  be the relative dimension of  $X/S$ . Since  $\pi$  is a submersion, relative coordinates near the finitely many image points identify a neighborhood in  $\mathcal{J}_D$  with a neighborhood in

$$S_0 \times \prod_i \mathbb{C}^{nm_i},$$

where  $S_0 \subset S$  is a sufficiently small neighborhood of the base point; the factors record the coefficients modulo  $w_i^{m_i}$  in local coordinates  $w_i$  centered at  $a_i$ . Thus  $\mathcal{J}_D \rightarrow S$  is smooth, including when  $D$  is nonreduced.

We next compute the obstruction, allowing both a moving base and a nonreduced jet condition. We regard the spectrum of a local Artinian  $\mathbb{C}$ -algebra as its associated one-point analytic space. Let  $A' \twoheadrightarrow A$  be a small extension of local Artinian  $\mathbb{C}$ -algebras with residue field  $\mathbb{C}$  and kernel  $I$ , with the maximal ideal of  $A'$  annihilating  $I$ . Fix a base point  $s_A : \text{Spec } A \rightarrow S$  and a deformation  $h_A : C \times \text{Spec } A \rightarrow X$  of  $h$  over  $s_A$ . Prescribe a base-point lift  $s_{A'} : \text{Spec } A' \rightarrow S$  and a jet lift  $j_{A'} : D \times \text{Spec } A' \rightarrow X$  over  $s_{A'}$ . These lifts must reduce to  $s_A$  and  $h_A|_D$ , respectively. Choose a finite Leray cover  $U_1, \dots, U_N$  of  $C$  by coordinate disks such that each  $h(U_i)$  lies in a relative coordinate chart for  $X \rightarrow S$ . For  $1 \leq i \leq \ell$ , take  $U_i$  centered at  $a_i$  and containing no other point of  $\text{supp } D$ ; the remaining disks avoid  $\text{supp } D$ .

In the relative charts chosen for the Leray cover, the component functions of the given local maps lift from  $A$  to  $A'$ . On a disk centered at  $a_i$ , the difference between such a lift and the prescribed map on  $m_i a_i$  is  $I$ -valued. The restriction

$$\mathcal{O}(U_i) \longrightarrow \mathbb{C}[w_i]/(w_i^{m_i})$$

is onto: a class is lifted by its Taylor polynomial. Let  $\delta_i$  be the  $I$ -valued discrepancy on  $m_i a_i$ , and let  $\tilde{\delta}_i$  be a Taylor-polynomial lift of  $\delta_i$  on  $U_i$ . Subtract  $\tilde{\delta}_i$  from the chosen local lift of  $h_A$ . The adjusted map agrees with  $j_{A'}$  on  $m_i a_i$  and still reduces to  $h_A$ , because  $\tilde{\delta}_i$  has coefficients in  $I$ .

On an overlap, two adjusted local lifts agree modulo  $I$ . Because  $I^2 = 0$ , the difference of the two lifts transforms linearly as a vertical derivation in  $E \otimes_{\mathbb{C}} I$ . Writing  $c_{ij}$  for the lift on  $U_j$  minus the lift on  $U_i$ , one has  $c_{ij} + c_{jk} + c_{ki} = 0$  on triple overlaps. Since both local maps agree with the prescribed map on  $D$ , the cocycle vanishes to order  $m_i$  at every  $a_i$  and hence has coefficients in  $E(-D) \otimes_{\mathbb{C}} I$ . Changing the local lifts changes  $(c_{ij})$  by a coboundary. The class

$$o = [(c_{ij})] \in H^1(C, E(-D)) \otimes_{\mathbb{C}} I$$

is therefore well defined; the Leray property identifies the displayed Čech group with sheaf cohomology. The equality  $o = 0$  means that  $c_{ij} = b_j - b_i$  for local sections  $b_i$  of  $E(-D) \otimes I$ . Subtracting  $b_i$  from the lift on  $U_i$  makes all adjusted lifts agree on overlaps. Conversely, global gluing makes the difference cocycle a coboundary. When a lift exists, all lifts with the same base point and  $D$ -restriction form a torsor under  $H^0(C, E(-D)) \otimes I$ . Thus  $o$  is a complete obstruction for every small extension. Taking dual numbers proves the tangent assertion, and taking  $D$  empty gives the unconstrained assertions.

The infinitesimal lifting property implies an analytic submersion as follows. If  $H^1(C, E(-D)) = 0$ , the preceding calculation gives the relative small-extension lifting property for  $\rho_D$ . Given an absolute lifting problem for  $(\mathcal{H}, [h])$ , first lift the composite of the given Artinian map with  $\rho_D$  to  $\mathcal{J}_D$ , which is possible since  $\mathcal{J}_D$  is nonsingular, and then use the relative lifting property. Thus  $(\mathcal{H}, [h])$  has the absolute small-extension lifting property. The absolute lifting property forces  $(\mathcal{H}, [h])$  to be nonsingular: write the local algebra of this germ in a minimal embedding of dimension  $e$  as

$$B = \mathbb{C}\{t_1, \dots, t_e\}/J, \quad J \subset \mathfrak{m}^2, \quad \mathfrak{m} = (t_1, \dots, t_e).$$

If  $J \neq 0$ , choose a relation of least order  $d \geq 2$  among all nonzero relations. The natural map from  $B$  to  $\mathbb{C}\{t\}/\mathfrak{m}^d$  would lift to  $\mathbb{C}\{t\}/\mathfrak{m}^{d+1}$ . Any such lift sends  $t_i$  to  $t_i$  plus a term of degree  $d$ , so substitution preserves the nonzero homogeneous degree- $d$  part of the chosen relation. This contradiction gives  $J = 0$ . Finally, lifting arbitrary dual-number points of  $\mathcal{J}_D$  through  $[h]$  says that  $d\rho_D$  is surjective. The holomorphic submersion theorem now gives a local section through  $[h]$ . Locally,  $\rho_D$  has the form  $V \times \mathbb{B}^k \rightarrow V$ , where  $\mathbb{B}^k$  is a ball in  $\mathbb{C}^k$  and  $V$  is open in  $\mathcal{J}_D$ . Pullback by  $\gamma$  gives  $\gamma^{-1}(V) \times \mathbb{B}^k \rightarrow \gamma^{-1}(V)$ , proving the asserted smoothness of  $\mathcal{H}_\gamma \rightarrow S'$ .  $\square$

*Remark 3.3.* Let  $p : X \rightarrow S$  have the relative first-jet sphere-family property. This property is preserved by restriction to an open subset of the base and by holomorphic base change  $g : S' \rightarrow S$  between complex manifolds. For  $X' = S' \times_S X$ , one has  $T_{X'/S'} \simeq S' \times_S T_{X/S}$ . Given one of its families  $G : \mathbb{P}^1 \times \Omega \rightarrow X$ , the pulled-back family is

$$G'(w; s', x, b) = (s', G(w; x, b)), \quad g(s') = p(x).$$

The domain of  $G'$  is  $\mathbb{P}^1 \times \Omega'_0$ , where

$$\Omega'_0 = \{(s', x, b) : (x, b) \in \Omega, g(s') = p(x)\} \subset T_{X'/S'}.$$

The identities for  $G'$  follow componentwise from (1.1).

For each fixed family  $G$  and each relatively compact parameter subdomain  $\Omega' \Subset \Omega$ , compactness of  $\mathbb{P}^1 \times \Omega'$  gives bounds for source and parameter derivatives of every fixed finite order. In local coordinates one covers this compact set by finitely many smaller source-parameter coordinate

neighborhoods whose closures lie in larger neighborhoods on which  $G$  takes values in fixed target charts; the coordinate expressions are holomorphic on the larger neighborhoods, so their derivatives are bounded on the smaller ones. The bounds may depend on the order and on the finite charts; no single holomorphic choice over an arbitrary compact jet set is asserted.

**Proposition 3.4.** *Let  $p : X \rightarrow S$  be a smooth projective morphism of complex manifolds. Suppose that, for every  $s \in S$ ,  $x \in X_s$ , and  $b \in T_{X_s, x}$ , there is a morphism  $h : \mathbb{P}^1 \rightarrow X_s$  with*

$$h(0) = x, \quad \partial_w h(0) = b, \quad H^1(\mathbb{P}^1, h^* T_{X_s}(-2[0])) = 0.$$

*Then  $p$  has the relative first-jet sphere-family property.*

*Proof.* Apply Theorem 3.2 with  $C = \mathbb{P}^1$  and  $D = 2[0]$ . In the fixed source coordinate  $w$ , the relative jet space  $\mathcal{J}_D$  is naturally  $T_{X/S}$ : a map from  $2[0]$  records its value and vertical derivative. At each chosen  $[h]$ , the displayed vanishing makes  $\rho_{2[0]}$  a holomorphic submersion, with respect to the base point, value, and derivative. It therefore has a local holomorphic section  $\sigma : \Omega \rightarrow \mathcal{H}$  through  $[h]$ . The universal evaluation map gives the jointly holomorphic family

$$G(w; x', b') = \text{ev}(w, \sigma(x', b')).$$

Since  $\rho_{2[0]} \circ \sigma = \text{Id}_\Omega$ , the restriction of  $G(\cdot; x', b')$  to  $2[0]$  has value  $x'$  and derivative  $b'$ . The evaluation map is over  $S$ , so  $p(G(w; x', b')) = p(x')$ . This verifies the three identities in (1.1). The vanishing hypothesis and submersion argument also apply at  $b = 0$ .  $\square$

**Proposition 3.5.** *Let  $p : X \rightarrow S$  be a smooth projective morphism of complex manifolds with rationally connected fibers. Then  $p$  has the relative first-jet sphere-family property. Explicitly, fix  $s_0 \in S$ ,  $x_0 \in X_{s_0}$ , and  $b_0 \in T_{X/S, x_0}$ . There are a neighborhood  $\Omega$  of  $(s_0, x_0, b_0)$  in the relative first-jet bundle and a jointly holomorphic map*

$$G : \mathbb{P}^1 \times \Omega \longrightarrow X$$

*such that*

$$p(G(w; s, x, b)) = s, \quad G(0; s, x, b) = x, \quad \partial_w G(0; s, x, b) = b.$$

*On relatively compact parameter subsets, the derivatives of  $G$  of every fixed finite order are uniformly bounded on  $\mathbb{P}^1$ , measured in fixed finite collections of parameter, source, and target charts.*

*Proof.* Theorem 3.1(i), applied in the fiber  $X_{s_0}$  to one jet of length two, gives a morphism  $h : \mathbb{P}^1 \rightarrow X_{s_0}$  with the prescribed value and derivative, including the zero derivative, and

$$H^1(\mathbb{P}^1, h^* T_{X_{s_0}}(-2[0])) = 0.$$

Proposition 3.4 gives the family  $G$ . The derivative bounds follow from Remark 3.3.  $\square$

*Proof of Theorem 1.5.* By Definition 1.1,  $\pi$  is proper and a holomorphic submersion; it is surjective by the nonempty-fiber convention. Proposition 3.5 gives the relative first-jet sphere-family property, and Theorem 3.1(ii) gives simple connectivity of the fibers. Together with their assumed connectedness, these are all the hypotheses of Theorem 1.4. That theorem gives the Oka-1 map property and the stated closed discrete interpolation conclusion.  $\square$

#### 4. THE FIXED SECTION EQUATION

**4.1. Vertical equations and jet twisting.** Fix a proper holomorphic submersion  $\pi : Z \rightarrow Y$ , a Riemann surface  $R$ , and a holomorphic map  $g : R \rightarrow Y$ . Put

$$X = R \times_Y Z, \quad p : X \longrightarrow R.$$

Then  $p$  is a proper holomorphic submersion. Write  $X_z = p^{-1}(z)$  for its fiber over  $z$ . Its vertical tangent bundle is

$$T_{X/R} = \ker(dp : TX \rightarrow p^* TR).$$

The correspondence between liftings and sections is

$$f : R \rightarrow Z \quad \longleftrightarrow \quad s_f : R \rightarrow X, \quad s_f(z) = (z, f(z)), \quad f = \text{pr}_Z \circ s_f.$$

Equip  $X$  with the metric induced from a product Hermitian metric on  $R \times Z$ . For points  $x_1, x_2 \in X$  over the same point of  $R$ ,

$$d_Z(\text{pr}_Z x_1, \text{pr}_Z x_2) \leq d_X(x_1, x_2),$$

since projection to  $Z$  does not increase lengths. The local constructions below also apply near a compact image of a section lying in the submersion locus of a holomorphic map.

To express nearby sections in one bundle, we use a local addition that stays in each fiber and is holomorphic in its vector parameter.

**Lemma 4.1.** *Let  $D \subset R$  be compact. There are an open neighborhood  $\mathcal{U} \subset T_{X/R}$  of the zero vectors over  $p^{-1}(D)$  and a smooth map*

$$\Psi : \mathcal{U} \longrightarrow X$$

*with the following properties:*

- (i)  $p(\Psi(x, v)) = p(x)$  and  $\Psi(x, 0) = x$ ;
- (ii) for fixed  $x$ , the map  $v \mapsto \Psi(x, v)$  is holomorphic and its derivative at zero is the identity of  $T_{X/R, x}$ ;
- (iii) after the fiber neighborhood is uniformly shrunk, the fiber map is a biholomorphism onto a neighborhood of  $x$  inside  $X_{p(x)}$ ;
- (iv) all derivatives through order two in the variables  $x$  and order three in the variables  $v$  are uniformly bounded on smaller fiber neighborhoods.

*Proof.* Put  $K_0 = p^{-1}(D)$ , which is compact because  $p$  is proper. Cover  $K_0$  by finitely many relative holomorphic coordinate charts

$$(z, y) : U_i \longrightarrow V_i \times W_i \subset \mathbb{C} \times \mathbb{C}^n, \quad p(z, y) = z,$$

where  $n$  is the fiber dimension. We average coordinate differences in the vertical tangent space at  $x$ , using weights that depend only on  $x$ . Choose a smooth partition of unity  $\rho_i$  on a neighborhood of  $K_0$ , subordinate to the relative coordinate cover, and put  $K_i = \text{supp } \rho_i \subseteq U_i$ . We first specify the domain of every summand. In the fiber product  $X \times_R X$ , put

$$\theta_i(x, y') = \rho_i(x) ((d_y \phi_i)_x)^{-1} (\phi_i(y') - \phi_i(x))$$

when  $x, y' \in U_i$ , where  $\phi_i$  is the vertical coordinate, and set  $\theta_i = 0$  when  $x \notin K_i$ . The two formulas agree on  $(U_i \times_R U_i) \cap ((X \setminus K_i) \times_R X)$  because  $\rho_i(x) = 0$  on that intersection. Thus  $\theta_i$  is smooth on the union

$$(U_i \times_R U_i) \cup ((X \setminus K_i) \times_R X).$$

This union contains the relative diagonal: if  $x \in K_i$ , then  $x \in U_i$ , and otherwise the zero definition applies. Intersecting these finitely many domains gives an open neighborhood  $\mathcal{V}$  of the relative diagonal over  $K_0$  on which

$$\Theta_x(y') = \sum_i \theta_i(x, y') \in T_{X/R, x}$$

is well defined and smooth. For fixed  $x$ ,  $\Theta_x$  is holomorphic in  $y'$ , and

$$\Theta_x(x) = 0, \quad (d_{y'} \Theta_x)_{y'=x} = \sum_i \rho_i(x) \text{Id} = \text{Id}.$$

Consider

$$\Gamma : \mathcal{V} \longrightarrow T_{X/R}, \quad \Gamma(x, y') = (x, \Theta_x(y')).$$

Along the relative diagonal, the derivative of  $\Gamma$  is block triangular: the first diagonal block is the identity in the  $x$ -variable and the second is the displayed identity in the vertical  $y'$ -variable. Hence

$\Gamma$  is a local diffeomorphism near every point of that diagonal. Compactness of  $K_0$ , together with the fact that  $\Gamma$  retains the first component  $x$ , allows one to shrink to a single neighborhood of the diagonal on which  $\Gamma$  is injective. Otherwise there would be pairs  $(x_j, y_j) \neq (x_j, y'_j)$  approaching the diagonal, with  $\Gamma(x_j, y_j) = \Gamma(x_j, y'_j)$ . After taking a subsequence,  $x_j \rightarrow x \in K_0$  and  $y_j, y'_j \rightarrow x$ . Both pairs then lie in a neighborhood where  $\Gamma$  is injective, a contradiction.

The inverse has the form  $(x, v) \mapsto (x, \Psi(x, v))$ . Shrinking once more, there is a number  $\rho > 0$  such that the closed metric ball bundle  $\{(x, v) : x \in K_0, |v| \leq 2\rho\}$  lies in the domain of  $\Gamma^{-1}$ . We work on  $|v| < \rho$ , leaving a fixed margin for the derivative estimates.

- (i) The map  $\Gamma$  preserves the first component, so its inverse satisfies  $p(\Psi(x, v)) = p(x)$ . Since  $\Theta_x(x) = 0$ , one also has  $\Psi(x, 0) = x$ .
- (ii) For fixed  $x$ , the holomorphic inverse-function theorem makes  $\Psi(x, \cdot)$  holomorphic. The identity  $(d_{y'}\Theta_x)_x = \text{Id}$  gives  $d_v\Psi(x, 0) = \text{Id}$ .
- (iii) On the neighborhood already chosen,  $\Gamma$  is injective and a local diffeomorphism. Its restriction to each fiber and the corresponding restriction of its inverse are holomorphic; thus  $\Psi(x, \cdot)$  is a biholomorphism onto a neighborhood of  $x$ .
- (iv) The inverse is smooth jointly in  $(x, v)$ . On smaller closed balls, a finite coordinate cover and compactness bound all mixed derivatives, including the orders in (iv).  $\square$

Given a local addition as in Lemma 4.1, let  $s$  be a smooth section of  $p$  on a neighborhood of a compact bordered Riemann surface. Put

$$E = s^*T_{X/R}.$$

For a small section  $u$  of  $E$ , put

$$\mathcal{S}_s(u) = (d_u\Psi(s(z), u))^{-1}\bar{\partial}[\Psi(s(z), u(z))]. \quad (4.1)$$

Here  $d_u\Psi(s(z), u(z))$  is an isomorphism  $E_z \rightarrow T_{X/R, \Psi(s(z), u(z))}$ . The derivative  $\bar{\partial}[\Psi(s(z), u(z))]$  is vertical because  $p(\Psi(s(z), u(z))) = z$ . Thus  $\mathcal{S}_s(u)$  is an  $E$ -valued  $(0, 1)$ -form on the domain of  $s$ .

**Lemma 4.2.** *In a smooth complex frame of  $E$ ,*

$$\mathcal{S}_s(u) = \bar{\partial}u + A(z, u),$$

where  $A$  is smooth in  $z$  and holomorphic in  $u$ , and  $A$  does not depend on derivatives of  $u$ . The linearization

$$D_s = d\mathcal{S}_s(0)$$

is a complex-linear Cauchy–Riemann operator on  $E$ . We write  $E_{D_s}$  for  $E$  endowed with the holomorphic structure defined by  $D_s$ . In a  $D_s$ -holomorphic frame,

$$A_u(z, 0) = 0.$$

If  $s$  is holomorphic on an open set  $U$ , then  $A(z, 0) = 0$  for  $z \in U$ .

*Proof.* In relative target coordinates, write the vertical coordinate expression of  $\Psi(s(z), q)$  as  $\Phi(z, q)$ . Let  $\partial_{\bar{z}}\Phi(z, q)$  denote differentiation with  $q$  fixed. Since  $\Phi$  is holomorphic in  $q$ ,

$$\partial_{\bar{z}}\Phi(z, u(z)) = \partial_{\bar{z}}\Phi(z, u) + \Phi_q(z, u)\partial_{\bar{z}}u.$$

Multiplication by  $\Phi_q^{-1}$  gives the displayed semilinear form with

$$A = \Phi_q^{-1}\partial_{\bar{z}}\Phi d\bar{z}.$$

The linearization is  $D_s = \bar{\partial} + A_u(z, 0)$  and satisfies the complex Leibniz rule; Lemma 2.1 supplies the holomorphic structure on  $E$ . Under a smooth frame change  $q = T(z)q'$ , the coefficient becomes

$$A'(z, q') = T^{-1}A(z, Tq') + T^{-1}(\bar{\partial}T)q'.$$

A  $D_s$ -holomorphic frame cancels the linear term. If  $s$  is holomorphic, then  $\mathcal{S}_s(0) = 0$ , so  $A(z, 0) = 0$  on every open set where  $s$  is holomorphic.  $\square$

In a general smooth frame, the coefficient  $A$  may include a linear term and obeys the inhomogeneous transformation law in the proof of Lemma 4.2. The intrinsic zero-order remainder is  $\mathcal{R}_s(u) = \mathcal{S}_s(u) - D_s u$ . In  $D_s$ -holomorphic frames,  $\mathcal{R}_s$  is represented by  $A$ , has zero fiber-linear term, and transforms tensorially. The auxiliary extension in the bordered construction applies a cutoff to the remainder  $\mathcal{R}_s$  in holomorphic frames. For a nonholomorphic section,  $D_s$  may depend on the chosen local addition; Lemma 4.3 identifies  $D_s$  canonically wherever  $s$  is holomorphic.

**Lemma 4.3.** *Let  $p : X \rightarrow \Sigma$  be holomorphic and let  $s$  be a smooth section whose image lies in the submersion locus. Choose a vertical local addition near  $s(\Sigma)$  and put  $D_s = d\mathcal{S}_s(0)$  by (4.1). If  $s$  is holomorphic on an open set  $U \subset \Sigma$ , then the Cauchy–Riemann operator  $D_s|_U$  is the natural Dolbeault operator on the holomorphic bundle  $s^*T_{X/\Sigma}|_U$ .*

*Proof.* Choose a source coordinate  $z$ , relative holomorphic target coordinates, and a holomorphic frame of  $s^*T_{X/\Sigma}$  on a smaller open subset of  $U$ . Let  $\Phi(z, q)$  denote the vertical-coordinate expression of  $\Psi(s(z), q)$  and put

$$B(z) = \partial_q \Phi(z, 0).$$

Since  $d_u \Psi(s(z), 0)$  is the identity of the vertical tangent space,  $B(z)$  is the coordinate matrix of the identity between two holomorphic frames. The matrix  $B(z)$  is therefore holomorphic and invertible. Since  $s$  is holomorphic,  $\Phi(z, 0)$  is holomorphic. The normalized equation is

$$\bar{\partial}q + A(z, q), \quad A(z, q) = (\partial_q \Phi(z, q))^{-1} \partial_{\bar{z}} \Phi(z, q) d\bar{z}.$$

Thus  $A(z, 0) = 0$ . On differentiating in  $q$  at zero, the derivative of the inverse matrix is multiplied by  $\partial_{\bar{z}} \Phi(z, 0) = 0$ , while the remaining term is  $B^{-1} \partial_{\bar{z}} B = 0$ . Hence  $A_q(z, 0) = 0$ , and the linearization in the chosen holomorphic frame is  $\bar{\partial}$ .  $\square$

Prescribed jets are enforced by requiring the vertical displacement to vanish along a divisor. The absence of constant and linear terms near the interpolation points makes the twisted equation smooth across that divisor.

**Lemma 4.4.** *Let  $A \subset \mathbb{R}$  be finite, let  $k_a \geq 0$  for  $a \in A$ , and set*

$$\Delta = \sum_{a \in A} (k_a + 1)a.$$

*Assume that  $\text{supp } \Delta$  is contained in an open set on which  $s$  is holomorphic. Set  $E_\Delta = E_{D_s}(-\Delta)$ , and let  $\iota : E_\Delta \rightarrow E_{D_s}$  be the canonical holomorphic morphism. Write  $\bar{\partial}_{E_\Delta}$  for the Dolbeault operator of  $E_\Delta$ . The operator*

$$\mathcal{S}_\Delta(v) = \iota^{-1} \mathcal{S}_s(\iota v),$$

*initially defined off  $\text{supp } \Delta$ , extends across  $\text{supp } \Delta$  to a smooth semilinear operator on a sufficiently small fixed fiber neighborhood in  $E_\Delta$ . In  $\bar{\partial}_{E_\Delta}$ -holomorphic frames  $\mathcal{S}_\Delta$  has the form*

$$\mathcal{S}_\Delta(v) = \bar{\partial}v + A_\Delta(z, v), \quad (A_\Delta)_v(z, 0) = 0. \quad (4.2)$$

*The coefficient  $A_\Delta(z, 0)$  vanishes on every open set on which  $s$  is holomorphic.*

*Proof.* Put  $n = \text{rank } E$ . Fix  $a \in \text{supp } \Delta$ , choose a coordinate  $z$  centered at  $a$ , and put  $m = k_a + 1$ . In compatible holomorphic frames,  $\iota$  is multiplication by  $z^m$ . Write  $\mathcal{S}_s(u) = \bar{\partial}u + \mathcal{A}(z, u)$  in these frames. Lemma 4.2 and the holomorphy of  $s$  near  $a$  give  $\mathcal{A}(z, 0) = \mathcal{A}_u(z, 0) = 0$ . Hence

$$\mathcal{A}(z, u) = \sum_{|\alpha| \geq 2} a_\alpha(z) u^\alpha$$

on a smaller fiber ball, where  $\alpha \in \mathbb{N}^n$  and  $a_\alpha$  is a smooth  $(0, 1)$ -form coefficient. Since  $\bar{\partial}z^m = 0$ , one obtains

$$z^{-m} \mathcal{S}_s(z^m v) = \bar{\partial}v + \sum_{|\alpha| \geq 2} a_\alpha(z) z^{m(|\alpha|-1)} v^\alpha. \quad (4.3)$$

The right-hand side is smooth at  $z = 0$  and holomorphic in  $v$ ; Cauchy estimates give locally uniform convergence with all mixed derivatives on a smaller fixed fiber ball. Every exponent  $m(|\alpha| - 1)$  is nonnegative, and every monomial has degree at least two in  $v$ . Thus the extended fiber-linear term vanishes at  $v = 0$ . Off  $\text{supp } \Delta$ , the identity  $D_s \iota = \iota \bar{\partial}_{E_\Delta}$  and  $\mathcal{A}_u(z, 0) = 0$  give  $(A_\Delta)_v(z, 0) = 0$  directly. Finally,  $\mathcal{S}_s(0) = 0$  wherever  $s$  is holomorphic, which proves the assertion about the constant term.  $\square$

In the reconstruction, the identity  $u = \iota v = z^{k_a+1}v$ , with  $v$  smooth and bounded, forces  $u = O(|z|^{k_a+1})$  near  $a$ . Once the reconstructed section is holomorphic, the bound for  $u$  gives equality of the prescribed jets, as shown at the end of the proof of Theorem 5.3; the parameter itself need not be holomorphic in an arbitrary smooth frame.

**4.2. Small solutions on a fixed bundle.** In both approximation constructions the bundle, norms, and right inverse are fixed before the disks are selected.

**Lemma 4.5.** *Let  $E$  be a holomorphic vector bundle on a compact Riemann surface  $\Sigma$ , with  $H^1(\Sigma, E) = 0$ . Fix smooth metrics, the resulting area form, a radius  $\rho > 0$ , and a positive number  $r_0$ . Write  $E(\rho) = \{v \in E : |v| < \rho\}$ . For  $0 < r < r_0$ , let*

$$A_r : E(\rho) \longrightarrow \Lambda^{0,1} T^* \Sigma \otimes E$$

*be smooth bundle maps, holomorphic in the fiber variable, with*

$$\sup_{E(\rho)} (|A_r| + |d_v A_r| + |d_v^2 A_r|) \leq M$$

*for a constant  $M$  independent of  $r$ . Put*

$$\mathbb{X} = W^{1,4}(\Sigma, E), \quad \mathbb{Y} = L^4(\Sigma, \Lambda^{0,1} T^* \Sigma \otimes E), \quad \mathcal{F}_r(v) = \bar{\partial}_E v + A_r(z, v(z)).$$

*The domain of  $\mathcal{F}_r$  is the open set  $\{v \in \mathbb{X} : \|v\|_{C^0} < \rho\}$ . Suppose that  $\varepsilon_r > 0$  tends to zero as  $r \downarrow 0$  and*

$$\|\mathcal{F}_r(0)\|_{\mathbb{Y}} \leq \varepsilon_r, \quad \|d\mathcal{F}_r(0) - \bar{\partial}_E\|_{\mathbb{X} \rightarrow \mathbb{Y}} \leq \varepsilon_r.$$

*For all sufficiently small  $r$ , the following hold:*

- (i) *There is a solution  $v_r \in \mathbb{X}$  of  $\mathcal{F}_r(v_r) = 0$  satisfying*

$$\|v_r\|_{W^{1,4}} \leq C\varepsilon_r, \quad \|v_r\|_{C^0} < \rho.$$

*The constant  $C$  and the required upper bound for  $\varepsilon_r$  depend only on the fixed bundle, metrics,  $\rho$ , and  $M$ .*

- (ii) *The solution  $v_r$  in (i) is smooth. No bounds for source derivatives of  $A_r$  that are uniform in  $r$  are required.*

*Proof.* (i) Dolbeault Fredholm theory and  $H^1(\Sigma, E) = 0$  give surjectivity of  $\bar{\partial}_E : \mathbb{X} \rightarrow \mathbb{Y}$ ; the corresponding regularity estimate is (2.4). Put

$$\mathbb{X}_0 = \{u \in \mathbb{X} : \langle u, h \rangle_{L^2} = 0 \text{ for every } h \in H^0(\Sigma, E)\}.$$

The restriction  $\bar{\partial}_E : \mathbb{X}_0 \rightarrow \mathbb{Y}$  is a bounded bijection between Banach spaces. The bounded inverse theorem gives  $T : \mathbb{Y} \rightarrow \mathbb{X}_0 \subset \mathbb{X}$  with  $\bar{\partial}_E T = I_{\mathbb{Y}}$ .

On  $\{v \in \mathbb{X} : \|v\|_{C^0} < \rho\}$ , the map  $\mathcal{F}_r$  is  $C^2$ . Indeed, Taylor expansion in each of finitely many fixed bundle frames gives

$$\begin{aligned} d\mathcal{F}_r(v)\xi &= \bar{\partial}_E \xi + (d_v A_r)(z, v(z))\xi(z), \\ d^2 \mathcal{F}_r(v)(\xi, \xi') &= (d_v^2 A_r)(z, v(z))[\xi(z), \xi'(z)]. \end{aligned}$$

Sobolev embedding  $W^{1,4} \hookrightarrow C^0$ , density, and the fiber bounds give the displayed first- and second-derivative formulas and continuity on the indicated open set. More explicitly, for fixed  $r$  and  $v$ , choose  $\|v\|_{C^0} < \rho' < \rho$ . On the closed  $\rho'$ -ball bundle,  $d_v^2 A_r$  has a modulus of continuity  $\omega_{r,\rho'}(t) \rightarrow 0$ .

For sufficiently small  $\|h\|_{\mathbb{X}}$ , the segments  $v + th$ ,  $0 \leq t \leq 1$ , lie in that ball bundle. Taylor's integral formula and  $W^{1,4} \hookrightarrow C^0$  give

$$\|A_r(\cdot, v + h) - A_r(\cdot, v) - d_v A_r(\cdot, v)h\|_{L^4} \leq C\|h\|_{C^0}\|h\|_{L^4} = O(\|h\|_{\mathbb{X}}^2).$$

Applying the integral formula to  $d_v A_r$  bounds the remainder after subtracting  $d_v^2 A_r(\cdot, v)[h, \xi]$  by

$$\omega_{r,\rho'}(\|h\|_{C^0})\|h\|_{C^0}\|\xi\|_{L^4}.$$

Taking the supremum over  $\|\xi\|_{\mathbb{X}} \leq 1$  proves Fréchet differentiability of the first differential in operator norm; the same modulus proves continuity of the second differential. No modulus uniform in  $r$  is needed. For

$$\mathcal{N}_r(v) = \mathcal{F}_r(v) - \mathcal{F}_r(0) - d\mathcal{F}_r(0)v,$$

the fiber bound and Sobolev embedding give a constant  $K$  independent of  $r$  such that

$$\|d^2 \mathcal{F}_r(u)(\xi, \xi')\|_{\mathbb{Y}} \leq K\|\xi\|_{\mathbb{X}}\|\xi'\|_{\mathbb{X}}.$$

Let  $C_S$  be a Sobolev embedding constant for  $\mathbb{X} \hookrightarrow C^0$ . Take  $\delta > 0$  with  $C_S\delta < \rho/2$ . For  $\|v\|_{\mathbb{X}}, \|w\|_{\mathbb{X}} \leq \delta$ , the segment  $w + t(v - w)$  stays in that ball, and

$$\mathcal{N}_r(v) - \mathcal{N}_r(w) = \int_0^1 (d\mathcal{F}_r(w + t(v - w)) - d\mathcal{F}_r(0))(v - w) dt.$$

The second-derivative bound therefore yields

$$\|\mathcal{N}_r(v) - \mathcal{N}_r(w)\|_{\mathbb{Y}} \leq C(\|v\|_{\mathbb{X}} + \|w\|_{\mathbb{X}})\|v - w\|_{\mathbb{X}}. \quad (4.4)$$

The constant in (4.4) depends on  $M$ , the finite frame changes, the Sobolev embedding constant, and the fixed area of  $\Sigma$ , and is independent of  $r$ .

Put  $P_r = d\mathcal{F}_r(0) - \bar{\partial}_E$ . For  $\varepsilon_r\|T\| < 1/2$ , one has  $\|P_r T\| < 1/2$ , so

$$R_r = T(I + P_r T)^{-1}$$

is a right inverse of  $d\mathcal{F}_r(0)$  with  $\|R_r\| \leq 2\|T\|$ . Indeed,  $d\mathcal{F}_r(0)T = I + P_r T$ , hence  $d\mathcal{F}_r(0)R_r = I$ . Consider the fixed-point equation

$$v = -R_r(\mathcal{F}_r(0) + \mathcal{N}_r(v)). \quad (4.5)$$

Increase  $K$ , if necessary, to bound the constant in (4.4), put  $C_0 = \max\{1, 4\|T\|\}$ , and take  $r$  small enough that

$$C_0\varepsilon_r < \delta, \quad 4\|T\|KC_0\varepsilon_r < \frac{1}{2}.$$

Write  $\mathcal{B}_r$  for the closed  $\mathbb{X}$ -ball of radius  $C_0\varepsilon_r$ , and write  $Q_r(v)$  for the right side of (4.5). Since  $\mathcal{N}_r(0) = 0$ ,

$$\begin{aligned} \|Q_r(v)\|_{\mathbb{X}} &\leq 2\|T\|(\varepsilon_r + KC_0^2\varepsilon_r^2) < C_0\varepsilon_r, \\ \|Q_r(v) - Q_r(w)\|_{\mathbb{X}} &\leq 4\|T\|KC_0\varepsilon_r\|v - w\|_{\mathbb{X}} < \frac{1}{2}\|v - w\|_{\mathbb{X}} \quad (v, w \in \mathcal{B}_r). \end{aligned}$$

The choice of  $\delta$  also gives  $C_S C_0\varepsilon_r < \rho/2$ . The Banach contraction theorem gives a fixed point  $v_r$ . Applying  $d\mathcal{F}_r(0)$  to (4.5) gives

$$d\mathcal{F}_r(0)v_r = -\mathcal{F}_r(0) - \mathcal{N}_r(v_r), \quad \mathcal{F}_r(v_r) = 0.$$

Membership in  $\mathcal{B}_r$  gives  $\|v_r\|_{\mathbb{X}} \leq C_0\varepsilon_r$ .

(ii) Fix a value of  $r$  for which the fixed point in (i) exists. In a local holomorphic frame the equation is  $\bar{\partial}v_r = -A_r(z, v_r)$ . The coefficient is smooth across all source charts. In a source coordinate, let  $\nabla$  denote the real gradient. Since  $v_r \in W^{1,4} \subset C^0$ , the Sobolev chain rule gives

$$\nabla[A_r(z, v_r)] = (\nabla_z A_r)(z, v_r) + (d_v A_r)(z, v_r)\nabla v_r \in L^4.$$

Local Dolbeault regularity yields  $v_r \in W^{2,4}$ . On successively smaller source charts, the Sobolev product and chain rules give  $A_r(z, v_r) \in W^{k,4}$  whenever  $v_r \in W^{k,4}$ , for  $k \geq 1$ . The local form of (2.4) then gives  $v_r \in W^{k+1,4}$ . Induction and the Sobolev embeddings prove smoothness. Source

derivatives enter only the regularity argument for a fixed  $r$ ; the bounds for those source derivatives may depend on  $r$ .  $\square$

## 5. ANALYTIC APPROXIMATION

**5.1. Prepared rational disks.** The remainder of the analytic construction assumes the relative first-jet sphere-family property, rather than projectivity or rational connectedness. Write  $n$  for the relative dimension,  $D_r = \Delta_r = \{z \in \mathbb{C} : |z| < r\}$ , and  $B_r = \{q \in \mathbb{C}^n : |q| < r\}$ . The substitution  $w = \sigma^2/z$  replaces the antiholomorphic first-order term of a smooth section by a holomorphic rational disk.

**Lemma 5.1.** *Let  $p : X \rightarrow D_{r_0}$  be a proper holomorphic submersion with the relative first-jet sphere-family property, and let*

$$\Phi : D_{r_0} \times B_{2d_0} \longrightarrow X, \quad p(\Phi(z, q)) = z,$$

*be smooth and holomorphic in  $q$ . Suppose that the image of  $\Phi$  near  $(0, 0)$  lies in a relative holomorphic chart  $(p, \phi)$ . Writing  $\hat{\Phi} = \phi \circ \Phi$ , we have, after shrinking the two balls,*

$$\hat{\Phi}(z, q) = m(q) + a(q)z + b(q)\bar{z} + R(z, q), \quad R = O_{C_q^3}(|z|^2), \quad (5.1)$$

where

$$m(q) = \hat{\Phi}(0, q), \quad a(q) = \partial_z \hat{\Phi}(0, q), \quad b(q) = \partial_{\bar{z}} \hat{\Phi}(0, q)$$

*are holomorphic in  $q$ . Here  $O_{C_q^3}(|z|^2)$  means that the  $q$ -derivatives through order three are bounded by  $C|z|^2$  on the fixed parameter ball. The notation  $\|\cdot\|_{C_q^j}$  denotes the supremum of all parameter derivatives through order  $j$  on the indicated ball. On  $|z| = \sigma$ , the identity  $\bar{z} = \sigma^2/z$  is the matching relation used to replace the antiholomorphic first-order term by a holomorphic rational disk.*

*There are  $d, s_0, C > 0$  such that, for every  $0 < \sigma < s_0/4$ , there is a jointly holomorphic map*

$$H_\sigma : D_{s_0} \times B_{2d} \longrightarrow X, \quad p(H_\sigma(z, q)) = z.$$

*On  $\sigma/2 \leq |z| \leq 2\sigma$ , the vertical coordinate of  $H_\sigma$  satisfies*

$$\phi(H_\sigma(z, q)) = m(q) + a(q)z + b(q)\frac{\sigma^2}{z} + R_\sigma(z, q), \quad (5.2)$$

*with, for  $|\alpha| \leq 3$ ,*

$$\sup_{|q| \leq d} |\partial_q^\alpha R_\sigma(z, q)| \leq C\sigma^2, \quad \sup_{|q| \leq d} |\partial_z \partial_q^\alpha R_\sigma(z, q)| \leq C\sigma. \quad (5.3)$$

*Moreover,  $d_q H_\sigma$  is uniformly bounded on  $D_{s_0} \times B_d$ , independently of  $\sigma$ . The constants are locally uniform when the source and target charts are fixed, the relevant first jets remain in a relatively compact subset of the domain of one fixed first-jet family, and the indicated mixed derivatives of  $\Phi$  are uniformly bounded. The additional compact-center uniformity needed below, including target-chart margins and inverse bounds, is stated separately in Lemma 5.2.*

*Proof.* Equation (5.1) is Taylor's formula in  $(\operatorname{Re} z, \operatorname{Im} z)$ . The functions  $m, a, b$  are holomorphic in  $q$ : smoothness permits commuting the source derivatives with  $\partial_{\bar{q}}$ , and  $\partial_{\bar{q}} \hat{\Phi} = 0$ .

Choose a family  $G$  from Definition 1.3 at the vertical first jet represented by  $(0, m(0), b(0))$ . In the relative chart, write the parameters of  $G$  as  $(t, x, \beta)$ . After shrinking  $s_0$  and  $d$ , the triple

$$(z, m(q) + a(q)z, b(q))$$

lies in a fixed compact subset  $\mathcal{L}$  of the parameter domain of  $G$  whenever  $|z| \leq s_0$  and  $|q| \leq 2d$ . By continuity of  $G$  and compactness of  $\mathcal{L}$ , there is  $\epsilon_G > 0$  such that  $G(w; \lambda)$  lies in the chosen target

chart for  $|w| \leq \epsilon_G$  and  $\lambda \in \mathcal{L}$ . Decrease  $s_0$  so that  $s_0/2 < \epsilon_G$ . Then on the annulus used below,  $|\sigma^2/z| \leq 2\sigma < s_0/2$ , so the coordinate expansion is defined. Set

$$H_\sigma(z, q) = G\left(\frac{\sigma^2}{z}; z, m(q) + a(q)z, b(q)\right), \quad (5.4)$$

where  $z \mapsto \sigma^2/z$  is viewed as a holomorphic map to  $\mathbb{P}^1$  taking 0 to  $\infty$ . For each fixed  $\sigma > 0$ , the coordinate  $w' = 1/w$  at infinity gives  $w' = z/\sigma^2$ , an expression that is holomorphic at zero and independent of  $q$ . Thus the formula in (5.4) extends holomorphically across  $z = 0$ , and the identity  $p(G(w; t, x, \beta)) = t$  gives  $p \circ H_\sigma = z$ . In particular,  $H_\sigma(0, q) = G(\infty; 0, m(q), b(q))$ .

The prescribed first jet of  $G$  gives a holomorphic map  $\mathcal{C}$  such that, for  $w$  near zero,

$$\phi(G(w; t, x, \beta)) = x + \beta w + w^2 \mathcal{C}(w; t, x, \beta).$$

Substitution into (5.4) gives (5.2), with

$$R_\sigma(z, q) = \frac{\sigma^4}{z^2} \mathcal{C}\left(\frac{\sigma^2}{z}; z, m(q) + a(q)z, b(q)\right).$$

On  $\sigma/2 \leq |z| \leq 2\sigma$  one has

$$\left|\frac{\sigma^4}{z^2}\right| \leq 4\sigma^2, \quad \left|\partial_z \frac{\sigma^4}{z^2}\right| \leq 16\sigma, \quad \left|\partial_z \frac{\sigma^2}{z}\right| \leq 4.$$

Cauchy estimates for  $\mathcal{C}$  and the fixed functions  $m, a, b$  give (5.3). The uniform bound for  $d_q H_\sigma$  follows from the parameter-derivative bound in Remark 3.3, since  $\sigma^2/z$  is independent of  $q$ . Differentiation in  $q$  therefore introduces no factor from the rational source map  $z \mapsto \sigma^2/z$ ; its  $z$ -derivatives need not be uniformly bounded near zero as  $\sigma \rightarrow 0$ . The local uniformity assertion follows by taking compact subsets of the chosen source, target, and parameter charts and applying the three displayed bounds and Cauchy estimates on those compact subsets.  $\square$

The value  $z = 0$  in this substitution corresponds to  $w = \infty$ . Thus the construction uses a jointly holomorphic family on all of  $\mathbb{P}^1$ , not merely germs near  $w = 0$ . Invertibility of the parameter derivative is needed on the transition annulus, not on the disk core.

## 5.2. Uniform matching on annuli.

**Lemma 5.2.** *Let  $p : X \rightarrow R$  be a proper holomorphic submersion with the relative first-jet sphere-family property, let  $P \subset R$  be compact, and let  $V$  be a neighborhood of  $P$ . Suppose that*

$$\Phi : V \times B_{2d_0} \longrightarrow X, \quad p(\Phi(z, q)) = z,$$

*is smooth, holomorphic in  $q$ , and*

$$d_q \Phi(z, 0) : \mathbb{C}^n \longrightarrow T_{X/R, \Phi(z, 0)}$$

*is an isomorphism for every  $z \in P$ . Then  $P$  has a finite cover by open sets  $V'_\nu \Subset V_\nu \Subset V$ , equipped with source coordinates, relative target charts, and first-jet families as in Definition 1.3, for which there are common constants*

$$d > 0, \quad \sigma_* > 0, \quad \mu > 0, \quad C > 0$$

*with the following properties.*

*Let  $c \in V'_\nu$ , put  $\zeta = \zeta_\nu(z) - \zeta_\nu(c)$ , and let  $0 < \sigma < \sigma_*$ , with*

$$h_\sigma = \sigma^{3/2} < \frac{\sigma}{4}.$$

Put

$$\begin{aligned}\Delta_{c,\sigma}^+ &= \{|\zeta| < \sigma + h_\sigma\}, \\ \Delta_{c,\sigma}^- &= \{|\zeta| < \sigma - h_\sigma\}, \\ \mathcal{T}_{c,\sigma} &= \Delta_{c,\sigma}^+ \setminus \overline{\Delta_{c,\sigma}^-}.\end{aligned}\tag{5.5}$$

There is a jointly holomorphic prepared family  $H_{c,\sigma}$  on the outer disk and, on  $\mathcal{T}_{c,\sigma} \times B_d$ , a smooth  $X$ -valued matched family  $M_{c,\sigma}$  with the following properties:

- (i) In the selected vertical target coordinate, write  $\hat{\Phi} = \phi_\nu \circ \Phi$  and  $\hat{H}_{c,\sigma} = \phi_\nu \circ H_{c,\sigma}$ . Then

$$\Xi_{c,\sigma} = (1 - \eta_{c,\sigma})\hat{\Phi} + \eta_{c,\sigma}\hat{H}_{c,\sigma}, \quad M_{c,\sigma}(z, q) = (p, \phi_\nu)^{-1}(z, \Xi_{c,\sigma}(z, q)),$$

where

$$\eta_{c,\sigma}(z) = \chi_0\left(\frac{|\zeta| - \sigma}{h_\sigma}\right),$$

and  $\chi_0 \in C^\infty(\mathbb{R}, [0, 1])$  is fixed, equal to one on  $(-\infty, -1/2]$  and zero on  $[1/2, \infty)$ . In particular, the displayed inverse target chart is defined for every indicated  $(z, q)$ .

- (ii) On the transition annulus,

$$\|\hat{H}_{c,\sigma} - \hat{\Phi}\|_{C_q^3} \leq Ch_\sigma, \quad \|(\partial_q \Xi_{c,\sigma})^{-1}\| \leq \mu^{-1}.\tag{5.6}$$

The coefficient

$$C_{c,\sigma}^M(z, q) = (\partial_q \Xi_{c,\sigma}(z, q))^{-1} \partial_{\bar{\zeta}} \Xi_{c,\sigma}(z, q) d\bar{\zeta}\tag{5.7}$$

satisfies

$$\sup_{|q| \leq d} |\partial_q^j C_{c,\sigma}^M(z, q)| \leq C, \quad 0 \leq j \leq 2,\tag{5.8}$$

uniformly in  $c$ ,  $\nu$ , and  $\sigma$ . The notation  $\partial_q^j$  denotes the collection of all order- $j$  derivatives in  $q$ .

- (iii) On the open outer collar  $\mathcal{T}_{c,\sigma} \cap \{|\zeta| > \sigma + h_\sigma/2\}$  the matched family equals  $\Phi$  and (5.7) equals

$$C^\Phi = (\partial_q \hat{\Phi})^{-1} \partial_{\bar{\zeta}} \hat{\Phi} d\bar{\zeta}.$$

On the open inner collar  $\mathcal{T}_{c,\sigma} \cap \{|\zeta| < \sigma - h_\sigma/2\}$  the family  $M_{c,\sigma}$  equals  $H_{c,\sigma}$  and  $C_{c,\sigma}^M = 0$ .

The inverse-matrix estimate is required only on the transition annulus. On the inner disk, reconstruction uses the joint holomorphy of  $H_{c,\sigma}$  and does not require  $d_q H_{c,\sigma}$  to be invertible. Only the  $q$ -derivatives in (5.8) are uniform as  $\sigma \rightarrow 0$ ; no uniform assertion is made about  $z$ -derivatives of the matched family or coefficient.

*Proof.* Fix  $c_0 \in P$ . Choose a source coordinate near  $c_0$  and a relative target chart whose image contains a product of the source neighborhood with a Euclidean ball about the vertical-coordinate image of  $\Phi(c_0, 0)$ . Shrink twice to obtain nested convex balls

$$P_{c_0}^0 \Subset P_{c_0}^1 \Subset P_{c_0}^2$$

inside the vertical target-coordinate image. Definition 1.3 at the antiholomorphic first jet of  $\Phi(\cdot, 0)$  at  $c_0$  gives a first-jet family on a parameter neighborhood. Shrink the source neighborhood and the  $q$ -ball so that, for every center in the smaller source neighborhood, the triples consisting of the moving base point, the value parameter, and the antiholomorphic first derivative lie in a relatively compact subdomain of that parameter neighborhood. Also arrange that the coordinate image of  $\Phi$  lies in  $P_{c_0}^0$  on the smaller source and parameter sets, and that  $d_q \Phi(z, 0)$  remains invertible throughout the smaller source set.

Compactness of  $P$  gives finitely many such smaller source sets  $V'_\nu$  covering  $P$ , with closures in the corresponding  $V_\nu$ . The mixed derivatives of  $\Phi$  through order two in the source and three in  $q$  have a common bound on the compact source and parameter sets just chosen. The evaluation maps of the finitely many first-jet families have common bounds for the required parameter derivatives on the

chosen relatively compact domains. The construction and proof of Lemma 5.1, after translation of the source coordinate to each center  $c$ , therefore give common  $d, \sigma_*$  and  $C$  and prepared families  $H_{c,\sigma}$ . Relabel the three nested target-coordinate balls belonging to the  $\nu$ th retained choice of local data as  $P_\nu^0 \Subset P_\nu^1 \Subset P_\nu^2$ . Decrease  $\sigma_*$  once more so that, for every  $c \in V'_\nu$  and  $0 < \sigma < \sigma_*$ , the closed outer disk  $\Delta_{c,\sigma}^+$  is contained in  $V_\nu$ . A common outer-disk radius exists because only finitely many inclusions  $\bar{V}'_\nu \Subset V_\nu$  occur.

(i) The finitely many inclusions  $P_\nu^0 \Subset P_\nu^1 \Subset P_\nu^2$  have a common positive coordinate margin. On the transition annulus the two Taylor expansions give

$$\hat{\Phi} - \hat{H}_{c,\sigma} = b_c(q) \left( \bar{\zeta} - \frac{\sigma^2}{\zeta} \right) + R_c(\zeta, q) - R_{c,\sigma}(\zeta, q).$$

Here

$$\left| \bar{\zeta} - \frac{\sigma^2}{\zeta} \right| = \frac{||\zeta|^2 - \sigma^2|}{|\zeta|} \leq h_\sigma \frac{2\sigma + h_\sigma}{\sigma - h_\sigma} \leq 3h_\sigma, \quad \sigma^2 \leq h_\sigma.$$

The factor  $\bar{\zeta} - \sigma^2/\zeta$  is independent of  $q$ . Combining the displayed bound with the uniform  $C_q^3$  bounds for  $b_c$  and the  $O_{C_q^3}(\sigma^2)$  remainders proves the first estimate in (5.6). After decreasing  $\sigma_*$ , the prepared family lies in  $P_\nu^1$  on the annulus. Since  $P_\nu^1$  is convex, the entire segment defining  $\Xi_{c,\sigma}$  remains in  $P_\nu^1 \Subset P_\nu^2$ . Thus  $M_{c,\sigma}$  takes values in  $X$ .

(ii) To prove the inverse bound in (5.6), use continuity and the assumed invertibility of  $d_q\Phi(z, 0)$  to obtain, after shrinking the common  $q$ -ball and source sets,

$$\|(\partial_q \hat{\Phi}(z, q))^{-1}\| \leq (2\mu)^{-1}$$

for one  $\mu > 0$  on all chosen source and parameter sets. The  $C_q^1$  part of the first estimate in (5.6) makes  $\|\partial_q \Xi_{c,\sigma} - \partial_q \hat{\Phi}\| < \mu$  after another decrease of  $\sigma_*$ . For  $L_0 = \partial_q \hat{\Phi}$  and  $L_1 = \partial_q \Xi_{c,\sigma}$ ,

$$L_1 = L_0(I + L_0^{-1}(L_1 - L_0)), \quad \|L_0^{-1}(L_1 - L_0)\| < \frac{1}{2}.$$

Hence  $\|L_1^{-1}\| \leq 2\|L_0^{-1}\| \leq \mu^{-1}$ , proving the second bound in (5.6).

For the following calculation, suppress the fixed indices  $c, \sigma$  in  $\eta, \Xi, \hat{H}$ . One has  $|\bar{\partial}\eta_{c,\sigma}| \leq Ch_\sigma^{-1}$ . Since  $\hat{H}$  is holomorphic, the numerator in the coefficient satisfies

$$\partial_{\bar{\zeta}}\Xi = (1 - \eta)\partial_{\bar{\zeta}}\hat{\Phi} + (\partial_{\bar{\zeta}}\eta)(\hat{H} - \hat{\Phi}).$$

The matching difference is  $O_{C_q^3}(h_\sigma)$ , so its product with the cutoff derivative is uniformly bounded through two  $q$ -derivatives. Set  $L_1 = \partial_q \Xi$ . The identity

$$\partial_{q_j}(L_1^{-1}) = -L_1^{-1}(\partial_{q_j}L_1)L_1^{-1}$$

and

$$\begin{aligned} \partial_{q_k}\partial_{q_j}L_1^{-1} &= L_1^{-1}(\partial_{q_k}L_1)L_1^{-1}(\partial_{q_j}L_1)L_1^{-1} \\ &\quad + L_1^{-1}(\partial_{q_j}L_1)L_1^{-1}(\partial_{q_k}L_1)L_1^{-1} \\ &\quad - L_1^{-1}(\partial_{q_j}\partial_{q_k}L_1)L_1^{-1} \end{aligned}$$

bound the first two derivatives of  $L_1^{-1}$  by the inverse bound and the derivatives of  $\Xi$  through order three. Applying the product rule to  $L_1^{-1}\partial_{\bar{\zeta}}\Xi d\bar{\zeta}$  proves (5.8).

(iii) On the outer collar,  $\eta = 0$  gives  $M_{c,\sigma} = \Phi$  and  $C_{c,\sigma}^M = C^\Phi$ . On the inner collar,  $\eta = 1$  gives  $M_{c,\sigma} = H_{c,\sigma}$  and  $C_{c,\sigma}^M = 0$ , because  $H_{c,\sigma}$  is jointly holomorphic.  $\square$

Lemma 5.2 also applies to finitely many local families  $\Phi_\alpha$  expressed in different holomorphic bundle frames, provided their smaller center sets cover the compact set under consideration. Apply the proof to each family and take minima of the finitely many radii and margins and maxima of the bounds. In a global construction, assign each selected disk to one family. Distinct disks are

disjoint, and on every outer collar the coefficient agrees with the same global equation. Thus no compatibility between different inner families, and no global holomorphic frame, is required.

The compact center sets, finite charts, and bundle frames are fixed first; then choose the common parameter radius, target margin, and  $\sigma_*$ . Only afterwards choose the global scale  $r$  and the finite disk family. For  $h_i = \sigma_i^{3/2}$  and  $\sigma_i \leq r$ , the two roles of  $h_\sigma = \sigma^{3/2}$  are

$$O(h_\sigma^{-1})O_{C_q^3}(h_\sigma) = O_{C_q^2}(1), \quad \sum_i \sigma_i h_i = \sum_i \sigma_i^2 \sigma_i^{1/2} \leq r^{1/2} \sum_i \sigma_i^2.$$

The first bound controls the matched coefficients; disjointness of the disks bounds  $\sum_i \sigma_i^2$  and gives the area estimate (5.14).

### 5.3. Approximation on bordered surfaces.

**Theorem 5.3.** *Let  $\pi : Z \rightarrow Y$  be a proper holomorphic submersion with connected fibers and the relative first-jet sphere-family property, let  $R$  be an open Riemann surface, let  $g : R \rightarrow Y$  be holomorphic, and let  $p : X = R \times_Y Z \rightarrow R$  be the pullback. Let  $D \subset R$  be a compact bordered Riemann surface, let  $K \subset \text{int } D$  be compact, and let  $A \subset \text{int } D$  be finite. Let  $s$  be a smooth section of  $p$  on a neighborhood of  $D$ , holomorphic on a neighborhood of  $K \cup A$ . Prescribe an integer  $k_a \geq 0$  for every  $a \in A$ . For every  $\epsilon > 0$  there is a section  $F$  of  $p$  holomorphic on a neighborhood of  $D$  such that*

$$\sup_K d_X(F, s) < \epsilon, \quad j_a^{k_a} F = j_a^{k_a} s \quad (a \in A).$$

*Proof.* The pullback  $p$  has the relative first-jet sphere-family property by Remark 3.3. Properness of  $p$  makes  $p(X)$  closed in  $R$ , and the local projection form of a holomorphic submersion makes  $p(X)$  open. The section  $s$  is defined near the nonempty bordered surface  $D$ , so  $p(X) \neq \emptyset$ . Connectedness of  $R$  therefore gives  $p(X) = R$ . If the fibers are zero-dimensional, every nonempty fiber is a finite discrete set and hence, by connectedness, a singleton. The proper holomorphic submersion  $p$  is then a bijective local biholomorphism, and hence a biholomorphism. The inverse  $p^{-1} : R \rightarrow X$  is holomorphic, and every local section of  $p$  equals the restriction of  $p^{-1}$ . Taking  $F = p^{-1}$  proves the zero-dimensional case. We may therefore assume that the fiber dimension is positive.

We proceed in seven steps.

*Step 1: embed a larger source neighborhood in a compact surface.* Choose connected compact bordered Riemann surfaces

$$D \Subset \text{int } D_0, \quad D_0 \Subset \text{int } D_1,$$

inside the neighborhood on which  $s$  is defined. Lemma 2.2 embeds a neighborhood  $W$  of  $D_1$  holomorphically into a compact Riemann surface  $\Sigma$ . We identify  $W$  with its image. Write  $g_\Sigma$  for the genus of  $\Sigma$ . Fix a smooth area form  $\omega_\Sigma$  on  $\Sigma$  and Hermitian metrics on the bundles used below. All areas and Sobolev norms refer to these fixed choices; the finitely many source-coordinate area forms are uniformly equivalent to  $\omega_\Sigma$ . By first replacing  $W$  by a smaller connected precompact neighborhood of  $D_1$ , we may assume that  $\overline{W}$  lies in the original domain of  $s$ . Fix the vertical local addition of Lemma 4.1 over  $p^{-1}(\overline{W})$ . All reconstructions of sections of  $X$  below will be performed in  $W$ ; auxiliary modifications will avoid the reconstruction neighborhood chosen in Step 4.

Set  $\Delta = \sum_{a \in A} (k_a + 1)a$ . Lemma 4.4 gives  $E_\Delta$  and the operator (4.2) on  $W$ . By the deformation retraction onto a graph in Subsection 2.2,  $W$  deformation retracts onto a countable locally finite graph  $\Gamma$ . Every complex vector bundle on  $\Gamma$  is trivial: choose frames at the vertices and extend them along the edges, which is possible because  $\text{GL}_n(\mathbb{C})$ , with  $n$  the bundle rank, is path connected; local finiteness makes the resulting frame continuous. Pullback along the deformation retraction therefore shows that every complex vector bundle on  $W$  is topologically trivial. By the Behnke–Stein theorem [6, Corollary 26.8],  $W$  is a Stein Riemann surface, and Grauert’s Oka principle [11] makes  $E_\Delta$  holomorphically trivial. Fix a holomorphic trivialization of  $E_\Delta$  on  $W$ .

*Step 2: extend the bundle to  $\Sigma$  with vanishing first cohomology.* After shrinking  $W$  slightly while retaining  $D_1$ , the complement of  $\overline{W}$  in  $\Sigma$  is nonempty. Choose an effective divisor  $Q$  on  $\Sigma$  supported in  $\Sigma \setminus \overline{W}$ , of degree greater than  $2g_\Sigma - 2$ , and put

$$\tilde{E} = \mathcal{O}_\Sigma(Q)^{\oplus n}, \quad n = \dim X_z.$$

The canonical section of  $\mathcal{O}_\Sigma(Q)$  is nonzero on  $W$ , and therefore trivializes  $\mathcal{O}_\Sigma(Q)|_W$ . Using the chosen trivialization of  $E_\Delta$ , identify

$$\tilde{E}|_W \cong E_\Delta.$$

Serre duality gives

$$H^1(\Sigma, \mathcal{O}_\Sigma(Q)) \cong H^0(\Sigma, K_\Sigma(-Q))^* = 0,$$

because  $\deg K_\Sigma(-Q) = 2g_\Sigma - 2 - \deg Q < 0$ . A nonzero holomorphic section of a line bundle has an effective zero divisor of degree equal to the bundle degree, so a negative-degree line bundle has no nonzero holomorphic section. Hence

$$H^1(\Sigma, \tilde{E}) = 0. \tag{5.9}$$

Take a smooth cutoff  $\chi$  supported in  $W$  and equal to one on a neighborhood of  $D_1$ . Under  $\tilde{E}|_W \cong E_\Delta$  from Step 2, extend the nonlinear coefficient by

$$\tilde{A}(z, v) = \chi(z)A_\Delta(z, v)$$

on a fixed small fiber ball and set  $\tilde{A} = 0$  outside  $W$ . Compact support of  $\chi$  makes  $\tilde{A}$  a smooth bundle-valued  $(0, 1)$ -form, holomorphic in  $v$ , and

$$\tilde{A}_v(z, 0) = 0.$$

On the compact surface, put

$$\tilde{\mathcal{S}}(v) = \bar{\partial}_{\tilde{E}}v + \tilde{A}(z, v). \tag{5.10}$$

The operator  $\tilde{\mathcal{S}}$  agrees with  $\mathcal{S}_\Delta$  on a neighborhood of  $D_1$ , while outside that neighborhood  $\tilde{\mathcal{S}} = 0$  is only an auxiliary equation. Since  $\tilde{A}_v(z, 0) = 0$ , the linearization  $d\tilde{\mathcal{S}}(0)$  is exactly  $\bar{\partial}_{\tilde{E}}$ .

By (5.9), fix a bounded right inverse  $T$  of  $\bar{\partial}_{\tilde{E}} : W^{1,4}(\Sigma, \tilde{E}) \rightarrow L^4_{0,1}(\Sigma, \tilde{E})$  as in the proof of Lemma 4.5. This choice precedes all disk selections.

*Step 3: prepare the rational disks.* Let

$$e_0 = \tilde{\mathcal{S}}(0).$$

The support of  $e_0$  is disjoint from a fixed neighborhood of  $K \cup A$ . Choose an open set  $W_1$  such that

$$D_1 \Subset W_1, \quad \overline{W}_1 \Subset \{z \in W : \chi(z) = 1\}.$$

After decreasing the fixed neighborhood of  $K \cup A$ , the compact set

$$P = \text{supp } e_0 \cap D_1$$

is disjoint from  $\text{supp } \Delta$ . Use the holomorphic trivialization of  $E_\Delta$  fixed in Step 1, and its identification with  $\tilde{E}$  over  $W$ , to regard  $q$  as a fixed holomorphic bundle coordinate. Compactness of  $\overline{W}_1$  and the domain of the fixed local addition give  $d_0 > 0$  such that, on  $W_1$ , the family

$$\Phi(z, q) = \Psi(s(z), \iota_z q)$$

is defined for every  $q \in B_{2d_0}$ . At  $q = 0$ ,

$$d_q \Phi(z, 0) = d_u \Psi(s(z), 0) \circ \iota_z = \iota_z.$$

The map  $\iota_z$  is an isomorphism for  $z \in P$ , because  $P \cap \text{supp } \Delta = \emptyset$ . Lemma 5.2 therefore supplies finitely many source and target charts, first-jet families, parameter domains, and common constants. The resulting holomorphic disk families are constructed from rational curves; we call the disks used in this part of the construction rational disks. Every such disk will have its center in  $\{e_0 \neq 0\} \cap W_1$

and its closed outer disk contained in the corresponding source set  $V_\nu$ . We use exactly the inner disk, transition annulus, cutoff, prepared family  $H_{c,\sigma}$ , target-valued matched family  $M_{c,\sigma}$ , and coefficient  $C_{c,\sigma}^M$  from Lemma 5.2.

The chain rule in the selected target coordinate gives, for every smooth  $q(z) \in B_d$  on the transition annulus of a rational disk,

$$\bar{\partial}(\phi_\nu \circ M_{c,\sigma}(z, q(z))) = \partial_q \Xi_{c,\sigma}(z, q(z))(\bar{\partial}q + C_{c,\sigma}^M(z, q(z))). \quad (5.11)$$

The formula is an identity of classical forms for smooth  $q$  and of distributional forms for  $q \in W^{1,4}$  with values in  $B_d$ , by the Sobolev chain rule. The target-coordinate inverse is legitimate by the positive chart margin in Lemma 5.2.

*Step 4: prepare auxiliary disks.* Choose once and for all an open neighborhood  $N_0$  of  $D$  with

$$D \subset N_0, \quad \overline{N}_0 \Subset \text{int } D_0.$$

Reconstruction will be restricted to  $N_0$ . For the auxiliary disks, we modify only the equation on  $\Sigma$  and define no  $X$ -valued family. The divisor  $\Delta$  records the interpolation orders and is distinct from the bordered source region  $D$ .

Since the sets  $V'_\nu$  cover  $P = \text{supp } e_0 \cap D_1$ , the compact set

$$P_0 = \text{supp } e_0 \setminus \bigcup_\nu V'_\nu$$

is disjoint from  $D_1$ , and hence from  $\overline{N}_0$ . Cover  $P_0$  by finitely many source-coordinate neighborhoods whose closures avoid  $\overline{N}_0$ , and choose a  $\bar{\partial}_{\tilde{E}}$ -holomorphic frame on each. In each such frame the auxiliary equation (5.10) is

$$\bar{\partial}v + C(z, v) = 0, \quad C_v(z, 0) = 0.$$

For a center  $c$  in such a frame and a sufficiently small  $\sigma$ , use the three regions in (5.5), the fixed cutoff  $\eta_{c,\sigma}$ , and set on the transition annulus

$$C_{c,\sigma}^\circ(z, v) = (1 - \eta_{c,\sigma}(z))C(z, v). \quad (5.12)$$

Thus the coefficient is exactly  $C$  on an open outer collar and exactly zero on an open inner collar. More precisely, use the zero coefficient on  $\{|\zeta| < \sigma - h_\sigma/2\}$ , the coefficient  $C_{c,\sigma}^\circ$  on  $\mathcal{T}_{c,\sigma}$ , and the original coefficient  $C$  on  $\{|\zeta| > \sigma + h_\sigma/2\}$ , with all sets intersected with the assigned frame domain. These three open sets cover that domain near the disk, and the formulas agree on both open overlaps. The three coefficient formulas therefore glue smoothly. The finite choice of frames and the fact that the cutoff is independent of  $v$  give uniform bounds through two fiber derivatives. Since  $C_v(z, 0) = 0$ , one has  $(C_{c,\sigma}^\circ)_v(z, 0) = 0$ , so the linearization of the modified auxiliary equation remains  $\bar{\partial}_{\tilde{E}}$ . Auxiliary disks carry no  $X$ -valued family.

*Step 5: select the disks.* If  $e_0$  vanishes identically, then on a neighborhood of  $D$  the identity  $\tilde{\mathcal{F}}(0) = \mathcal{S}_\Delta(0) = 0$  says that  $s$  is already holomorphic. Taking  $F = s$  proves the theorem. We henceforth assume that  $U = \{e_0 \neq 0\}$  is nonempty.

Choose a finite smooth triangulation of  $\Sigma$ , whose existence follows from [19, Theorem 10.6], and subdivide the triangulation so finely that every closed triangle meeting  $\text{supp } e_0$  is contained either in one smaller center set  $V'_\nu$  inside  $W_1$  or in one of the auxiliary frame neighborhoods chosen in Step 4. The center sets from Step 3 and frame neighborhoods from Step 4 form a finite open cover of  $\text{supp } e_0$ . Compactness gives a positive Lebesgue number for this cover on a neighborhood of  $\text{supp } e_0$ ; subdivision with smaller mesh gives the stated containment. Denote these closed triangles by  $T_1, \dots, T_M$ . Assign one such choice of local data and one of the two types to each  $T_j$ . The union of the triangle edges has area zero. In particular, every triangle assigned the auxiliary type, and hence every auxiliary disk selected in that triangle, is disjoint from  $\overline{N}_0$ .

Before choosing the disk scale, fix the parameter domain of the global equation. Let  $d_* > 0$  be smaller than all coordinate-parameter radii used above. Since the finitely many assigned frames are fixed on compact source sets, there is a constant  $C_F \geq 1$  such that the coordinate vector  $q$  of any fiber vector  $v$  satisfies  $|q| \leq C_F|v|$  in each applicable frame. Choose a fixed metric radius  $\rho > 0$  with  $C_F\rho < d_*$ , and also within the domain of  $\tilde{A}$ . These choices depend only on the fixed local data, before  $r$  and the disks are selected.

Choose  $0 < r < 1/16$ , to be decreased in Step 6. Then  $h_\sigma = \sigma^{3/2} < \sigma/4$  whenever  $0 < \sigma \leq r$ . For  $1 \leq j \leq M$  and  $c \in U \cap \text{int } T_j$ , take as candidate sets all disks  $\Delta_{c,\sigma}^+$  from (5.5) such that

$$0 < \sigma \leq r, \quad \overline{\Delta_{c,\sigma}^+} \subset U \cap \text{int } T_j,$$

and such that the closed disk remains in the corresponding larger set  $V_\nu$  or in the assigned auxiliary frame. For a disk assigned to Step 3, also require  $\sigma < \sigma_*$  from Lemma 5.2. Every condition is open at the center, so these candidates contain arbitrarily small disks and form a fine Vitali cover of  $U$  away from the null set of triangle edges.

In the assigned coordinate for  $T_j$ , restrict the fixed area measure to  $T_j$  and extend it by zero to  $\mathbb{R}^2$ . This is a finite Radon measure. Apply the covering theorem [5, Section 1.5.2, Theorem 1.28] to the closures of the candidate disks with this measure. It gives a countable pairwise disjoint family covering  $U \cap \text{int } T_j$  up to a null set. In each  $T_j$ , take a finite initial subfamily for which the remaining uncovered area is at most  $r^{1/2}/M$ . The combined selection is finite, each closed outer disk lies in its assigned triangle interior, and the closed outer disks are pairwise disjoint. Index the selected disks by  $i$ , and let  $\zeta_i$  be the assigned source coordinate centered at  $c_i$ . Moreover,

$$\text{area} \left( U \setminus \bigcup_i \Delta_{c_i,\sigma_i}^+ \right) \leq r^{1/2}. \quad (5.13)$$

Write  $h_i = \sigma_i^{3/2}$  and let  $\mathcal{T}_r$  be the union of the transition annuli. The actual outer radius is  $\sigma_i + h_i$ , which lies between  $\sigma_i$  and  $5\sigma_i/4$ . In each assigned coordinate, the outer disk has Euclidean area  $\pi(\sigma_i + h_i)^2 \geq \pi\sigma_i^2$ , and the annulus has area

$$\pi((\sigma_i + h_i)^2 - (\sigma_i - h_i)^2) = 4\pi\sigma_i h_i.$$

Comparability with the fixed area form and disjointness imply  $\sum_i \sigma_i^2 \leq C \text{area}(\Sigma)$ . Absorbing  $\text{area}(\Sigma)$  into the fixed constant gives

$$\sum_i \sigma_i^2 \leq C, \quad \text{area}(\mathcal{T}_r) \leq C \sum_i \sigma_i h_i = C \sum_i \sigma_i^2 \sigma_i^{1/2} \leq C r^{1/2}, \quad (5.14)$$

after enlarging  $C$ . Since every closed outer disk is contained in  $U$ , no selected disk meets the fixed neighborhood of  $K \cup A$  on which  $e_0 = 0$ . For later gluing, put

$$\mathcal{I}_i = \{z \in \Delta_{c_i,\sigma_i}^+ : |\zeta_i| < \sigma_i - h_i/2\}, \quad \mathcal{O}_r = \Sigma \setminus \bigcup_i \{z \in \Delta_{c_i,\sigma_i}^+ : |\zeta_i| \leq \sigma_i + h_i/2\}.$$

The sets  $\mathcal{O}_r$ ,  $\mathcal{I}_i$ , and the individual transition annuli form an open cover of  $\Sigma$ . Every selected outer disk is contained in  $\mathcal{I}_i \cup \mathcal{T}_{c_i,\sigma_i}$ .

Define the global modified coefficient as follows. On the ball bundle  $\tilde{E}(\rho) = \{\xi \in \tilde{E} : |\xi| < \rho\}$ , use  $\tilde{A}$  from (5.10) over  $\mathcal{O}_r$ , use the zero coefficient over every  $\mathcal{I}_i$ , use  $C_{c_i,\sigma_i}^M$  over the transition annulus of a rational disk in the fixed holomorphic frame from Step 1, and use  $C_{c_i,\sigma_i}^\circ$  over an auxiliary transition annulus in its assigned  $\bar{\partial}_{\tilde{E}}$ -holomorphic frame. Call the bundle map obtained by gluing these formulas

$$\tilde{A}_r : \tilde{E}(\rho) \longrightarrow \Lambda^{0,1} T^* \Sigma \otimes \tilde{E}.$$

On the outer collar of a rational disk,  $C_{c_i,\sigma_i}^M = C^\Phi$  by Lemma 5.2. Lemmas 4.2 and 4.4 show that  $C^\Phi$  is exactly the coefficient  $A_\Delta$  of the twisted equation; because its assigned source set lies where

$\chi = 1$ , one has  $C^\Phi = A_\Delta = \tilde{A}$  on that outer collar. On its inner collar the coefficient is exactly zero. The corresponding equalities for auxiliary disks follow from (5.12). Thus all local definitions agree on open collars. Since the disks are disjoint and every modification equals the original coefficient near its outer boundary,  $\tilde{A}_r$  is a globally defined smooth bundle map. Moreover,

$$\sup_{\tilde{E}(\rho)} (|\tilde{A}_r| + |d_v \tilde{A}_r| + |d_v^2 \tilde{A}_r|) \leq C, \quad (5.15)$$

with  $C$  independent of the number and radii of the disks. For each fixed  $r$ , the source derivatives of  $\tilde{A}_r$  are bounded, by the open-collar gluing, but no source-derivative bound uniform in  $r$  is asserted.

At  $v = 0$ ,  $\tilde{A}_r(z, 0)$  vanishes on every  $\mathcal{I}_i$ , while outside the selected outer disks  $\tilde{A}_r(z, 0)$  equals the original coefficient. Since every selected outer disk lies in  $U$ , the residual of the modified equation is therefore zero outside the measurable set

$$\mathcal{E}_r = \left( U \setminus \bigcup_i \Delta_{c_i, \sigma_i}^+ \right) \cup \mathcal{T}_r.$$

Only  $\text{area}(\mathcal{E}_r)$  is estimated; no estimate for  $\text{area}(\overline{\mathcal{E}_r})$  is used.

Set

$$\mathcal{X} = W^{1,4}(\Sigma, \tilde{E}), \quad \mathcal{Y} = L_{0,1}^4(\Sigma, \tilde{E}).$$

Sobolev embedding makes

$$\mathcal{B} = \{v \in \mathcal{X} : \|v\|_{C^0} < \rho\}$$

an open neighborhood of zero. Put

$$\mathcal{S}_r(v) = \bar{\partial}_{\tilde{E}} v + \tilde{A}_r(z, v(z)), \quad \mathcal{S}_r : \mathcal{B} \longrightarrow \mathcal{Y}. \quad (5.16)$$

The fiber bounds give the  $C^2$  property and uniform nonlinear estimate of Lemma 4.5. No derivative of a grafted  $X$ -valued map occurs in (5.16).

*Step 6: uniform estimates and solution.* By Step 5,  $\mathcal{S}_r(0)$  vanishes outside  $\mathcal{E}_r$ . The residual vanishes on every  $\mathcal{I}_i$  and on the unmodified part of  $\{e_0 = 0\}$ . The pointwise norm of  $\mathcal{S}_r(0)$  is uniformly bounded by (5.15). Hence (5.13)–(5.14) give

$$\|\mathcal{S}_r(0)\|_{\mathcal{Y}}^4 \leq C(r^{1/2} + \text{area}(\mathcal{T}_r)) \leq Cr^{1/2},$$

so

$$\|\mathcal{S}_r(0)\|_{\mathcal{Y}} \leq Cr^{1/8}. \quad (5.17)$$

On  $\mathcal{O}_r$  and every  $\mathcal{I}_i$ , the linearization at zero is  $\bar{\partial}_{\tilde{E}}$ . On auxiliary annuli, the linearization also equals  $\bar{\partial}_{\tilde{E}}$  by (5.12) and  $C_v(z, 0) = 0$ . Consequently  $d\mathcal{S}_r(0) - \bar{\partial}_{\tilde{E}}$  is multiplication by a uniformly bounded zero-order coefficient vanishing outside the transition annuli of the rational disks. For  $\xi \in \mathcal{X}$ ,

$$\|(d\mathcal{S}_r(0) - \bar{\partial}_{\tilde{E}})\xi\|_{L^4} \leq C \text{area}(\mathcal{T}_r)^{1/4} \|\xi\|_{C^0}.$$

The Sobolev embedding  $W^{1,4} \hookrightarrow C^0$  and (5.14) therefore give

$$\|d\mathcal{S}_r(0) - \bar{\partial}_{\tilde{E}}\|_{\mathcal{X} \rightarrow \mathcal{Y}} \leq Cr^{1/8}. \quad (5.18)$$

Apply Lemma 4.5 to  $\tilde{E}$  and  $\tilde{A}_r$ , using (5.9), (5.15), (5.17), and (5.18), with  $\varepsilon_r = C_1 r^{1/8}$  for a fixed sufficiently large  $C_1$ . Lemma 4.5 gives a smooth solution  $v_r$  satisfying

$$\mathcal{S}_r(v_r) = 0, \quad \|v_r\|_{W^{1,4}} \leq Cr^{1/8}. \quad (5.19)$$

For each selected disk, let  $q_i(z)$  be the coordinate vector of  $v_r(z)$  in its assigned frame. Let  $C_S$  be the norm of a fixed Sobolev embedding  $\mathcal{X} \hookrightarrow C^0$ . Then

$$|q_i(z)| \leq C_F |v_r(z)| \leq C_F C_S Cr^{1/8} < d_*$$

after  $r$  is decreased. Thus every coordinate parameter lies in the ball used to construct its assigned disk. Smoothness uses the open-collar gluing for each fixed  $r$ , without uniform source-derivative estimates as  $r \rightarrow 0$ .

*Step 7: reconstruct the section.* Put  $N = N_0$ . By the choice made in Steps 4–5,  $N$  meets no auxiliary disk. For each selected rational disk, put

$$U_i^- = N \cap \mathcal{I}_i, \quad U_i^0 = N \cap \mathcal{T}_{c_i, \sigma_i}.$$

Also set

$$U^+ = N \cap \mathcal{O}_r.$$

The sets  $U^+$ ,  $U_i^-$ , and  $U_i^0$  cover  $N$ . Let  $F_r$  be the section given by

$$F_r(z) = \begin{cases} H_{c_i, \sigma_i}(z, q_i(z)), & z \in U_i^-, \\ M_{c_i, \sigma_i}(z, q_i(z)), & z \in U_i^0, \\ \Psi(s(z), \iota v_r(z)), & z \in U^+ \end{cases}. \quad (5.20)$$

On  $U_i^- \cap U_i^0$  the cutoff is identically one, so  $M_{c_i, \sigma_i} = H_{c_i, \sigma_i}$  for every  $q$ . On  $U_i^0 \cap U^+$  the cutoff is identically zero, so  $M_{c_i, \sigma_i} = \Phi$ . Thus the formulas agree on open collars, not merely on boundary circles. The three formulas give a smooth map  $F_r : N \rightarrow X$ . Each branch satisfies  $p(F_r(z)) = z$ , so  $F_r$  is a section of  $p$ .

We check holomorphy on each type of open region. On an outer region the definitions of  $\mathcal{S}_s$  and  $\mathcal{S}_\Delta$  give

$$\bar{\partial} F_r = d_u \Psi(s, \iota v_r) \circ \iota \mathcal{S}_\Delta(v_r) = 0,$$

because  $\mathcal{S}_r = \mathcal{S}_\Delta$  on  $U^+$ . On a transition annulus, (5.11) and  $\mathcal{S}_r(v_r) = 0$  give

$$\bar{\partial}(\phi_\nu \circ F_r) = \partial_q \Xi_{c_i, \sigma_i}(z, q_i(z))(\bar{\partial} q_i + C_{c_i, \sigma_i}^M(z, q_i(z))) = 0.$$

On an inner region the local equation is  $\bar{\partial} q_i = 0$ . Since  $H_{c_i, \sigma_i}$  is jointly holomorphic,

$$\bar{\partial} F_r = d_q H_{c_i, \sigma_i}(z, q_i(z)) \bar{\partial} q_i = 0.$$

The core reconstruction uses only joint holomorphy of  $H_{c_i, \sigma_i}$ . All three displayed  $\bar{\partial} F_r = 0$  identities are classical because  $v_r$  is smooth. These open regions cover  $N$ , and their formulas agree on the open collars, so the identities prove that  $F_r$  is holomorphic on  $N$ .

Let  $V_{K,A} \subset N$  be a fixed neighborhood of  $K \cup A$  on which  $e_0 = 0$ , as chosen before the disk selection. Since every selected disk is contained in  $U = \{e_0 \neq 0\}$ , it is disjoint from  $V_{K,A}$ . On  $V_{K,A}$ ,

$$F_r(z) = \Psi(s(z), \iota v_r(z)).$$

On a smaller compact neighborhood of  $K$  inside  $V_{K,A}$ , the uniform derivative bound for  $\Psi$ , Sobolev embedding, and (5.19) yield

$$\sup_K d_X(F_r, s) \leq C_\Psi \sup_K \|\iota z\| \|v_r\|_{C^0} \leq C r^{1/8}.$$

Choose  $r$  so small that  $C r^{1/8} < \epsilon$ . Near  $a \in A$ , choose a local coordinate  $z$  centered at  $a$ . In compatible holomorphic frames, the local-addition parameter is

$$\iota v_r = z^{k_a+1} v_r$$

and  $\Psi(s, u) = s + O(u)$  in target coordinates. Since  $F_r$  and  $s$  are holomorphic near  $a$ , the difference  $h(z) = \phi(F_r(z)) - \phi(s(z))$  in a common target chart satisfies  $|h(z)| \leq C|z|^{k_a+1}$ . Hence  $h(z)/z^{k_a+1}$  is bounded and holomorphic on the punctured disk and extends across the origin. The Taylor coefficients of  $h$  through degree  $k_a$  vanish, so  $j_a^{k_a} F_r = j_a^{k_a} s$ . This proves the theorem.  $\square$

The sets  $K$  and  $A$  need not contain one another. Either may be empty; when  $K$  is empty the approximation condition is omitted.

#### 5.4. Exhaustion and the lifting homotopy.

**Proposition 5.4.** *Let  $\pi : Z \rightarrow Y$  be a surjective proper holomorphic submersion with connected and simply connected fibers and the relative first-jet sphere-family property. Let  $R$  be an open Riemann surface, let  $K \subset R$  be compact, let  $A \subset R$  be closed discrete, and let  $g : R \rightarrow Y$  be holomorphic. Suppose that  $f : R \rightarrow Z$  is a continuous lifting of  $g$  and is holomorphic near  $K \cup A$ . Given  $\epsilon > 0$  and  $q : A \rightarrow \mathbb{N}$ , there is a holomorphic lifting  $F : R \rightarrow Z$  such that*

$$\sup_K d_Z(F, f) < \epsilon, \quad j_a^{q(a)} F = j_a^{q(a)} f \quad (a \in A).$$

*Proof.* Put  $k_a = q(a)$ . Work in the pullback  $p : X = R \times_Y Z \rightarrow R$ , with the metric fixed in Section 4. Let  $s_f(z) = (z, f(z))$  be the section corresponding to  $f$ . The pullback has the sphere-family property by Remark 3.3. Choose a connected exhaustion by compact bordered Riemann surfaces

$$C_1 \Subset \text{int } C_2 \Subset \cdots, \quad \bigcup_j C_j = R,$$

whose boundaries avoid  $A$ , with  $K \subset \text{int } C_1$ . The exhaustion statement in Subsection 2.2 supplies this exhaustion for an arbitrary compact  $K$ ; neither the  $C_j$  nor  $K$  need be Runge. At each step below, Theorem 5.3 likewise permits an arbitrary compact set for approximation. Put  $A_j = A \cap C_j$ , which is finite. Choose positive numbers  $\epsilon_j$  with

$$\sum_{j \geq 1} \epsilon_j < \epsilon.$$

Use  $\epsilon_j$  as the error bound at the  $j$ th approximation step.

We construct sections  $F_j$ , holomorphic on neighborhoods of  $C_j$ , such that

$$\sup_K d_X(F_1, s_f) < \epsilon_1, \quad j_a^{k_a} F_j = j_a^{k_a} s_f \quad (a \in A_j), \quad (5.21)$$

and

$$\sup_{C_j} d_X(F_{j+1}, F_j) < \epsilon_{j+1}. \quad (5.22)$$

For the first step, use the relative smoothing statement in Subsection 2.2 to replace the given continuous section  $s_f$  by a smooth section on a neighborhood of  $C_1$ , without changing  $s_f$  near  $K \cup A_1$ , and apply Theorem 5.3.

Assume  $F_j$  has been constructed. Choose pairwise disjoint coordinate disks about the finitely many points of  $A_{j+1} \setminus A_j$ , with their closures in  $\text{int } C_{j+1}$  and disjoint from a neighborhood of  $C_j$ , on which the original section  $s_f$  is holomorphic. Such disks exist because the boundaries of the exhaustion avoid  $A$ . Lemma 2.3, applied to  $C_j \subset C_{j+1}$  and these disks, extends the prescribed sections  $F_j$  near  $C_j$  and  $s_f$  near the new interpolation points to a smooth section  $s_{j+1}$  near  $C_{j+1}$ .  $s_{j+1}$  is holomorphic near  $C_j \cup A_{j+1}$ . Apply Theorem 5.3 with approximation set  $C_j$ , interpolation set  $A_{j+1}$ , and error  $\epsilon_{j+1}$ . The resulting section is  $F_{j+1}$  and satisfies (5.21)–(5.22).

For every fixed  $m$  and all  $n > \ell \geq m$ , the nesting of the exhaustion and (5.22) give

$$\sup_{C_m} d_X(F_n, F_\ell) \leq \sum_{j=\ell}^{n-1} \epsilon_{j+1}.$$

Thus  $(F_j)$  is uniformly Cauchy on every compact subset. For fixed  $m$ , all later sections over  $C_m$  take values in the compact set  $p^{-1}(C_m)$ , since  $p$  is proper. The compact metric space  $p^{-1}(C_m)$  is complete. Pointwise limits therefore define a map  $F_\infty$ , and the uniform convergence on each  $C_m$  makes  $F_\infty$  continuous. Passing to the limit in  $p(F_j(z)) = z$ , for  $z \in C_m$  and  $j \geq m$ , gives  $p(F_\infty(z)) = z$ . Since  $\bigcup_m C_m = R$ , the map  $F_\infty : R \rightarrow X$  is a section.

Fix  $x \in R$  and choose a relative target chart  $U$  about  $F_\infty(x)$ . By continuity of  $F_\infty$ , there are a source coordinate disk  $\Delta$  about  $x$  and an open set  $U' \Subset U$  such that  $F_\infty(\bar{\Delta}) \subset U'$ . The compact

set  $F_\infty(\bar{\Delta})$  has positive distance from  $X \setminus U$ . Uniform convergence on  $\bar{\Delta}$  therefore puts  $F_j(\bar{\Delta})$  in  $U$  for all sufficiently large  $j$ . In the vertical coordinates of  $U$ , the maps  $F_j|_\Delta$  are holomorphic and converge uniformly to the coordinate expression of  $F_\infty$ . The Weierstrass theorem proves that  $F_\infty$  is holomorphic near  $x$ , and hence on  $R$ .

Summing the estimates on  $K$  in (5.21)–(5.22) gives

$$\sup_K d_X(F_\infty, s_f) < \sum_{j \geq 1} \epsilon_j < \epsilon.$$

If  $a \in A$ , choose  $m$  with  $a \in A_m$ . For every  $j \geq m$ , the jet  $j_a^{k_a} F_j$  equals the prescribed jet  $j_a^{k_a} s_f$ . In particular,  $F_j(a) = s_f(a)$  for  $j \geq m$ , so  $F_\infty(a) = s_f(a)$ . Choose a small source disk about  $a$  and a relative target chart containing the images of  $s_f$ ,  $F_\infty$ , and every sufficiently late  $F_j$  on that disk, as in the preceding paragraph. In a source coordinate centered at  $a$ , choose a circle  $|\zeta| = \tau$  inside that disk. For a vertical coordinate map  $\phi$  and  $0 \leq k \leq k_a$ ,

$$\partial_z^k(\phi \circ F_j)(0) = \frac{k!}{2\pi i} \int_{|\zeta|=\tau} \frac{\phi(F_j(\zeta))}{\zeta^{k+1}} d\zeta \longrightarrow \partial_z^k(\phi \circ F_\infty)(0).$$

Thus the derivatives of  $F_\infty$  agree with those of  $s_f$  through the prescribed order. This equality is independent of the chosen coordinates by the holomorphic chain rule, so  $j_a^{k_a} F_\infty = j_a^{k_a} s_f$ . Put  $F_Z = \text{pr}_Z \circ F_\infty : R \rightarrow Z$ . Then

$$\pi \circ F_Z = g, \quad \sup_K d_Z(F_Z, f) \leq \sup_K d_X(F_\infty, s_f) < \epsilon, \quad j_a^{k_a} F_Z = j_a^{k_a} f.$$

The map  $F_Z$  is the required holomorphic lifting.  $\square$

*Proof of Theorem 1.4.* The map  $\pi$  is a proper surjective holomorphic submersion. By Ehresmann's theorem [4],  $\pi$  is a Serre fibration. Given a lifting problem from an open Riemann surface, Proposition 5.4 supplies the holomorphic lifting with the required approximation and closed discrete jet interpolation.

Every fiber is connected and simply connected by hypothesis. The pullback  $p : X = R \times_Y Z \rightarrow R$  is a locally trivial bundle with the same fibers. Lemma 2.4 gives a section homotopy  $H : R \times [0, 1] \rightarrow X$  between the sections associated with  $f$  and  $F$ . Set  $f_t(z) = \text{pr}_Z H(z, t)$ . Then

$$f_0 = f, \quad f_1 = F, \quad \pi \circ f_t = g \quad (0 \leq t \leq 1).$$

These identities prove the homotopy conclusion of the strengthened assertion. Specializing to Runge  $K$  and finite  $A \subset K$  gives all conditions in Definition 1.2, exactly the Oka-1 map condition of [1, Definition 7.7].  $\square$

## 6. ALGEBRAIC PREPARATION

The algebraic proof requires a fixed bundle with vanishing first cohomology and a smooth identification with the bundle carrying the section equation. We first show how to make such an identification holomorphic near the prescribed data. We then construct the required algebraic section by elementary transforms and smoothing of combs.

**6.1. Bundle isomorphisms near prescribed sets.** Lemma 4.3 identifies the natural holomorphic structure along a holomorphic section. We first construct a smooth bundle identification and then make it holomorphic near the prescribed sets.

**Lemma 6.1.** *Let  $p : X \rightarrow \Sigma$  be a holomorphic map and let*

$$H : \Sigma \times [0, 1] \longrightarrow X$$

*be a smooth homotopy through sections whose image lies in the submersion locus of  $p$ . Here  $H_t(z) = H(z, t)$  and  $p \circ H_t = \text{id}_\Sigma$ . Then the complex vector bundles*

$$H_0^* T_{X/\Sigma} \quad \text{and} \quad H_1^* T_{X/\Sigma}$$

are smoothly isomorphic. If  $H$  is independent of the homotopy parameter on a neighborhood of a set  $Q \subset \Sigma$ , the isomorphism may be chosen to be the natural identity on that neighborhood of  $Q$ .

*Proof.* The vertical tangent bundle is a smooth complex vector bundle on a neighborhood of the image of  $H$ . Choose a Hermitian connection on  $T_{X/\Sigma}$  over that neighborhood. Parallel transport along  $t \mapsto H(z, t)$  gives

$$J_z : T_{X/\Sigma, H(z, 0)} \longrightarrow T_{X/\Sigma, H(z, 1)}.$$

In local frames, transport solves a linear ordinary differential equation on  $[0, 1]$ ; the coefficients depend smoothly on  $z$ . Thus  $J_z$  is complex linear, invertible, and smooth in  $z$ . If  $H(z, t)$  is constant in  $t$ , the transport equation gives  $J_z = \text{id}$ .  $\square$

Parallel transport gives a smooth bundle isomorphism, but the approximation argument requires holomorphy near the prescribed data. The next lemma achieves that local holomorphy while retaining the original isomorphism near a disjoint compact set.

**Lemma 6.2.** *Let  $V$  and  $E$  be holomorphic vector bundles of the same rank over a compact Riemann surface  $\Sigma$ , and let  $J_0 : V \rightarrow E$  be a smooth complex vector-bundle isomorphism. Let  $P, Q \subset \Sigma$  be compact sets with disjoint neighborhoods. Assume that  $P$  is contained in a relatively compact open Riemann surface  $W \subset \Sigma \setminus Q$  and that  $J_0$  is holomorphic on a neighborhood of  $Q$ . Then there is a smooth bundle isomorphism  $J : V \rightarrow E$  which is holomorphic on a neighborhood of  $P \cup Q$  and agrees with  $J_0$  outside a relatively compact open neighborhood of  $P$  disjoint from  $Q$ .*

*Proof.* A smooth isomorphism on the compact base need not be holomorphic. We construct a holomorphic isomorphism on the open subsurface  $W$  and use a cutoff to retain  $J_0$  near  $Q$ . After shrinking  $W$ , choose

$$P \subset W_0 \Subset W \Subset \Sigma \setminus Q.$$

Every component of  $W$  is an open Riemann surface and retracts onto a graph. Hence every complex vector bundle on  $W$  is topologically trivial, and Grauert's Oka principle [11, Satz I] makes  $V|_W$  and  $E|_W$  holomorphically trivial. Put  $n = \text{rank } V = \text{rank } E$ . In fixed holomorphic trivializations,  $J_0|_W$  is a smooth map

$$g_0 : W \longrightarrow \text{GL}_n(\mathbb{C}).$$

The complex Lie group  $\text{GL}_n(\mathbb{C})$  is a homogeneous complex manifold and hence an Oka manifold. The Oka principle for maps from the Stein manifold  $W$  [7, Theorem 1.1] therefore gives a homotopy  $g_t : W \rightarrow \text{GL}_n(\mathbb{C})$  from  $g_0$  to a holomorphic map  $g_1$ . Reparametrize the homotopy to be constant near both endpoints. Relative smooth approximation (Subsection 2.2), relative to both endpoints, applied to the trivial bundle with fiber  $\text{GL}_n(\mathbb{C})$  over a compact subsurface times  $[0, 1]$ , makes  $(z, t) \mapsto g_t(z)$  smooth on the region used for the cutoff construction. The relative condition preserves both  $g_0$  and  $g_1$ .

Choose  $\chi \in C^\infty(\Sigma, [0, 1])$  supported in  $W$  and equal to one near  $\overline{W}_0$ . On  $W$  represent  $J$  by

$$z \longmapsto g_{\chi(z)}(z),$$

and put  $J = J_0$  outside  $W$ . Near  $\partial W$  the two definitions agree because  $\chi = 0$  and therefore  $g_{\chi(z)}(z) = g_0(z)$ . The resulting  $J$  is smooth and invertible, is represented by the holomorphic matrix  $g_1$  near  $P$ , and equals  $J_0$  near  $Q$ .  $\square$

**6.2. Positive elementary transforms.** Positive elementary transforms will remove the first-cohomology obstruction. Given a vector bundle  $E$  of rank  $r$  on a smooth projective curve  $B$ , distinct points  $b_i \in B$ , and lines  $\xi_i \subset E_{b_i}$ , its *positive elementary transform* is the vector bundle  $E'$  fitting into

$$0 \longrightarrow E \longrightarrow E' \longrightarrow \bigoplus_i \mathbb{C}_{b_i} \longrightarrow 0$$

with the indicated local pole directions. Explicitly, if  $x$  is a local parameter at  $b_i$  and a frame  $e_1, \dots, e_r$  satisfies  $\xi_i = \mathbb{C}e_1(b_i)$ , the transformed sheaf is locally generated by  $x^{-1}e_1, e_2, \dots, e_r$ . Here  $\mathbb{C}_{b_i}$  denotes a skyscraper sheaf of length one; the notation does not specify a canonical trivialization of its one-dimensional fiber.

We now make the parameter space of elementary transforms explicit. We use the convention that  $\mathbb{P}_B(E)$  parametrizes one-dimensional subspaces of the fibers of  $E$ . For  $m \geq 1$ , put

$$\mathcal{H}_m(E) = \{((b_1, \xi_1), \dots, (b_m, \xi_m)) \in \mathbb{P}_B(E)^m : b_i \neq b_j \text{ for } i \neq j\}. \quad (6.1)$$

$\mathcal{H}_m(E)$  is the ordered parameter space of simple positive elementary transforms. If  $O \subset B$  is Euclidean open, let  $\mathcal{H}_m(E; O)$  denote the subset on which every  $b_i$  belongs to  $O$ .

**Lemma 6.3.** *Let  $B$  be a smooth projective curve, let  $E$  be a vector bundle of positive rank on  $B$ , and fix an integer  $m \geq 1$ . The variety  $\mathcal{H}_m(E)$  is smooth and irreducible, and there is a vector bundle*

$$\mathcal{E}_m^+ \longrightarrow B \times \mathcal{H}_m(E)$$

whose restriction over  $\eta = ((b_1, \xi_1), \dots, (b_m, \xi_m))$  is the positive elementary transform  $E_\eta^+$  of  $E$  in the indicated directions. Consequently, for every line bundle  $M$  on  $B$ , the locus

$$\{\eta \in \mathcal{H}_m(E) : H^1(B, E_\eta^+ \otimes M) = 0\} \quad (6.2)$$

is Zariski open.

*Proof.* Put  $r = \text{rank } E$  and  $S = \mathcal{H}_m(E)$ . Let  $\rho : B \times S \rightarrow B$  be projection and  $\beta_i : S \rightarrow B$  the  $i$ th base-point map. Write  $\Gamma_i \subset B \times S$  for the graph of  $\beta_i$ . The divisors  $\Gamma_i$  are pairwise disjoint, and we put

$$\mathcal{D} = \Gamma_1 + \dots + \Gamma_m.$$

On  $\Gamma_i \cong S$  there is a tautological line subbundle

$$\mathcal{L}_i \subset \rho^* E|_{\Gamma_i}.$$

If  $j_i : \Gamma_i \hookrightarrow B \times S$  is the inclusion, put

$$\mathcal{E}_m^+ = \ker \left( \rho^* E(\mathcal{D}) \longrightarrow \bigoplus_{i=1}^m (j_i)_* \frac{\rho^* E(\mathcal{D})|_{\Gamma_i}}{\mathcal{L}_i \otimes \mathcal{O}(\mathcal{D})|_{\Gamma_i}} \right). \quad (6.3)$$

The arrow in (6.3) first restricts to each  $\Gamma_i$  and then takes the indicated quotient. The kernel requires the polar part at  $\Gamma_i$  to lie in the chosen tautological line.

We check local freeness, which also identifies the fibers. Near a point of  $\Gamma_i$ , choose a local equation  $x = 0$  for  $\Gamma_i$  and a frame  $e_1, \dots, e_r$  of  $\rho^* E$  in which  $\mathcal{L}_i$  is generated by  $e_1$  along  $\Gamma_i$ . Since the other graphs are disjoint from this neighborhood, the kernel in (6.3) is freely generated by

$$x^{-1}e_1, e_2, \dots, e_r.$$

Away from the graphs,  $\mathcal{E}_m^+$  equals  $\rho^* E$ . Thus  $\mathcal{E}_m^+$  is a vector bundle. The displayed local basis remains a basis after arbitrary base change on  $S$ ; hence formation of the kernel commutes with restriction to every parameter fiber. The restriction of  $\mathcal{E}_m^+$  to  $B \times \{\eta\}$  is therefore precisely the sheaf of rational sections of  $E$  with at most a simple pole at  $b_i$ , whose polar direction belongs to  $\xi_i$ .

The projective bundle  $\mathbb{P}_B(E)$  is smooth and irreducible. The nonempty open subset  $\mathcal{H}_m(E) \subset \mathbb{P}_B(E)^m$  is therefore smooth and irreducible. Finally,  $\mathcal{E}_m^+ \otimes \rho^* M$  is locally free. Since  $B \times S \rightarrow S$  is smooth, the sheaf  $\mathcal{E}_m^+ \otimes \rho^* M$  is flat over  $S$ ; the projection is also proper. Cohomology semicontinuity therefore makes the function

$$\eta \longmapsto h^1(B, E_\eta^+ \otimes M)$$

upper semicontinuous; equivalently, the locus (6.2) is Zariski open [22, Tag 0BDN].  $\square$

The cohomology-vanishing construction is due to Graber–Harris–Starr [9, Lemma 2.5]. We give a universal-family construction and a collision argument that separates coincident elementary modifications. These make the construction’s compatibility with localization in a Euclidean open set and with additional open conditions on the directions explicit.

**Lemma 6.4.** *Let  $B$  be a smooth projective curve, let  $E$  be a vector bundle of positive rank on  $B$ , let  $O \subset B$  be a nonempty Euclidean open set, and let  $N \geq 0$  be an integer. There are an integer  $m \geq 1$  and a nonempty Zariski-open subset  $\mathcal{U}_m \subset \mathcal{H}_m(E)$  such that every  $\eta \in \mathcal{U}_m$  satisfies*

$$H^1(B, E_\eta^+(-W)) = 0 \quad (6.4)$$

*for every effective divisor  $W$  of degree  $N$ . Moreover, if  $\mathcal{G} \subset \mathcal{H}_m(E)$  is any Zariski-open dense subset, then*

$$\mathcal{U}_m \cap \mathcal{G} \cap \mathcal{H}_m(E; O) \neq \emptyset.$$

*In particular, the transform may be chosen with distinct support points in  $O$  while satisfying any additional Zariski-open dense conditions on the ordered points and pole directions.*

*Proof.* We first obtain vanishing of the first cohomology after one fixed twist  $-R_0$ . Riemann–Roch will then give the assertion simultaneously for all  $W$  of degree  $N$ , since  $\deg(R_0 - W)$  will equal the genus of  $B$ . Write  $g$  for the genus of  $B$  and put  $r = \text{rank } E$ . Fix an effective divisor  $R_0$  of degree  $g + N$  and put

$$F = E(-R_0).$$

Tensoring by a line bundle canonically identifies the projective bundles of lines  $\mathbb{P}_B(F)$  and  $\mathbb{P}_B(E)$ , and positive elementary transforms commute with tensoring: the local generators  $x^{-1}e_1, e_2, \dots, e_r$  are all tensored by the same local line-bundle generator. Thus, for corresponding configurations  $\eta$ ,

$$F_\eta^+ \cong E_\eta^+(-R_0). \quad (6.5)$$

We first exhibit a configuration supported in  $O$  for which the right side of (6.5) has vanishing first cohomology. Choose  $t \in O$ , shrink a Zariski-open neighborhood  $U \subset B$  of  $t$  so that  $F|_U$  has an algebraic frame  $e_1, \dots, e_r$ , and choose a small Euclidean disk  $\Delta \Subset U \cap O$  centered at  $t$ . Since  $\mathcal{O}_B(t)$  is ample, Serre vanishing gives an integer  $\mu \geq 1$  such that

$$H^1(B, F(\mu t)) = 0. \quad (6.6)$$

Set  $m = r\mu$ .

To allow the support points to coincide, use the parameter space

$$Z = U^{r\mu}, \quad z = (z_{k,j})_{1 \leq k \leq r, 1 \leq j \leq \mu}.$$

For each pair  $(k, j)$  let  $\Gamma_{k,j} \subset B \times Z$  be the graph of the coordinate map  $z_{k,j} : Z \rightarrow U \hookrightarrow B$ , and put

$$\mathcal{D}_k = \sum_{j=1}^{\mu} \Gamma_{k,j}.$$

Each graph is an effective Cartier divisor relative to  $Z$ . Locally on the smooth relative curve  $B \times Z \rightarrow Z$ , the equation for  $\mathcal{D}_k$  is the product of the equations for  $\Gamma_{k,j}$ . After arbitrary base change on  $Z$ , each factor remains a non-zero-divisor, and hence so does their product, even when coordinates coincide; repeated factors record their multiplicities. All the  $\mathcal{D}_k$  are supported in  $U \times Z$ .

We construct a vector bundle  $\mathcal{F}^+$  on  $B \times Z$ . On  $U \times Z$  set

$$\mathcal{F}^+|_{U \times Z} = \bigoplus_{k=1}^r \mathcal{O}_{U \times Z}(\mathcal{D}_k) e_k. \quad (6.7)$$

On

$$(B \times Z) \setminus \bigcup_{k,j} \Gamma_{k,j}$$

use the pullback of  $F$ . These two open sets cover  $B \times Z$ , since all of the graphs lie over  $U$ . On  $(U \times Z) \setminus \bigcup_{k,j} \Gamma_{k,j}$  the canonical section of  $\mathcal{O}(\mathcal{D}_k)$  is invertible, and hence canonically identifies the  $k$ th summand of (6.7) with the line generated by  $e_k$  in the pullback of  $F$ . The identifications glue (6.7) to a vector bundle on  $B \times Z$ . Since  $B \times Z \rightarrow Z$  is smooth, the locally free sheaf  $\mathcal{F}^+$  is flat over  $Z$ .

Let  $z_0 \in Z$  be the point at which every coordinate  $z_{k,j}$  equals  $t$ . Then  $\mathcal{D}_k|_{B \times \{z_0\}} = \mu t$  for every  $k$ . On  $U$ , the restriction of the special fiber is

$$(\mathcal{F}^+|_{B \times \{z_0\}})|_U = \bigoplus_{k=1}^r \mathcal{O}_U(\mu t) e_k.$$

The gluing with  $F$  off  $t$  is the canonical one used in (6.7), so globally

$$\mathcal{F}^+|_{B \times \{z_0\}} \cong F(\mu t).$$

Write  $\mathcal{F}_z^+ = \mathcal{F}^+|_{B \times \{z\}}$ . By (6.6) and cohomology semicontinuity for the proper projection  $B \times Z \rightarrow Z$ , there is a Zariski-open neighborhood  $Z^0$  of  $z_0$  such that

$$H^1(B, \mathcal{F}_z^+) = 0 \quad (z \in Z^0). \quad (6.8)$$

Let  $Z^* \subset Z$  be the complement of all diagonals  $z_{k,j} = z_{k',j'}$  for  $(k,j) \neq (k',j')$ .  $Z^*$  is a nonempty Zariski-open dense subset of the irreducible variety  $Z = U^{r\mu}$ . The polydisk

$$Z(\Delta) = \Delta^{r\mu} \subset Z^{\text{an}}$$

is a Euclidean-open neighborhood of  $z_0$ . Hence

$$Z^0 \cap Z^* \cap Z(\Delta) \neq \emptyset.$$

Indeed,  $Z^0 \cap Z(\Delta)$  is a nonempty Euclidean-open set containing  $z_0$ , whereas the finite union of diagonals  $Z \setminus Z^*$  is a proper algebraic subset and has empty Euclidean interior.

Choose  $z \in Z^0 \cap Z^* \cap Z(\Delta)$  and order the  $m = r\mu$  coordinates of  $z$  lexicographically by  $(k,j)$ . The points  $z_{k,j}$  are distinct and belong to  $O$ . At the point  $z_{k,j}$ , formula (6.7) permits a simple pole precisely in the line spanned by  $e_k(z_{k,j})$ . Thus the ordered configuration

$$\eta(z) = ((z_{k,j}, \mathbb{C}e_k(z_{k,j})))_{k,j} \in \mathcal{H}_m(F; O) \cong \mathcal{H}_m(E; O)$$

has

$$\mathcal{F}_z^+ \cong F_{\eta(z)}^+.$$

Equations (6.5) and (6.8) give

$$H^1(B, E_{\eta(z)}^+(-R_0)) = 0.$$

By Lemma 6.3, the locus

$$\mathcal{U}_m = \left\{ \eta \in \mathcal{H}_m(E) : H^1(B, E_\eta^+(-R_0)) = 0 \right\} \quad (6.9)$$

is therefore a nonempty Zariski-open subset.

We next pass from the single divisor  $R_0$  to all effective divisors of degree  $N$ , keeping the same configuration  $\eta \in \mathcal{U}_m$ . Let  $W$  be any such divisor. The line bundle  $\mathcal{O}_B(R_0 - W)$  has degree  $g$ . Writing  $K_B$  for the canonical line bundle of  $B$ , Riemann–Roch gives

$$h^0(B, \mathcal{O}_B(R_0 - W)) - h^0(B, K_B \otimes \mathcal{O}_B(-R_0 + W)) = 1,$$

so  $\mathcal{O}_B(R_0 - W)$  has a nonzero section. Hence there is an effective divisor  $A_W$  with

$$R_0 - W \sim A_W.$$

For every  $\eta \in \mathcal{U}_m$  we have

$$E_\eta^+(-W) \cong E_\eta^+(-R_0)(A_W).$$

The exact sequence

$$0 \longrightarrow E_\eta^+(-R_0) \longrightarrow E_\eta^+(-R_0)(A_W) \longrightarrow \mathcal{Q}_{\eta,W} \longrightarrow 0$$

has torsion quotient  $\mathcal{Q}_{\eta,W}$ , so  $H^1(B, \mathcal{Q}_{\eta,W}) = 0$ . The cohomology sequence gives a surjection

$$H^1(B, E_\eta^+(-R_0)) \longrightarrow H^1(B, E_\eta^+(-W)) \longrightarrow 0.$$

The first group vanishes by (6.9). This proves (6.4) for every  $W$ , with the same fixed configuration  $\eta \in \mathcal{U}_m$ .

It remains to impose localization and any further generality conditions. The variety  $\mathcal{H}_m(E)$  is irreducible, and  $\mathcal{H}_m(E; O)$  is a nonempty Euclidean-open subset: choose  $m$  distinct points of  $O$  and arbitrary lines over them. A proper algebraic subset of an irreducible complex variety has empty Euclidean interior, so  $\mathcal{H}_m(E; O)$  is Zariski dense. If  $\mathcal{G} \subset \mathcal{H}_m(E)$  is Zariski open and dense, then  $\mathcal{U}_m \cap \mathcal{G}$  is a nonempty Zariski-open subset of  $\mathcal{H}_m(E)$  and therefore meets  $\mathcal{H}_m(E; O)$ . Thus  $\mathcal{U}_m \cap \mathcal{G} \cap \mathcal{H}_m(E; O) \neq \emptyset$ , which proves the final assertion.  $\square$

**6.3. Combs and sections with vanishing cohomology.** Attaching a rational tooth realizes a positive elementary transform on the handle. We also need to control the normal bundle on the tooth: a nonzero projection of the modification direction to the trivial summand gives the splitting below.

**Lemma 6.5.** *Let  $A$  be a globally generated vector bundle on  $\mathbb{P}^1$ , let  $x \in \mathbb{P}^1$ , and put  $F = A \oplus \mathcal{O}_{\mathbb{P}^1}$ . If  $\lambda \subset F_x$  is a line whose projection to  $\mathcal{O}_{\mathbb{P}^1, x}$  is nonzero, then the positive elementary transform of  $F$  at  $x$  in the direction  $\lambda$  is isomorphic to*

$$A \oplus \mathcal{O}_{\mathbb{P}^1}(x).$$

*In particular, if  $A$  is ample, this transform is globally generated and has vanishing first cohomology.*

*Proof.* After scaling a generator, write  $\lambda = \mathbb{C}(a_x, 1)$  with  $a_x \in A_x$ . Global generation gives a section  $a \in H^0(\mathbb{P}^1, A)$  with  $a(x) = a_x$ . The bundle automorphism

$$(u, c) \longmapsto (u - ca, c)$$

of  $A \oplus \mathcal{O}_{\mathbb{P}^1}$  has inverse  $(u, c) \mapsto (u + ca, c)$  and sends  $(a_x, 1)$  to  $(0, 1)$ . Positive elementary transforms are functorial under bundle automorphisms, and the transform in the direction  $\mathbb{C}(0, 1)$  is  $A \oplus \mathcal{O}_{\mathbb{P}^1}(x)$ . If  $A$  is ample, the splitting theorem gives  $A \cong \bigoplus_\nu \mathcal{O}_{\mathbb{P}^1}(a_\nu)$  with  $a_\nu \geq 1$ . Each summand and  $\mathcal{O}_{\mathbb{P}^1}(x) \cong \mathcal{O}_{\mathbb{P}^1}(1)$  is globally generated and has zero  $H^1$ , proving the last assertion.  $\square$

The handle and teeth meet along a tree. Surjective evaluation at each node allows componentwise global generation and cohomology vanishing to pass to the whole curve.

**Lemma 6.6.** *Let  $C$  be a connected nodal curve whose irreducible components are smooth and whose dual graph is a tree. Let  $E$  be a vector bundle on  $C$ . If, for every irreducible component  $C_\alpha \subset C$ , the bundle  $E|_{C_\alpha}$  is globally generated and*

$$H^1(C_\alpha, E|_{C_\alpha}) = 0,$$

*then  $E$  is globally generated and  $H^1(C, E) = 0$ .*

*Proof.* We argue by induction on the number of irreducible components. The one-component case is the hypothesis. Otherwise choose a leaf component  $L$  of the dual tree, let  $C'$  be the closure of  $C \setminus L$ , and write  $x = L \cap C'$ . Let  $j_{C'} : C' \hookrightarrow C$  and  $j_L : L \hookrightarrow C$  be the inclusions, and let  $E_x$  also denote the skyscraper sheaf with that fiber. The normalization sequence at  $x$  is

$$0 \longrightarrow E \longrightarrow (j_{C'})_*(E|_{C'}) \oplus (j_L)_*(E|_L) \longrightarrow E_x \longrightarrow 0,$$

where the last map is the difference of the two evaluations at  $x$ . By induction,  $E|_{C'}$  is globally generated and has vanishing first cohomology; global generation and  $H^1(L, E|_L) = 0$  are hypotheses for  $L$ . The map on global sections to  $E_x$  is surjective, since either component has surjective evaluation at  $x$ . The exact segment

$$H^0(C', E|_{C'}) \oplus H^0(L, E|_L) \longrightarrow E_x \longrightarrow H^1(C, E) \longrightarrow 0$$

therefore gives  $H^1(C, E) = 0$ .

To prove global generation at  $y \in C' \setminus \{x\}$ , prescribe  $v \in E_y$ . Choose  $s' \in H^0(C', E|_{C'})$  with  $s'(y) = v$  and  $s_L \in H^0(L, E|_L)$  with  $s_L(x) = s'(x)$ . The normalization sequence glues  $(s', s_L)$  to a section of  $E$  with value  $v$  at  $y$ . For  $y \in L \setminus \{x\}$ , first prescribe the section on  $L$  and then match on  $C'$ . For  $y = x$ , prescribe the same value  $v \in E_x$  on both components and glue. These cases cover  $C$  and complete the induction.  $\square$

For the comb, these normal-bundle conditions must persist under smoothing with a prescribed line-bundle twist. The degree-one projection then identifies the smoothed curve with the base and converts its normal bundle into a vertical tangent bundle.

**Lemma 6.7.** *Let  $Y$  be a smooth projective variety, let  $B$  be a smooth projective curve, and let  $p : Y \rightarrow B$  be a morphism. Let  $C \subset Y$  be a connected nodal curve which is a local complete intersection in  $Y$ , and write  $q = p|_C$ . Assume that*

$$q_*[C] = [B],$$

*that the normal bundle  $N_{C/Y}$  is globally generated, and that*

$$H^1(C, N_{C/Y}) = 0.$$

*Let  $M$  be a line bundle on  $B$  and suppose in addition that*

$$H^1(C, N_{C/Y} \otimes q^*M) = 0.$$

*Then  $C$  has an embedded smoothing  $C_t \subset Y$  over a pointed smooth curve  $(T, 0)$ , with  $C_0 = C$ , such that, for all sufficiently small nonzero  $t$ ,*

$$H^1(C_t, N_{C_t/Y} \otimes (p|_{C_t})^*M) = 0.$$

*The curve  $C_t$  is connected and smooth, the morphism  $p|_{C_t} : C_t \rightarrow B$  has degree one, and hence  $p|_{C_t}$  is an isomorphism. If  $h_t : B \rightarrow Y$  is the resulting section, then*

$$N_{C_t/Y} \cong h_t^*T_{Y/B}.$$

*Proof.* Global generation supplies a direction smoothing every node, and the untwisted  $H^1$  vanishing makes the chosen smoothing direction unobstructed. The twisted vanishing then persists by semicontinuity, while  $q_*[C] = [B]$  recovers a section from the smooth curve.

At every node  $x$  of  $C$ , the local embedded deformation space has a one-dimensional smoothing quotient

$$N_{C/Y}|_x \longrightarrow T_{C,x}^1 \cong \mathbb{C}.$$

Here  $T_{C,x}^1$  is the space of first-order deformations of the node. The map  $N_{C/Y}|_x \rightarrow T_{C,x}^1$  is onto: in local coordinates in which the two branches are the coordinate axes, the deformation  $uv = t$  smooths the node. For each node  $x$ , compose evaluation with the smoothing quotient to obtain a surjection  $\lambda_x : H^0(C, N_{C/Y}) \rightarrow T_{C,x}^1 \cong \mathbb{C}$ . Global generation ensures surjectivity of evaluation. Since the node set is finite, one can choose

$$\sigma \in H^0(C, N_{C/Y}) \setminus \bigcup_{x \in \text{Sing } C} \ker \lambda_x.$$

Then  $\lambda_x(\sigma) \neq 0$  at every node, so  $\sigma$  smooths all nodes to first order.

Since  $C$  is a local complete intersection in the smooth variety  $Y$ , the embedded first-order deformations of  $C$  have tangent space  $H^0(C, N_{C/Y})$  and obstructions in  $H^1(C, N_{C/Y})$  by [13, Theorem 6.2(b) and Remark 6.2.1]. The local lifting hypothesis in that theorem holds because  $C$  is a local complete intersection in a smooth ambient variety. Since  $H^1(C, N_{C/Y}) = 0$ , the Hilbert scheme is smooth at  $[C]$ , and the tangent vector  $\sigma$  integrates to a one-parameter embedded deformation

$$\mathcal{C} \subset Y \times T$$

over a smooth pointed curve  $(T, 0)$ . At each original node  $x$ , a local smoothing parameter  $a_x(t)$  satisfies  $a_x(0) = 0$  and  $a'_x(0) = \lambda_x(\sigma) \neq 0$ . Thus  $a_x(t) \neq 0$  for small  $t \neq 0$ . Away from node neighborhoods, openness of smoothness and compactness of  $C$  exclude new singularities after shrinking  $T$ . Moreover,  $h^0(C, \mathcal{O}_C) = 1$ . Upper semicontinuity gives  $h^0(C_t, \mathcal{O}_{C_t}) \leq 1$ , while constants give the opposite inequality; hence every sufficiently small nonzero fiber  $C_t$  is connected and smooth.

After another shrinking, every fiber  $C_t \subset Y$  is a local complete intersection. Since  $\mathcal{C} \rightarrow T$  is flat and locally of finite presentation and  $Y \times T \rightarrow T$  is smooth, the fiberwise criterion for relative regular immersions shows that

$$\mathcal{C} \hookrightarrow Y \times T$$

is a regular immersion which remains regular after arbitrary base change on  $T$  [22, Tags 063U and 063W]. Consequently

$$\mathcal{N} = N_{\mathcal{C}/(Y \times T)}$$

is locally free. Write  $\mathcal{I}$  for the ideal of  $\mathcal{C}$  in  $Y \times T$  and  $I_t$  for the ideal of  $C_t$  in  $Y$ . Then

$$(\mathcal{I}/\mathcal{I}^2)|_{C_t} \cong I_t/I_t^2.$$

Dualizing these locally free conormal sheaves gives

$$\mathcal{N}|_{C_t} \cong N_{C_t/Y}$$

for every  $t$ . Put

$$\mathcal{M} = (p \circ \text{pr}_Y|_{\mathcal{C}})^* M.$$

The sheaf  $\mathcal{N} \otimes \mathcal{M}$  is locally free on the flat family  $\mathcal{C} \rightarrow T$ , hence flat over  $T$ . Upper semicontinuity and the assumed vanishing on  $C = C_0$  give

$$H^1(C_t, N_{C_t/Y} \otimes (p|_{C_t})^* M) = 0$$

for all sufficiently small  $t$ .

Finally, let  $L$  be an ample line bundle on  $B$ . The pullback of  $L$  and the structure sheaf on the proper flat family  $\mathcal{C} \rightarrow T$  are flat over  $T$ . After shrinking to a connected neighborhood of 0, both Euler characteristics are constant [22, Tag 0B9T]. Riemann–Roch gives

$$\deg((p|_{C_t})^* L) = \chi(C_t, (p|_{C_t})^* L) - \chi(C_t, \mathcal{O}_{C_t}).$$

Thus  $\deg((p|_{C_t})^* L)$  is constant and equals  $\deg L > 0$ , because  $q_*[C] = [B]$ . A constant map has pullback degree zero, so  $p_t = p|_{C_t}$  is nonconstant. For a nonconstant map of projective curves,

$$\deg(p_t^* L) = \deg(p_t) \deg L.$$

It follows that  $\deg(p_t) = 1$ . A degree-one morphism between smooth connected projective curves is an isomorphism. Let  $h_t : B \rightarrow Y$  be the inverse of  $p_t$  followed by the inclusion  $C_t \hookrightarrow Y$ . Then  $dp \circ dh_t = \text{Id}_{TB}$ , so  $p$  is a submersion along  $h_t(B)$ . The map

$$\Pi : u \longmapsto u - dh_t(dp(u))$$

projects  $h_t^* TY$  onto  $h_t^* T_{Y/B}$  with kernel  $dh_t(TB)$ . The map  $\Pi$  identifies the normal quotient  $h_t^* TY/dh_t(TB)$  with  $h_t^* T_{Y/B}$ , proving the normal-bundle assertion.  $\square$

We can now combine localized elementary transforms with the normal-bundle calculations. Attaching teeth over a chosen Euclidean open set and smoothing the comb produces the required section with vanishing twisted first cohomology. For a smooth proper variety over an algebraically closed field of characteristic zero, rational connectedness is understood in the equivalent senses of [17, Definition–Theorem 2.1].

**Proposition 6.8.** *Let*

$$p : \mathcal{X} \longrightarrow B$$

*be a projective morphism from a smooth projective complex variety to a smooth projective curve. Assume that the geometric generic fiber is smooth and rationally connected. Let  $R \subset B$  be an open set over which  $p$  is smooth with rationally connected fibers. Given an effective divisor  $D$  supported in  $R$  and a nonempty Euclidean open set*

$$O \Subset R \setminus \text{supp } D,$$

*there is an algebraic section  $h : B \rightarrow \mathcal{X}$  such that*

$$H^1(B, h^*T_{\mathcal{X}/B}(-D)) = 0.$$

*The comb construction can be localized over  $O$ : after the stabilization  $\mathcal{X} \times \mathbb{P}^3 \rightarrow B$ , every tooth may be attached in a fiber over a point of  $O$ .*

*Proof.* The theorem of Graber–Harris–Starr [9, Theorem 1.1] gives a section  $h_0 : B \rightarrow \mathcal{X}$ . Since  $dp \circ dh_0 = \text{Id}_{T_B}$ ,  $p$  is a submersion along  $h_0(B)$ , even over a singular fiber. Thus  $h_0^*T_{\mathcal{X}/B}$  is a vector bundle on all of  $B$ . For embedded teeth in every relative dimension, put

$$\tilde{\mathcal{X}} = \mathcal{X} \times \mathbb{P}^3, \quad \tilde{p} = p \circ \text{pr}_{\mathcal{X}},$$

choose  $y_0 \in \mathbb{P}^3$ , and let

$$\tilde{h}_0 = (h_0, y_0) : B \longrightarrow \tilde{\mathcal{X}}.$$

The projective factor provides nonzero projected attachment directions and explicit embedded teeth, even if their projections to  $\mathcal{X}$  are not immersed. Each tooth meets the handle, whose projective component is constant, only at its attaching point. Finally, the direct-sum decomposition of the relative tangent bundle transfers the required  $H^1$  vanishing to the projected section  $h$ . The fibers of  $\tilde{p}$  over  $R$  are smooth rationally connected varieties of dimension at least three. Along  $\tilde{h}_0(B)$ , the splitting induced by  $d\tilde{p} \circ d\tilde{h}_0 = \text{id}_{T_B}$  gives

$$\tilde{E} := N_{\tilde{h}_0(B)/\tilde{\mathcal{X}}} \cong h_0^*T_{\mathcal{X}/B} \oplus (T_{\mathbb{P}^3, y_0} \otimes \mathcal{O}_B).$$

Apply Lemma 6.4 to  $\tilde{E}$  with  $N = \deg D + 1$ . For the integer  $m$  supplied by Lemma 6.4, let  $\mathcal{G} \subset \mathcal{H}_m(\tilde{E})$  be the locus on which every chosen line has nonzero projection to the summand  $T_{\mathbb{P}^3, y_0}$ . The locus  $\mathcal{G}$  is Zariski open and dense: its complement is the union, over the  $m$  entries, of the loci where the chosen line is contained in  $h_0^*T_{\mathcal{X}/B}$ . Apply the final assertion of Lemma 6.4 with this choice of  $\mathcal{G}$ . We obtain points  $b_i \in O$ , tangent directions  $\xi_i \subset \tilde{E}_{b_i}$ , and a positive elementary transform  $\tilde{E}'$  such that

$$H^1(B, \tilde{E}'(-D - b)) = 0 \quad (b \in B).$$

For each  $b \in B$ , twisting upward by  $b$ , by  $D$ , or by  $D + b$  gives

$$H^1(B, \tilde{E}'(-D)) = 0, \quad H^1(B, \tilde{E}'(-b)) = 0, \quad H^1(B, \tilde{E}') = 0.$$

For either  $F = \tilde{E}'(-D)$  or  $F = \tilde{E}'$ , denote the skyscraper sheaf with fiber  $F|_b$  by  $F_b$ . The exact sequence

$$0 \longrightarrow F(-b) \longrightarrow F \longrightarrow F_b \longrightarrow 0$$

and  $H^1(B, F(-b)) = 0$  imply that  $H^0(B, F) \rightarrow F_b$  is onto for every  $b \in B$ . Thus both bundles are globally generated.

The projective-space factor gives explicit embedded teeth. Identify

$$\tilde{E}_{b_i} = T_{\mathcal{X}_{b_i}, h_0(b_i)} \oplus T_{\mathbb{P}^3, y_0}.$$

Let  $\ell_i \subset T_{\mathbb{P}^3, y_0}$  be the projection of  $\xi_i$ ;  $\ell_i$  is a line because  $\xi_i$  has nonzero projective-space projection. Let

$$\nu_i : \mathbb{P}^1 \hookrightarrow \mathbb{P}^3$$

be the projective line through  $y_0$  with tangent line  $\ell_i$ , and choose its parameter so that  $\nu_i(0) = y_0$ . Put  $w_i = \nu_i'(0)$ . The projection maps  $\xi_i$  isomorphically onto  $\ell_i$ , so there is a unique  $v_i \in T_{\mathcal{X}_{b_i}, h_0(b_i)}$  such that  $(v_i, w_i) \in \xi_i$ . Theorem 3.1(i), applied in the smooth rationally connected fiber  $\mathcal{X}_{b_i}$ , gives a morphism

$$u_i : \mathbb{P}^1 \longrightarrow \mathcal{X}_{b_i}$$

with  $u_i(0) = h_0(b_i)$ ,  $u_i'(0) = v_i$ , and

$$H^1(\mathbb{P}^1, u_i^* T_{\mathcal{X}_{b_i}}(-2[0])) = 0.$$

Write  $u_i^* T_{\mathcal{X}_{b_i}} \cong \bigoplus_{\nu} \mathcal{O}(a_{\nu})$ . Since  $h^1(\mathbb{P}^1, \mathcal{O}(a_{\nu} - 2)) = \max\{1 - a_{\nu}, 0\}$ , the displayed vanishing implies  $a_{\nu} \geq 1$  for every summand. Put

$$T_i = (u_i, \nu_i)(\mathbb{P}^1) \subset \mathcal{X}_{b_i} \times \mathbb{P}^3.$$

The map  $(u_i, \nu_i)$  is an embedding because  $\nu_i$  is an embedding, and its tangent line at the attachment point is  $\xi_i$ . Moreover,

$$(u_i, \nu_i)^* T_{\tilde{\mathcal{X}}_{b_i}} = u_i^* T_{\mathcal{X}_{b_i}} \oplus \nu_i^* T_{\mathbb{P}^3}$$

is ample: the first summand is ample by  $a_{\nu} \geq 1$ , and  $\nu_i^* T_{\mathbb{P}^3} \cong \mathcal{O}(2) \oplus \mathcal{O}(1)^{\oplus 2}$ . Hence the quotient  $N_{T_i/\tilde{\mathcal{X}}_{b_i}}$  is ample. Since the points  $b_i$  are distinct, the teeth are pairwise disjoint. The handle has fixed  $\mathbb{P}^3$  coordinate  $y_0$ , and the embedding  $\nu_i$  satisfies  $\nu_i^{-1}(y_0) = \{0\}$ . Thus each tooth meets the handle at exactly its attachment point. The two tangent lines there are distinct because the tooth's tangent is vertical and the handle maps isomorphically to  $B$ . Hence

$$C = \tilde{h}_0(B) \cup T_1 \cup \cdots \cup T_m \subset \tilde{\mathcal{X}}$$

is a connected nodal curve which is a local complete intersection in  $\tilde{\mathcal{X}}$  and whose dual graph is a tree. Write  $N_C = N_{C/\tilde{\mathcal{X}}}$  and  $q = \tilde{p}|_C$ .

The normal-sheaf calculation of [9, Lemma 2.6] identifies

$$N_C \otimes \mathcal{O}_{\tilde{h}_0(B)} \cong \tilde{E}'.$$

On a tooth  $T_i$ , the exact sequence

$$0 \longrightarrow N_{T_i/\tilde{\mathcal{X}}_{b_i}} \longrightarrow N_{T_i/\tilde{\mathcal{X}}} \longrightarrow \mathcal{O}_{T_i} \longrightarrow 0$$

splits because

$$\text{Ext}^1(\mathcal{O}_{T_i}, N_{T_i/\tilde{\mathcal{X}}_{b_i}}) = H^1(T_i, N_{T_i/\tilde{\mathcal{X}}_{b_i}}) = 0;$$

the last vanishing follows from ampleness on  $T_i \cong \mathbb{P}^1$ . By [14, Proposition 23], applied with ambient variety  $\tilde{\mathcal{X}}$ , smooth divisor  $\tilde{\mathcal{X}}_{b_i}$ , curve  $T_i$  in that divisor, and the handle as the transverse branch, restriction of the normal bundle of the nodal union to  $T_i$  is the positive elementary transform of this split bundle in the direction of the handle. The handle direction has nonzero projection to the last trivial summand, since the handle is transverse to the fiber. Lemma 6.5 therefore gives

$$N_C \otimes \mathcal{O}_{T_i} \cong N_{T_i/\tilde{\mathcal{X}}_{b_i}} \oplus \mathcal{O}_{\mathbb{P}^1}(1),$$

which is globally generated and has vanishing first cohomology.

The handle maps isomorphically to  $B$  and every tooth is vertical, so  $q_*[C] = [B]$ . Let

$$\mathcal{L} = q^* \mathcal{O}_B(-D).$$

The restriction of  $\mathcal{L}$  to the handle is  $\mathcal{O}_B(-D)$ ; the restriction of  $\mathcal{L}$  to each vertical tooth is trivial. Consequently both  $N_C$  and  $N_C \otimes \mathcal{L}$  restrict to globally generated bundles with vanishing first cohomology on every irreducible component. By Lemma 6.6,

$$\begin{aligned} N_C \text{ is globally generated,} & \quad H^1(C, N_C) = 0, \\ N_C \otimes \mathcal{L} \text{ is globally generated,} & \quad H^1(C, N_C \otimes \mathcal{L}) = 0. \end{aligned}$$

Apply Lemma 6.7 with  $Y = \tilde{\mathcal{X}}$  and  $M = \mathcal{O}_B(-D)$ . This gives a nearby smooth curve which is the image of a section

$$\tilde{h} = (h, \ell) : B \longrightarrow \mathcal{X} \times \mathbb{P}^3$$

and satisfies

$$H^1(B, \tilde{h}^* T_{\tilde{\mathcal{X}}/B}(-D)) = 0.$$

Since

$$\tilde{h}^* T_{\tilde{\mathcal{X}}/B}(-D) \cong h^* T_{\mathcal{X}/B}(-D) \oplus \ell^* T_{\mathbb{P}^3}(-D),$$

the first cohomology of each direct summand vanishes. In particular,

$$H^1(B, h^* T_{\mathcal{X}/B}(-D)) = 0,$$

as required. Every tooth used above lies over one of the chosen points  $b_i \in O$ .  $\square$

The localization conclusion concerns the attachment fibers; the smoothed section need not equal  $h_0$  outside  $O$ .

## 7. COMPACT APPROXIMATION AND ALGEBRAIC LIFTINGS

**7.1. Approximation on a compact source.** The compact approximation argument transports the equation by a smooth bundle isomorphism  $J$ , represented locally by  $G$ . This introduces the linear coefficient  $L = G^{-1} \bar{\partial} G$  in addition to the residual  $e$ . The disk construction must make the residual small in  $L^4$  and the multiplication operator defined by the linear coefficient small from  $W^{1,4}$  to  $L^4$  before Lemma 4.5 applies. Near the prescribed data, holomorphy of the section makes  $e = 0$ , and holomorphy of  $J$  makes  $L = 0$ ; formula (7.2) records these two roles.

**Theorem 7.1.** *Let  $\Sigma$  be a compact Riemann surface, let  $X$  be a complex manifold, and let  $p : X \rightarrow \Sigma$  be a proper holomorphic map. Let  $\Omega \subset \Sigma$  be open such that  $p^{-1}(\Omega) \rightarrow \Omega$  is a holomorphic submersion with the relative first-jet sphere-family property. Let  $s : \Sigma \rightarrow X$  be a smooth section whose image lies in the submersion locus of  $p$ . Choose a vertical local addition near  $s(\Sigma)$ , and let  $E_{D_s}$  be  $s^* T_{X/\Sigma}$  with the holomorphic structure given by  $D_s = d\mathcal{S}_s(0)$  in Lemma 4.2. Let  $K \subset \Sigma$  be compact, let  $A \subset \Sigma$  be finite, and prescribe an integer  $k_a \geq 0$  for each  $a \in A$ . Put*

$$D = \sum_{a \in A} (k_a + 1) a.$$

*Assume that there is an open set  $U_h \subset \Sigma$  such that*

$$K \cup A \cup (\Sigma \setminus \Omega) \subset U_h,$$

*$s$  is holomorphic on  $U_h$ , and there are a holomorphic vector bundle  $V \rightarrow \Sigma$  with  $H^1(\Sigma, V) = 0$  and a smooth complex bundle isomorphism*

$$J : V \longrightarrow E_D := E_{D_s}(-D)$$

*which is holomorphic on  $U_h$ . Then, for every  $\epsilon > 0$ , there is a holomorphic section  $F : \Sigma \rightarrow X$  of  $p$  such that*

$$\sup_K d_X(F, s) < \epsilon, \quad j_a^{k_a} F = j_a^{k_a} s \quad (a \in A).$$

*Proof.* Put  $n = \text{rank } V = \text{rank } E_D$ . Fix smooth metrics and an area form on  $\Sigma$ , and a bounded right inverse  $T$  for  $\bar{\partial}_V$ , using  $H^1(\Sigma, V) = 0$  and Lemma 4.5. These choices precede the disk construction. Let  $C_S$  denote the Sobolev embedding constant for  $W^{1,4}(\Sigma, V) \hookrightarrow C^0(\Sigma, V)$ . We divide the proof into six steps.

*Step 1: the equation on the fixed bundle with vanishing first cohomology.* The local addition in the statement can be obtained by the finite-chart argument of Lemma 4.1, applied near the compact image  $s(\Sigma)$  in the submersion locus. Denote this local addition by  $\Psi$ . Its domain contains a fixed neighborhood of the zero section in  $s^*T_{X/\Sigma}$ . Applying Lemma 4.4 with jet divisor  $D$  gives the twisted section equation  $\mathcal{S}_D$  on a fixed fiber neighborhood in  $E_D$ .

Pull  $\mathcal{S}_D$  back through  $J$ :

$$\mathcal{T}(v) = J^{-1}\mathcal{S}_D(Jv).$$

In a holomorphic frame of  $V$ ,  $\mathcal{T}$  has the form

$$\mathcal{T}(v) = \bar{\partial}_V v + \mathcal{A}(z, v), \quad (7.1)$$

where  $\mathcal{A}$  is smooth in  $z$ , holomorphic in  $v$ , and contains no derivative of  $v$ . Put

$$e = \mathcal{A}(\cdot, 0), \quad L = d_v \mathcal{A}(\cdot, 0).$$

In holomorphic frames of  $V$  and  $E_D$ , if  $J$  is represented by  $G(z) \in \text{GL}_n(\mathbb{C})$  and the coefficient of  $\mathcal{S}_D$  is  $A_D$ , then

$$\mathcal{A}(z, q) = G^{-1}(\bar{\partial}G)q + G^{-1}A_D(z, Gq).$$

Since  $(A_D)_q(z, 0) = 0$ , evaluating at  $q = 0$  and differentiating in  $q$  give

$$e = G^{-1}A_D(z, 0), \quad L = G^{-1}\bar{\partial}G. \quad (7.2)$$

On  $U_h$ , holomorphy of  $s$  gives  $A_D(z, 0) = 0$  and hence  $e = 0$ ; holomorphy of  $J$  gives  $\bar{\partial}G = 0$  and hence  $L = 0$ . Hence

$$U = \{z \in \Sigma : |e(z)| + |L(z)| > 0\}$$

satisfies

$$\bar{U} \subset \Sigma \setminus U_h \Subset \Omega \setminus \text{supp } D.$$

In particular,  $\bar{U}$  is compact and avoids the jet divisor. If  $U = \emptyset$ , then  $e = L = 0$  and  $\mathcal{T}(0) = 0$ . The original section  $s$  is already holomorphic, so  $F = s$  proves the theorem. We henceforth assume that  $U \neq \emptyset$ .

*Step 2: local families for rational disks.* Cover  $\bar{U}$  by finitely many source coordinates, relative target charts, and holomorphic frames  $\tau_z : \mathbb{C}^n \rightarrow V_z$ . In such a frame, put

$$\Phi(z, q) = \Psi(s(z), \iota_z J_z \tau_z q), \quad (7.3)$$

where  $\iota : E_D \rightarrow E_{D_s}$  is the divisor morphism. For  $z \in \bar{U}$ , the maps  $\iota_z$ ,  $J_z$ , and  $\tau_z$  are isomorphisms, and

$$d_q \Phi(z, 0) = d_u \Psi(s(z), 0) \circ \iota_z J_z \tau_z = \iota_z J_z \tau_z.$$

Thus  $d_q \Phi(z, 0)$  is an isomorphism. Lemma 5.2 and its finite-local-family formulation therefore provide finitely many source neighborhoods, relative target charts, parameter balls, and prepared and matched families with common constants. Each source neighborhood retains its chosen holomorphic frame of  $V$ , which will be used for every disk selected there.

We verify the outer-collar coefficient. Let  $\phi$  be the vertical coordinate map of a selected relative target chart, and write  $\hat{\Phi} = \phi \circ \Phi$ . For a smooth map  $q = q(z)$  with values in the chosen parameter ball,

$$(\partial_q \hat{\Phi})^{-1} \bar{\partial}[\hat{\Phi}(z, q(z))] = \bar{\partial}q + C^\Phi(z, q),$$

where

$$C^\Phi = (\partial_q \hat{\Phi})^{-1} \partial_{\bar{z}} \hat{\Phi} d\bar{z}.$$

By the definitions of the normalized equation, the divisor twist, and the gauge, the left side equals  $\tau^{-1}J^{-1}\mathcal{S}_D(J\tau q)$ , the coordinate expression in the frame  $\tau$ . Comparing with (7.1) gives

$$C^\Phi = \mathcal{A} \quad (7.4)$$

in the chosen frame. Thus the coefficient of the matched family agrees with the gauged global coefficient on the outer collar of each selected disk and is exactly zero on the inner collar of that disk.

*Step 3: a finite Vitali selection.* Choose a finite smooth triangulation sufficiently fine that every closed triangle meeting  $\bar{U}$  is contained in one of the smaller source neighborhoods chosen in Step 2. Enumerate the closed triangles meeting  $U$  as  $T_1, \dots, T_M$ . Each  $T_j$  inherits the source coordinate, frame, and local families from its assigned neighborhood.

Before choosing  $r$ , fix a common fiber neighborhood for the equation. For the finitely many frames  $\tau$ , choose  $C_F \geq 1$  with  $|\tau_z^{-1}v| \leq C_F|v|$  on their compact source sets. Let  $d_* > 0$  be smaller than all parameter radii fixed in Step 2. Choose  $\rho > 0$  so that  $C_F\rho < d_*$  and the ball bundle  $V(\rho) = \{v \in V : |v| < \rho\}$  lies in the domain of  $\mathcal{A}$ . These choices depend only on the fixed local data, before the scale  $r$  and the disks are selected.

Given  $0 < r < \min\{1/16, \sigma_*\}$ , where  $\sigma_*$  is the common threshold from Lemma 5.2, use in each  $T_j$  the candidate disks  $\Delta_{c,\sigma}^+$  from (5.5) with

$$c \in U \cap \text{int } T_j, \quad 0 < \sigma \leq r, \quad \overline{\Delta_{c,\sigma}^+} \subset U \cap \text{int } T_j.$$

The fixed smooth area form defines a Radon measure. Apply the Vitali covering theorem for this measure as in Step 5 of Theorem 5.3; the triangle edges have area zero. In each  $T_j$ , truncate the countable family so that the uncovered area in  $U \cap \text{int } T_j$  is at most  $r^{1/2}/M$ . The resulting finite family of pairwise disjoint closed disks satisfies

$$\text{area} \left( U \setminus \bigcup_i \Delta_{c_i, \sigma_i}^+ \right) \leq r^{1/2}. \quad (7.5)$$

Put  $h_i = \sigma_i^{3/2}$  and let  $\mathcal{T}_r$  be the union of the transition annuli. As in (5.14), disjointness gives

$$\sum_i \sigma_i^2 \leq C, \quad \text{area}(\mathcal{T}_r) \leq Cr^{1/2}. \quad (7.6)$$

With  $\zeta_i$  the centered source coordinate assigned to disk  $i$ , put

$$\begin{aligned} \mathcal{I}_i &= \{z \in \Delta_{c_i, \sigma_i}^+ : |\zeta_i| < \sigma_i - h_i/2\}, \\ \mathcal{O}_r &= \Sigma \setminus \bigcup_i \{z \in \Delta_{c_i, \sigma_i}^+ : |\zeta_i| \leq \sigma_i + h_i/2\}. \end{aligned}$$

The regions  $\mathcal{O}_r, \mathcal{I}_i, \mathcal{T}_{c_i, \sigma_i}$  form an open cover of  $\Sigma$ .

*Step 4: the modified operator and the two small-area estimates.* On the fixed ball bundle  $V(\rho)$  chosen in Step 3, let  $\mathcal{A}_r$  be the coefficient obtained by using  $\mathcal{A}$  over  $\mathcal{O}_r$ , the zero coefficient over every  $\mathcal{I}_i$ , and the coefficient of the matched family over the corresponding transition annulus. Equation (7.4) and the open-collar properties make  $\mathcal{A}_r$  a smooth bundle map. On overlaps of holomorphic frames, the zero-order coefficients transform tensorially, so the formulas represent one coefficient on  $V(\rho)$ . The uniform estimates of Lemma 5.2 give

$$\sup_{V(\rho)} (|\mathcal{A}_r| + |d_v \mathcal{A}_r| + |d_v^2 \mathcal{A}_r|) \leq C, \quad (7.7)$$

with  $C$  independent of the number and radii of the disks. Put

$$\mathbb{X} = W^{1,4}(\Sigma, V), \quad \mathbb{Y} = L_{0,1}^4(\Sigma, V), \quad \mathcal{T}_r(v) = \bar{\partial}_V v + \mathcal{A}_r(z, v(z)), \quad \|v\|_{C^0} < \rho.$$

Put

$$e_r = \mathcal{A}_r(\cdot, 0), \quad L_r = d_v \mathcal{A}_r(\cdot, 0)$$

and

$$\mathcal{E}_r = \left( U \setminus \bigcup_i \Delta_{c_i, \sigma_i}^+ \right) \cup \mathcal{T}_r.$$

Both  $e_r$  and  $L_r$  vanish outside the measurable set  $\mathcal{E}_r$ . Indeed, the entire coefficient is zero on each inner region; on the unmodified outer region  $\mathcal{A}_r = \mathcal{A}$  has constant and linear terms  $e$  and  $L$ , which vanish off  $U$ ; and on the annuli the two terms are uniformly bounded by (7.7). Equations (7.5)–(7.6) give

$$\text{area}(\mathcal{E}_r) \leq Cr^{1/2}.$$

Consequently

$$\|\mathcal{T}_r(0)\|_{\mathbb{Y}} = \|e_r\|_{L^4} \leq Cr^{1/8}. \quad (7.8)$$

Moreover, for every  $\xi \in \mathbb{X}$ ,

$$\begin{aligned} \|(d\mathcal{T}_r(0) - \bar{\partial}_V)\xi\|_{L^4} &= \|L_r \xi\|_{L^4(\mathcal{E}_r)} \\ &\leq C \text{area}(\mathcal{E}_r)^{1/4} \|\xi\|_{C^0} \\ &\leq Cr^{1/8} \|\xi\|_{W^{1,4}}, \end{aligned}$$

where the last step uses  $W^{1,4} \hookrightarrow C^0$ . Hence

$$\|d\mathcal{T}_r(0) - \bar{\partial}_V\|_{\mathbb{X} \rightarrow \mathbb{Y}} \leq Cr^{1/8}. \quad (7.9)$$

The linear gauge defect need not be pointwise small. After the disk replacements, both  $e_r$  and  $L_r$  vanish outside  $\mathcal{E}_r$ , whose area is  $O(r^{1/2})$ . The resulting  $O(r^{1/8})$  estimates control the residual in  $L^4$  and the linear perturbation from  $W^{1,4}$  to  $L^4$ . In the bordered construction, the residual can be nonzero only on the uncovered defect set and the transition annuli, whereas the linear perturbation vanishes outside the transition annuli of the rational disks. These estimates concern the measurable nonzero loci, not their closed supports. No small-area bound is asserted for the set on which the full nonlinear coefficient  $\mathcal{A}_r(z, \cdot)$  differs from  $\mathcal{A}(z, \cdot)$ .

*Step 5: solution on the fixed bundle.* Apply Lemma 4.5 to  $V$  and  $\mathcal{A}_r$ . The cohomology vanishing is a hypothesis, the uniform fiber bounds are (7.7), and the two smallness assumptions follow from (7.8) and (7.9), with  $\varepsilon_r = C_1 r^{1/8}$  for a fixed sufficiently large  $C_1$ . Thus, for small  $r$ , there is a smooth solution  $v_r$  satisfying

$$\mathcal{T}_r(v_r) = 0, \quad \|v_r\|_{W^{1,4}} \leq Cr^{1/8}. \quad (7.10)$$

For every assigned frame,

$$|\tau_z^{-1} v_r(z)| \leq C_F C_S \|v_r\|_{W^{1,4}} \leq Cr^{1/8} < d_*$$

for small  $r$ . Thus each coordinate parameter belongs to its assigned ball. Smoothness uses the open-collar construction for each fixed  $r$ , as in the solution lemma.

*Step 6: global target-valued reconstruction.* Let  $q_i = \tau_i^{-1} v_r$ , where  $\tau_i$  is the holomorphic frame assigned to the  $i$ th selected disk. On all of  $\Sigma$ , put

$$F_r(z) = \begin{cases} H_{c_i, \sigma_i}(z, q_i(z)), & z \in \mathcal{I}_i, \\ M_{c_i, \sigma_i}(z, q_i(z)), & z \in \mathcal{T}_{c_i, \sigma_i}, \\ \Psi(s(z), \iota_z J_z v_r(z)), & z \in \mathcal{O}_r. \end{cases}$$

On each inner collar, the cutoff  $\eta_{c_i, \sigma_i} = 1$  gives  $M_{c_i, \sigma_i} = H_{c_i, \sigma_i}$ ; on each outer collar,  $\eta_{c_i, \sigma_i} = 0$  gives  $M_{c_i, \sigma_i} = \Phi$ . Thus the formulas glue smoothly. Each branch projects to  $z$  under  $p$ . On  $\mathcal{I}_i$ , the equation is  $\bar{\partial}q_i = 0$ , so

$$\bar{\partial}F_r = d_q H_{c_i, \sigma_i}(z, q_i(z)) \bar{\partial}q_i = 0.$$

On  $\mathcal{T}_{c_i, \sigma_i}$ , equation (5.11) gives

$$\bar{\partial}(\phi \circ F_r) = \partial_q \Xi_{c_i, \sigma_i}(\bar{\partial}q_i + C_{c_i, \sigma_i}^M(z, q_i)) = 0.$$

On  $\mathcal{O}_r$ , use the identity

$$\mathcal{S}_s(\iota J v_r) = \iota \mathcal{S}_D(J v_r)$$

which extends smoothly across  $\text{supp } D$  by Lemma 4.4. Since  $\mathcal{T}_r = \mathcal{T}$  on  $\mathcal{O}_r$ , the outer branch satisfies

$$\bar{\partial}F_r = d_s \Psi(s, \iota J v_r) \iota J \mathcal{T}_r(v_r) = 0.$$

The equality holds also at points of  $D$ , by the smooth extension. These open regions cover  $\Sigma$ , so  $F_r$  is a global holomorphic section of  $p$ .

Every selected disk lies in  $U$  and is disjoint from  $U_h$ , which contains  $K \cup A$ . The approximation estimate uses only  $U_h$ . On a neighborhood of  $K \cup A$ ,

$$F_r = \Psi(s, \iota J v_r).$$

Let  $C_\Psi$  bound the fiber derivative of  $\Psi$  on the fixed neighborhood of the zero section used over  $K$ . The bounds for  $\iota$  and  $J$  over a compact neighborhood of  $K$ , together with Sobolev embedding and (7.10), give

$$\sup_K d_X(F_r, s) \leq C_\Psi \sup_K \|\iota_z J z\| \|v_r\|_{C^0} \leq C r^{1/8}.$$

Near  $a \in A$ , choose a source coordinate  $z$  centered at  $a$  and compatible holomorphic frames. Then  $\iota$  is multiplication by  $z^{k_a+1}$  and  $J$  is holomorphic. Boundedness of  $J v_r$  gives  $\iota J v_r = O(|z|^{k_a+1})$ . For a common target chart  $\phi$ ,

$$h(z) = \phi(F_r(z)) - \phi(s(z)) = O(|z|^{k_a+1}).$$

The function  $h$  is holomorphic, so  $h(z)/z^{k_a+1}$  extends holomorphically across 0. Therefore  $j_a^{k_a} F_r = j_a^{k_a} s$ . Choose  $r$  with  $C r^{1/8} < \epsilon$  to obtain the required approximation.  $\square$

**7.2. Completion of the algebraic proof.** To apply the compact approximation theorem to an affine source, we first complete the family without changing the morphism over that source.

**Lemma 7.2.** *Let  $p_R : X_R \rightarrow R$  be a projective morphism from a smooth variety to a smooth affine curve. There are a smooth projective completion  $R \subset B$  and a projective morphism  $p : \mathcal{X} \rightarrow B$  with  $\mathcal{X}$  smooth projective such that  $\mathcal{X}|_R \cong X_R$  over  $R$ .*

*Proof.* Choose a smooth projective completion  $R \subset B$ . Embed  $X_R$  as a closed subvariety of  $R \times \mathbb{P}^N$  and let  $\bar{X}_R$  be its Zariski closure in  $B \times \mathbb{P}^N$ . Since  $X_R$  is closed over  $R$ , the restriction of  $\bar{X}_R$  to  $R$  is exactly  $X_R$ . Take a projective resolution  $r : \mathcal{X} \rightarrow \bar{X}_R$  which is an isomorphism over the already smooth open subset  $X_R$ , using resolution of singularities in characteristic zero [15]. The morphism  $p = p_R \circ r : \mathcal{X} \rightarrow B$  has the required properties; no dominance assumption on  $p_R$  is needed.  $\square$

*Proof of Theorem 1.7.* The analytification  $\pi^{\text{an}}$  is a proper smooth submersion and hence a Serre fibration by Ehresmann's theorem [4]. It remains to prove the stronger lifting assertion in the theorem. Work with the algebraic pullback

$$p_R : X_R = R \times_Y Z \longrightarrow R.$$

Let  $s_f(z) = (z, f(z))$  be the continuous section of  $p_R^{\text{an}}$  associated with the original lifting  $f$ . By Lemma 7.2, choose a smooth projective completion  $R \subset B$  and a projective model  $p : \mathcal{X} \rightarrow B$

with smooth total space which equals  $X_R \rightarrow R$  over  $R$ . The morphism  $p$  may have bad fibers over  $B \setminus R$ .

We verify the hypothesis on the geometric generic fiber in Proposition 6.8 before applying it. Put  $\bar{\eta} = \text{Spec } \overline{\mathbb{C}(R)}$ , where  $\overline{\mathbb{C}(R)}$  is an algebraic closure of the function field of  $R$ , and write  $X_{\bar{\eta}} = X_R \times_R \bar{\eta}$ . This fiber is smooth and projective by base change. Since the complex fibers are connected,  $h^0((X_R)_b, \mathcal{O}_{(X_R)_b}) = 1$  for every  $b \in R(\mathbb{C})$ . Upper semicontinuity and compatibility of fiber cohomology with field extension give  $h^0(X_{\bar{\eta}}, \mathcal{O}_{X_{\bar{\eta}}}) = 1$  [22, Tag 0BDN]; nonemptiness and constant functions give the lower bound. Thus  $X_{\bar{\eta}}$  is connected and, being smooth over an algebraically closed field, integral.

In relative dimension zero,  $X_{\bar{\eta}}$  is a point and is rationally connected. In positive relative dimension, fix  $b \in R(\mathbb{C})$ . Holomorphic maps from  $\mathbb{P}^1$  to the projective fiber  $(X_R)_b$  are algebraic [23, Proposition 15], so its rational connectedness agrees with the algebraic definition. Definition–Theorem 2.1(3) of [17] gives a morphism  $u : \mathbb{P}^1 \rightarrow (X_R)_b$  with  $u^*T_{(X_R)_b}$  ample. It is nonconstant, hence very free in the sense of [14, Definition 15]; the fiber is therefore separably rationally connected. The restriction of  $p_R$  to  $\text{Spec } \mathcal{O}_{R,b}$  is a smooth proper local model with this special fiber. Thus  $b$  is a place of very good reduction in the sense of [14, Remark 28], which implies that the generic fiber is geometrically separably rationally connected over  $\mathbb{C}(R)$ . In particular,  $X_{\bar{\eta}}$  is rationally connected. Since  $\mathcal{X}|_R \cong X_R$ , this is also the geometric generic fiber of  $p : \mathcal{X} \rightarrow B$ , as required.

Put

$$D = \sum_{a \in A} (k_a + 1)a.$$

Since  $R$  is noncompact and  $K \cup A$  is compact, choose a nonempty Euclidean open set  $O \Subset R \setminus (K \cup A)$ . Proposition 6.8 gives an algebraic section  $h : B \rightarrow \mathcal{X}$  with

$$V = h^*T_{\mathcal{X}/B}(-D), \quad H^1(B, V) = 0.$$

Over  $R$ , the map  $p_R$  is a smooth proper bundle with connected simply connected fibers. Lemma 2.4 gives a homotopy  $H$  through sections from  $h|_R$  to  $s_f$ . Choose a smooth function  $\chi : R \rightarrow [0, 1]$  which is one near  $K \cup A$  and has compact support, and put

$$s_0(z) = H(z, \chi(z)).$$

The map  $s_0$  is a continuous section of  $p_R$ , equal to  $s_f$  near  $K \cup A$  and to  $h$  outside a compact subset of  $R$ . Apply the relative smoothing statement in Subsection 2.2 to the smooth bundle  $p_R : X_R \rightarrow R$ , relative to smaller neighborhoods of  $K \cup A$  and of the complement of a large compact set. We obtain a smooth section  $s_R : R \rightarrow X_R$  equal to  $s_f$  near  $K \cup A$  and to  $h$  near the ends of  $R$ , homotopic to  $s_0$  through sections by a homotopy fixed near  $K \cup A$  and near the ends of  $R$ . Since  $s_R = h$  near the ends of  $R$ , extending  $s_R$  by  $h$  gives a smooth section  $s : B \rightarrow \mathcal{X}$ . The formula

$$H^x(z, t) = H(z, t\chi(z))$$

gives a homotopy from  $h|_R$  to  $s_0$  which is fixed outside a compact subset of  $R$ . Concatenate  $H^x$  with the homotopy supplied by relative smoothing from  $s_0$  to  $s_R$ , using the first and second halves of the time interval. Extend that concatenation by the constant section  $h$  near  $B \setminus R$ . Denote the resulting homotopy from  $h$  to  $s$  by  $H^B$ . The homotopy  $H^B$  is fixed near  $B \setminus R$ . Thus the cutoff produces the homotopy on the compact curve; the section-homotopy lemma is applied only over the open curve  $R$ .

Reparametrize  $H^B$  to be constant near  $t = 0, 1$ . Regard the restriction of  $H^B$  to  $R \times [0, 1]$  as a section of the pullback of  $p_R$  by  $R \times [0, 1] \rightarrow R$ . The relative smoothing statement in Subsection 2.2, relative to the endpoints and the fixed region near  $B \setminus R$ , gives a smooth homotopy, still denoted  $H^B$ . The image of  $H^B$  lies in the submersion locus: over  $R$  the family is smooth, and near  $B \setminus R$  the homotopy equals the constant section  $h$ , along which  $dp \circ dh = \text{id}_{T_B}$  makes  $dp$  surjective even

over a bad fiber. Fix a vertical local addition near  $s(B)$  and let  $D_s = d_{\mathcal{S}_s}(0)$  be its linearized operator on  $s^*T_{\mathcal{X}/B}$ . Lemma 6.1, followed by tensoring with  $\mathcal{O}_B(-D)$ , gives a smooth complex bundle isomorphism

$$J_0 : h^*T_{\mathcal{X}/B}(-D) \longrightarrow E_{D_s}(-D).$$

$J_0$  supplies the global smooth identification and is the natural identity near  $B \setminus R$ . Lemma 4.3 shows that this identity is holomorphic there. Choose an open subsurface  $W \Subset R$  containing  $K \cup A$ ; this verifies the open-surface hypothesis of Lemma 6.2. Apply Lemma 6.2 with

$$P = K \cup A, \quad Q = B \setminus R.$$

We obtain a smooth isomorphism

$$J : V \longrightarrow E_{D_s}(-D)$$

which is holomorphic near  $K \cup A$  and agrees with the identity near  $B \setminus R$ .

Choose an open neighborhood  $U_h$  of  $K \cup A \cup (B \setminus R)$  with closure contained in the region where both  $s$  and  $J$  are holomorphic, and put  $\Omega = R$ . The map  $p$  is proper, and  $p|_R$  has the relative first-jet sphere-family property by Proposition 3.5. The image of  $s$  lies in the submersion locus,  $H^1(B, V) = 0$ , and  $s, J$  are holomorphic on  $U_h$ . Thus Theorem 7.1 applies. To transfer the metric estimate from  $\mathcal{X}$  to  $Z$ , write  $\text{pr}_Z : \mathcal{X}|_R \cong R \times_Y Z \rightarrow Z$ . Choose a compact neighborhood  $C_K$  of  $s(K)$  in  $\mathcal{X}|_R$ . First choose  $\delta_0 > 0$  so that the  $\delta_0$ -neighborhood of  $s(K)$  is contained in  $C_K$ . Uniform continuity of  $\text{pr}_Z|_{C_K}$  then gives  $0 < \delta < \delta_0$  such that

$$z \in K, \quad p(x) = z, \quad d_{\mathcal{X}}(x, s(z)) < \delta \implies d_Z(\text{pr}_Z(x), f(z)) < \epsilon.$$

Apply Theorem 7.1 with error  $\delta$  to obtain a holomorphic section

$$\tilde{F} : B \longrightarrow \mathcal{X}$$

approximating  $s$  in  $d_{\mathcal{X}}$  on  $K$  and having the prescribed jets at  $A$ . Since  $B$  and  $\mathcal{X}$  are projective, the graph form of Chow's theorem gives algebraicity of  $\tilde{F}$  [23, Proposition 15]. Restricting to  $R$  and projecting by  $\text{pr}_Z$  gives an algebraic lifting  $F : R \rightarrow Z$  with the required  $d_Z$  error. The relation  $s|_R = s_f$  holds near  $A$ , and holomorphic composition preserves jet equality. Hence

$$j_a^{k_a}(\text{pr}_Z \circ \tilde{F}) = j_a^{k_a}(\text{pr}_Z \circ s_f) = j_a^{k_a} f \quad (a \in A).$$

Finally, the maps

$$z \longmapsto (z, F^{\text{an}}(z)), \quad z \longmapsto (z, f(z))$$

are sections of  $X_R^{\text{an}} \rightarrow R^{\text{an}}$ . The fibers are connected and simply connected, so Lemma 2.4 gives a homotopy of these graph sections. Projection to  $Z^{\text{an}}$  gives a homotopy from  $f$  to  $F^{\text{an}}$  through continuous liftings of  $g^{\text{an}}$ .  $\square$

## AI USE DISCLOSURE

The central ideas, mathematical insights, and overall approach of this work originated with the authors. AI tools were used in a supporting role to assist in manuscript preparation and revision. The authors take full responsibility for the paper's content and the accuracy of its references.

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